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Lab Title: Lab 1 — Digital Forensics Case Handling, Autopsy and Sleuth Kit Analysis

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Introduction

This laboratory practical involved the digital forensic examination of the authorised disk image Ch01InChap01.dd using both Autopsy and The Sleuth Kit command-line tools. The examination was carried out in a controlled forensic environment to ensure that the original evidence was properly protected throughout the investigation. A dedicated folder structure was first created to organise the original evidence, working copy, recovered files, screenshots, reports, hash values, logs, and investigation notes. After downloading the forensic image, its file size and storage location were recorded. MD5 and SHA-256 hash values were then calculated, and the original image was set to read-only before a separate working copy was created and verified for analysis.

The working copy was examined using Autopsy. A new forensic case and host were created, and the disk image was added and recognised as a FAT12 volume. The integrity of the image was checked using its MD5 hash value. Autopsy was then used to locate deleted files, examine INCOME.XLS, recover relevant evidence, and perform keyword searches using terms such as income, billing, and client. Evidence notes were also added during the examination, and a forensic report was generated at the end of the Autopsy analysis.

The investigation was then repeated and verified using The Sleuth Kit. Commands such as `img_stat`, `fsstat`, `fls`, `istat`, `icat`, `blkcat`, `tsk_recover`, and `blkls` were used to examine the forensic image, identify deleted files, determine the metadata and data sectors linked to INCOME.XLS, and recover the spreadsheet using different methods. The recovered copies were compared using cryptographic hash values and byte-for-byte comparison to confirm that they were identical. Finally, the recovered spreadsheet was successfully opened in LibreOffice Calc, showing that the recovered file was complete and usable.

Objectives

By completing this lab, trainees will demonstrate their ability to apply the digital forensic investigation process in a controlled practical environment. Students will preserve and hash a forensic disk image, perform structured analysis using Autopsy, examine the image using The Sleuth Kit command-line tools, identify and recover deleted files, analyse file metadata and data units, perform keyword searches, verify recovered evidence using cryptographic hashes, and prepare a professional forensic report containing repeatable findings.

Students are expected to demonstrate practical competence with tools and commands including Autopsy, `img_stat`, `mmls`, `fsstat`, `fls`, `istat`, `icat`, `blkcat`, `blkls`, `tsk_recover`, `md5sum` and `sha256sum`.

Laboratory / Tools Used

Laboratory / Tools Used Tool / Environment	Use in the investigation
Kali Linux	Main forensic examination and command-line environment
VMware Workstation	Environment in which the Kali Linux system was operated
Autopsy Forensic Browser 2.24C	Case creation, evidence examination, deleted-file analysis, keyword searching, evidence notes, file export and reporting
The Sleuth Kit 4.12.1	Command-line forensic examination and recovery
Md5sum	MD5 integrity verification
Sha256sum	SHA-256 integrity verification
Cmp	Byte-for-byte comparison of recovered files
Img_stat	Identification of forensic image characteristics
fsstat	FAT12 filesystem examination
fls	File and deleted-file listings
istat	Metadata and data-sector examination
icat	Recovery of INCOME.XLS using metadata address 13
blkcat	Sector-based recovery of INCOME.XLS
Tsk_recover	Recovery of deleted and allocated files
Blklis	Extraction/examination of unallocated space
file	Verification of recovered file type
String /grep	Searching extracted unallocated data
LibreOffice Calc	Local examination of the recovered spreadsheet

Key Findings

Evidence integrity

The evidence examined was Ch01InChap01.dd, with a recorded size of 1,474,560 bytes. MD5 and SHA-256 hashes were calculated for the original evidence, and the original image was made read-only before a working copy was created. The working copy was hashed and compared against the original, with the investigation recording that the files were identical.

The recorded original image hashes were:

MD5: a117773bcf1fc88ec0ab8e0a349fbbcb

SHA-256: 3ce8053e4f3d9c8ab98b3aadb2480685efb8e4980d34297b83bd5a09b1a7b122

Filesystem

Autopsy and fsstat identified the filesystem as FAT12. The image was successfully attached to the Autopsy case after being treated as a volume image, and the Autopsy MD5 integrity check passed.

Deleted files identified

The investigation identified the following deleted files:

File	Metadata address
Billing Letter.doc	8
confirmation.txt	11
letter1.txt	15
Regrets.doc	17

These deleted entries were identified in Autopsy and independently confirmed using the recursive fls deleted-file listing.

INCOME.XLS

INCOME.XLS was identified as an allocated file with:

Metadata address: 13

File size: 13,824 bytes

Data sectors: 285–311

MD5: 6a2e65afc5af4fc5f9da2859df134eac

Its metadata and sectors were examined using both Autopsy and istat.

Deleted Billing Letter.doc

The deleted Billing Letter.doc was examined and recovered. The work records:

Metadata address: 8

Status: Not Allocated

File size: 24,064 bytes

Data sectors: 237–283

MD5: 9fe241d0dde27e83442010b3eee5ad32

Its recovery was verified using hashes.

Keyword searches

Autopsy searches were performed for:

income

billing

client

The screenshots show results being returned for these searches, including multiple occurrences associated with the search terms.

Multiple recovery methods

INCOME.XLS was recovered using:

Autopsy export

icat

blkcat

tsk_recover

The recovered copies produced identical hashes and were subsequently confirmed to be byte-for-byte identical using cmp.

The common recovered-file hashes recorded in the investigation were:

MD5: 6a2e65afc5af4fc5f9da2859df134eac

SHA-256: 8d7cd7204d3dae161a8fa879e184865b3bc4a57a4e688abd522a9ff03f62252d

tsk_recover

The default tsk_recover operation recovered the deleted files, while allocated recovery was also performed to obtain Income.xls. The recovered Income.xls was compared with the icat copy.

Unallocated-space analysis

blkl was used to extract unallocated data from the evidence image. The resulting unallocated-space file was subsequently examined using keyword searching.

Successful spreadsheet examination

The recovered INCOME.XLS was successfully opened using LibreOffice Calc. The final screenshot shows the workbook displaying a January Cash Flow worksheet with January, February and March worksheet tabs. This confirms that the recovered spreadsheet was readable and could be examined locally.

Task 1: Creating a dedicated forensic working directory

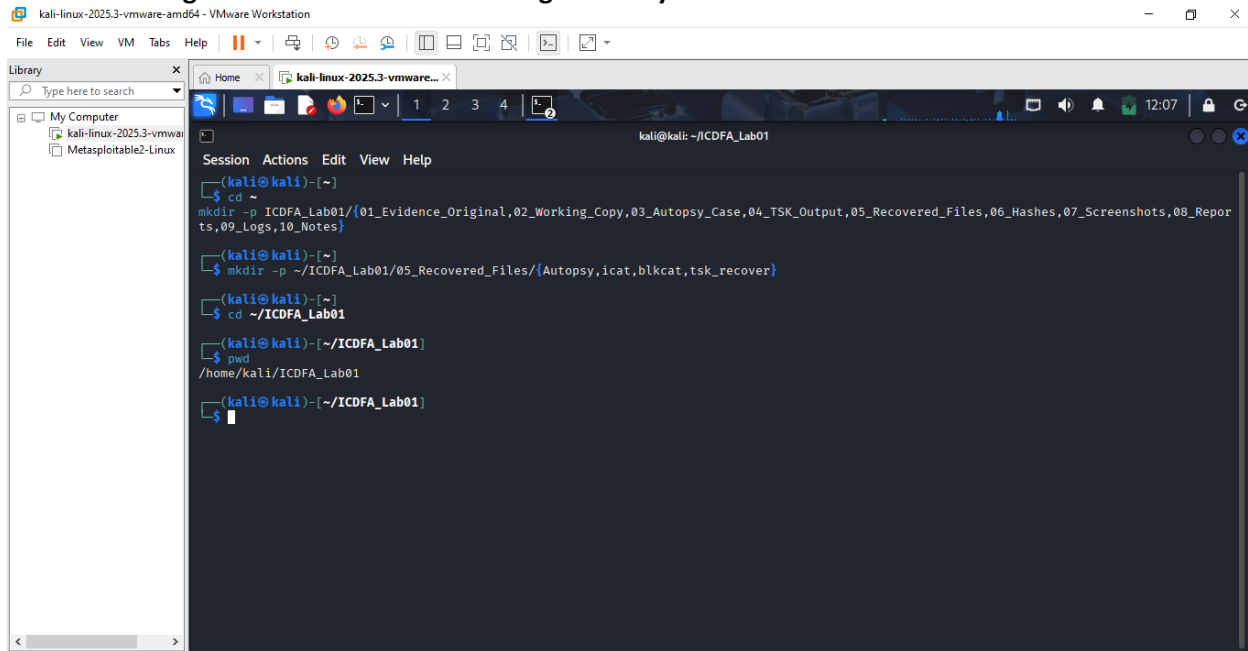


Fig 1. Creating a dedicated forensic working directory.

Explanation

I created separate folders to keep the forensic evidence and analysis files properly organised.

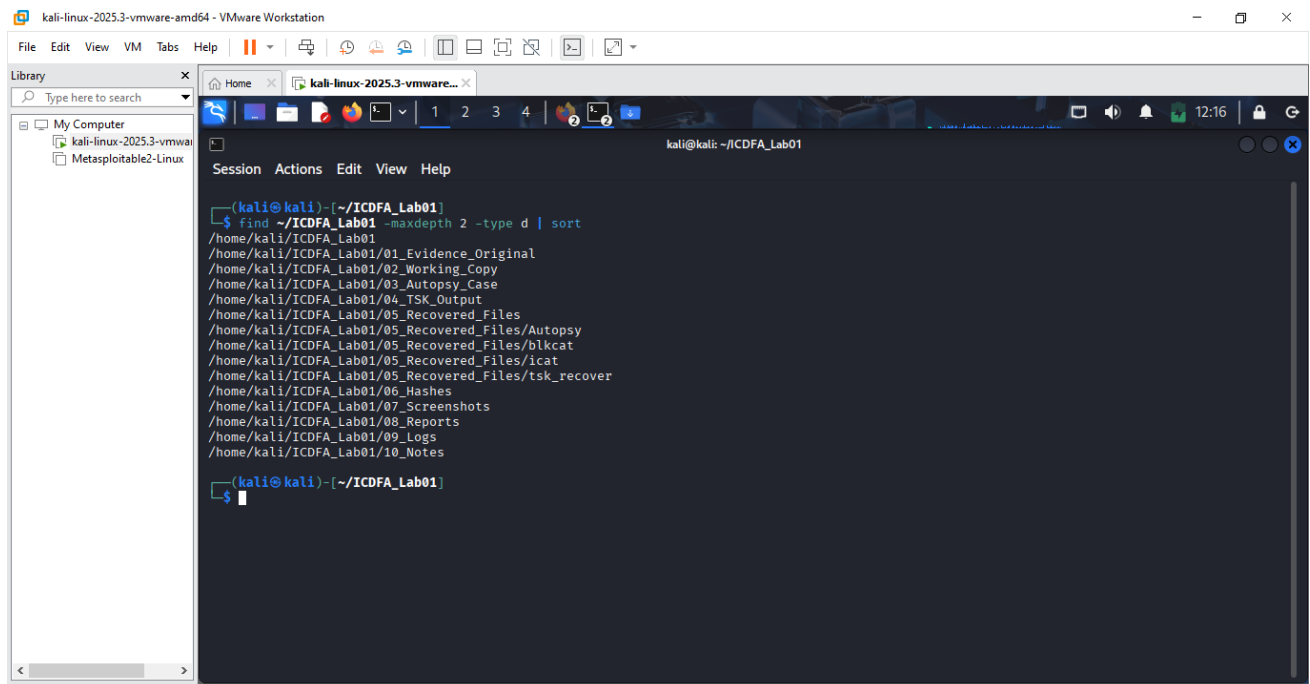


Fig 1.1 Lab folder structure.

Explanation

This confirms that all the required lab folders were created successfully.

Part A — Autopsy Analysis

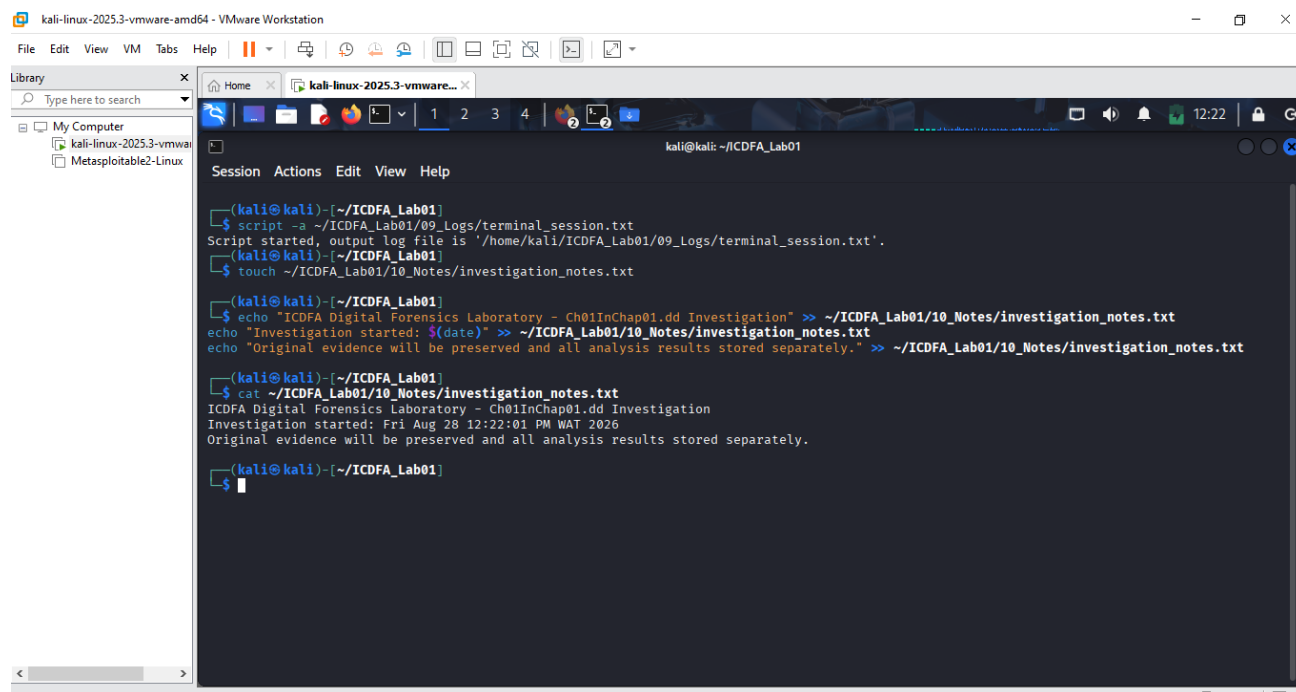


Fig 1.2 Investigation setup and logging.

Explanation

I started recording my investigation steps and terminal activity for proper documentation.

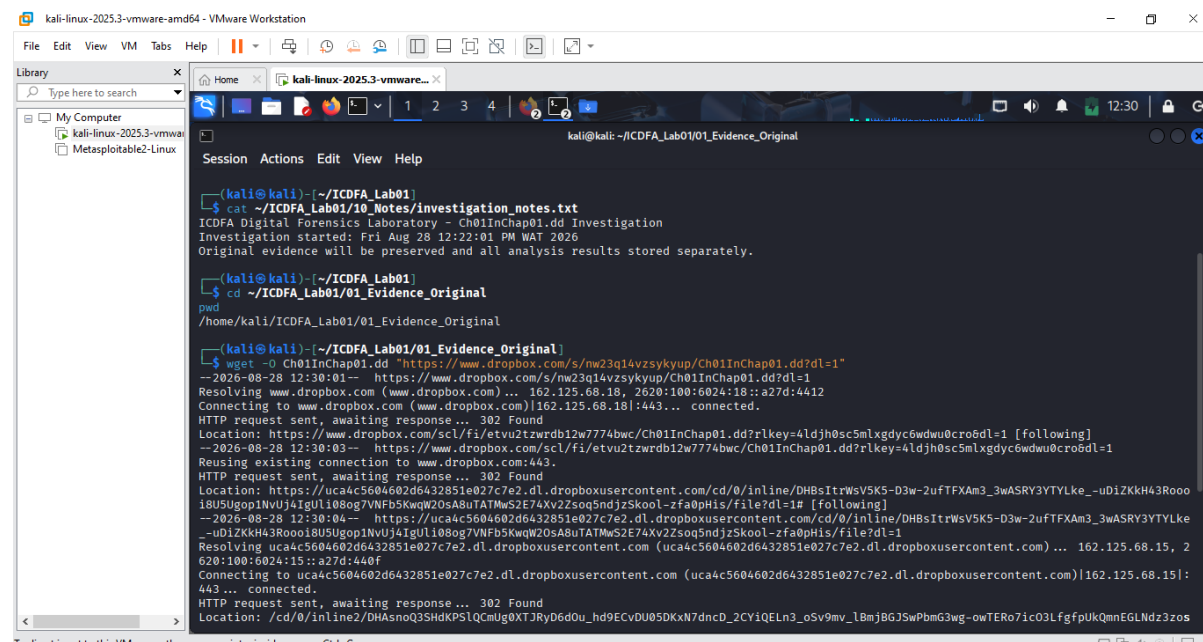


Fig 1.3 Downloading the authorized evidence image.

Explanation

I downloaded the authorised Ch01InChap01.dd forensic image into the evidence folder.

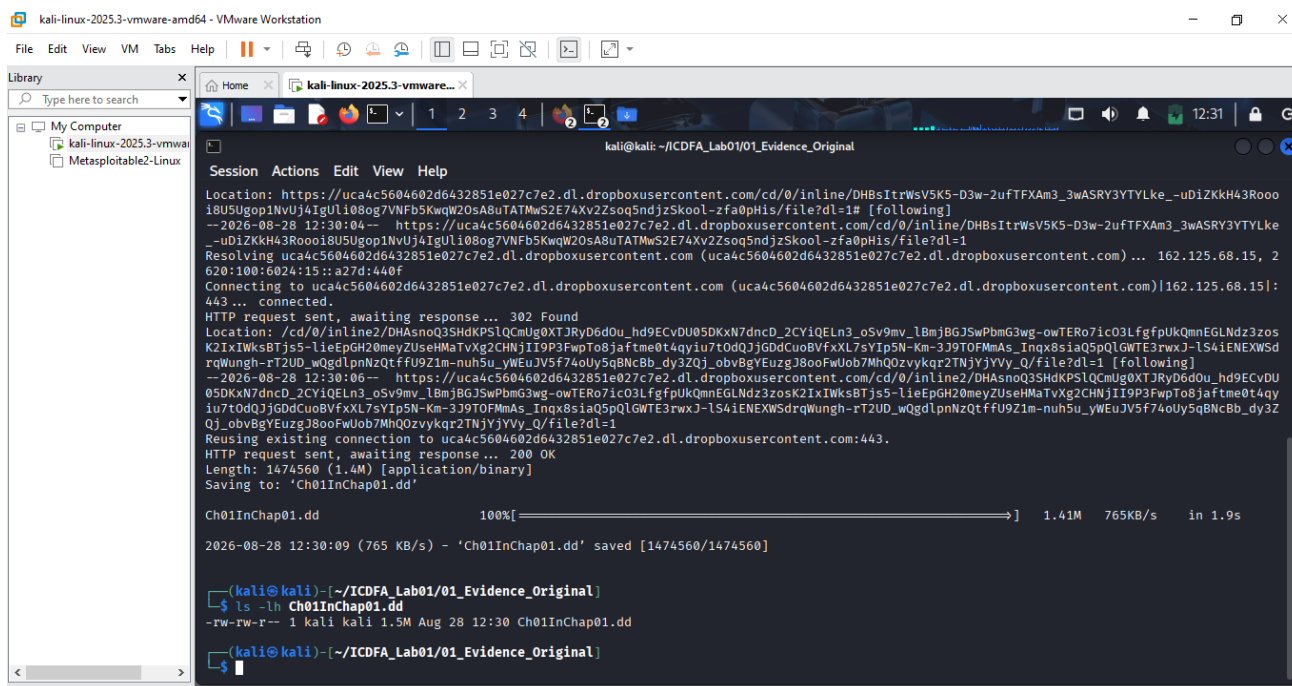


Fig 1.4 Confirming the download.

Explanation

This confirms that the evidence image finished downloading successfully.

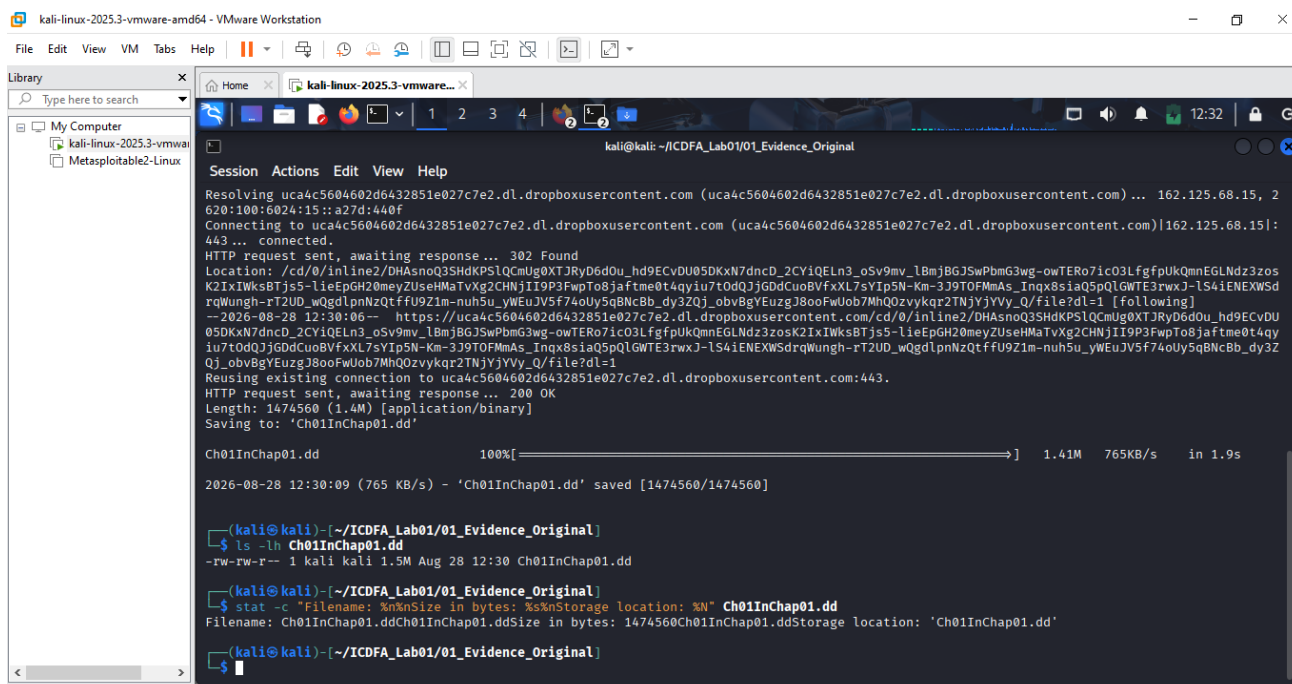


Fig 1.5 Original Evidence Information.

Explanation

I recorded the basic information about the original forensic image.

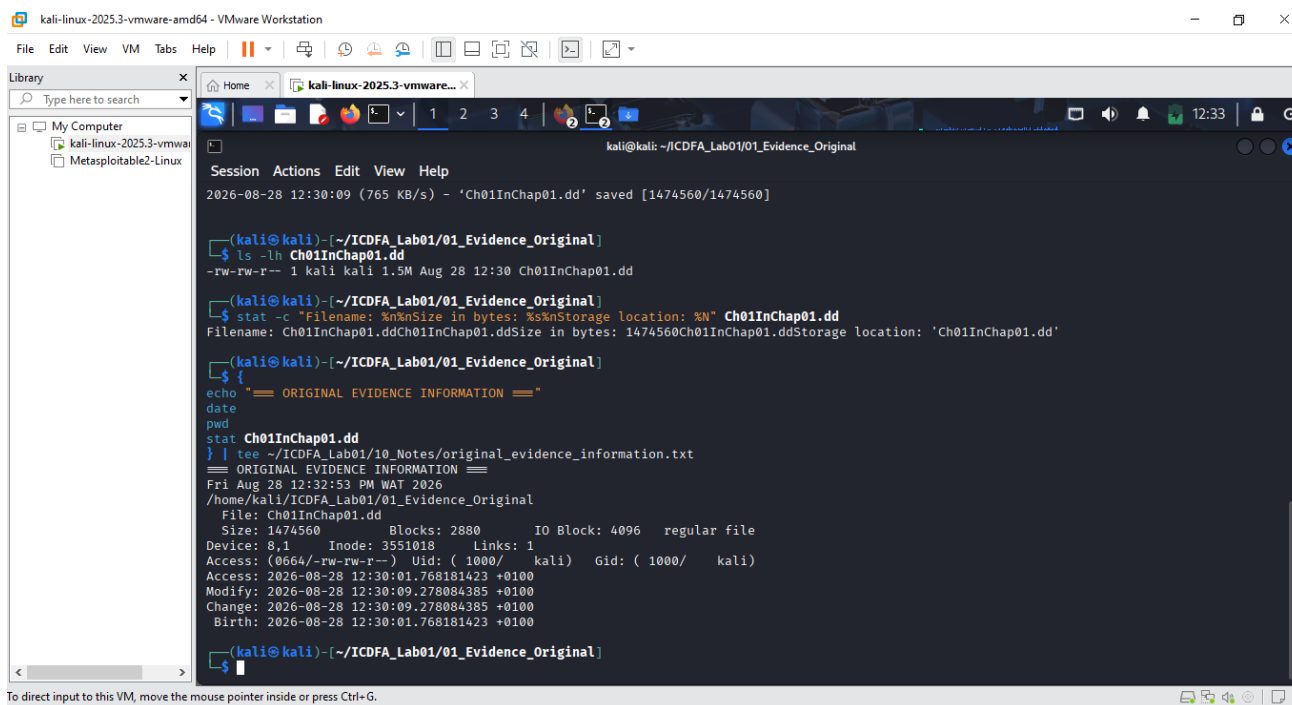


Fig 1.6 Filename, size and storage location.

Explanation

This shows the evidence filename, exact size and where it was stored.

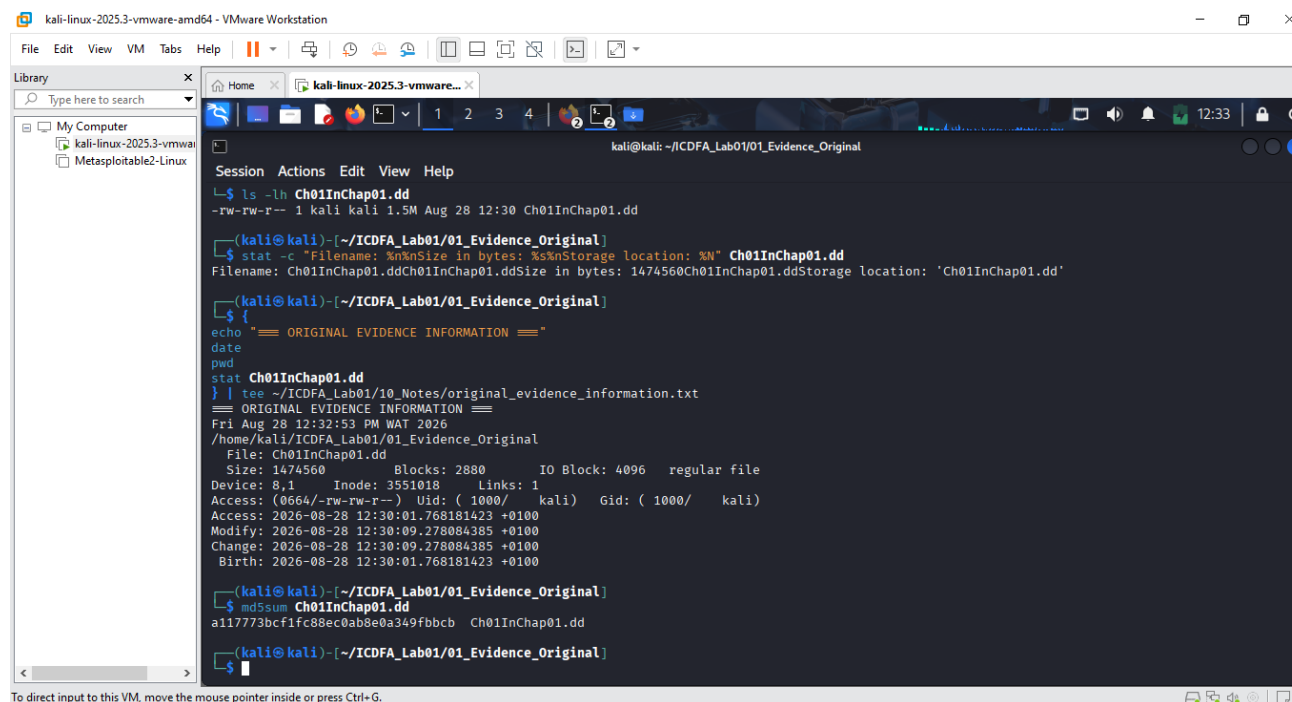
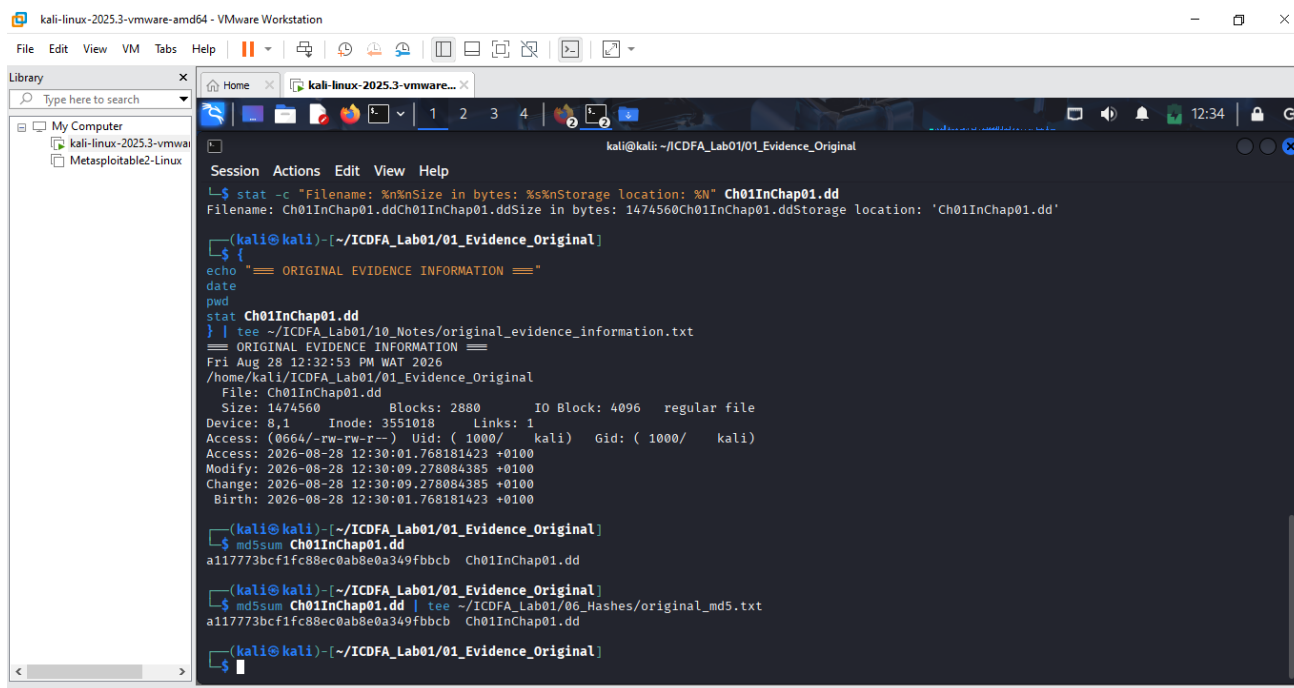


Fig 1.7 Calculating the original MD5 hash.

Explanation

I calculated the MD5 hash of the original image to establish its integrity.



```
kali@kali: ~/ICDFA_Lab01/01_Evidence_Original
$ stat -c "Filename: %n\nSize in bytes: %s\nStorage location: %N" Ch01InChap01.dd
Filename: Ch01InChap01.ddCh01InChap01.ddSize in bytes: 1474560Ch01InChap01.ddStorage location: 'Ch01InChap01.dd'

(kali@kali)~/ICDFA_Lab01/01_Evidence_Original
$ {
echo " == ORIGINAL EVIDENCE INFORMATION == "
date
pwd
stat Ch01InChap01.dd
} | tee ~/ICDFA_Lab01/10_Notes/original_evidence_information.txt
== ORIGINAL EVIDENCE INFORMATION ==
Fri Aug 28 12:32:53 PM WAT 2026
/home/kali/ICDFA_Lab01/01_Evidence_Original
File: Ch01InChap01.dd
Size: 1474560      Blocks: 2880      IO Block: 4096   regular file
Device: 8,1      Inode: 3551018    Links: 1
Access: (0664/-rw-rw-r--)  Uid: ( 1000/   kali)   Gid: ( 1000/   kali)
Access: 2026-08-28 12:30:01.768181423 +0100
Modify: 2026-08-28 12:30:09.278084385 +0100
Change: 2026-08-28 12:30:09.278084385 +0100
Birth: 2026-08-28 12:30:01.768181423 +0100

(kali@kali)~/ICDFA_Lab01/01_Evidence_Original
$ md5sum Ch01InChap01.dd
a117773bcf1fc88ec0ab8e0a349fbcb  Ch01InChap01.dd

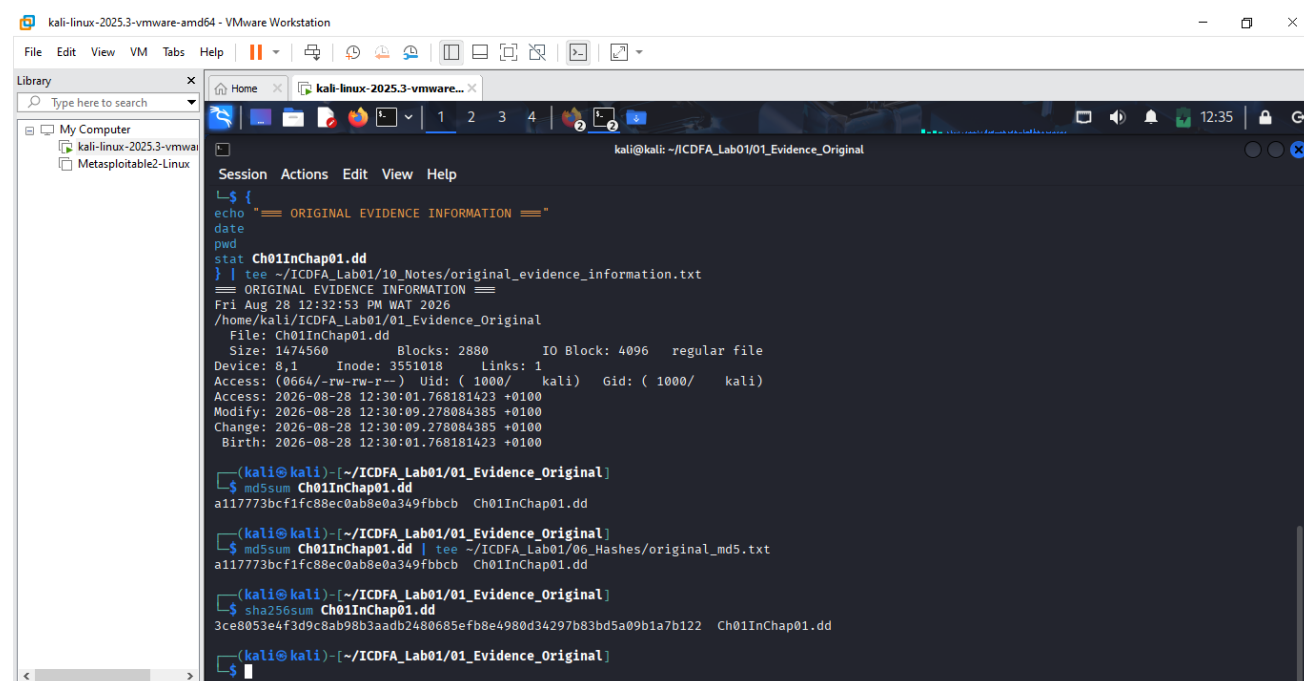
(kali@kali)~/ICDFA_Lab01/01_Evidence_Original
$ md5sum Ch01InChap01.dd | tee ~/ICDFA_Lab01/06_Hashes/original_md5.txt
a117773bcf1fc88ec0ab8e0a349fbcb  Ch01InChap01.dd

(kali@kali)~/ICDFA_Lab01/01_Evidence_Original
$
```

Fig 1.8 Saving the result of md5sum.

Explanation

The MD5 result was saved so it could be referred to later.



```
kali@kali: ~/ICDFA_Lab01/01_Evidence_Original
$ {
echo " == ORIGINAL EVIDENCE INFORMATION == "
date
pwd
stat Ch01InChap01.dd
} | tee ~/ICDFA_Lab01/10_Notes/original_evidence_information.txt
== ORIGINAL EVIDENCE INFORMATION ==
Fri Aug 28 12:32:53 PM WAT 2026
/home/kali/ICDFA_Lab01/01_Evidence_Original
File: Ch01InChap01.dd
Size: 1474560      Blocks: 2880      IO Block: 4096   regular file
Device: 8,1      Inode: 3551018    Links: 1
Access: (0664/-rw-rw-r--)  Uid: ( 1000/   kali)   Gid: ( 1000/   kali)
Access: 2026-08-28 12:30:01.768181423 +0100
Modify: 2026-08-28 12:30:09.278084385 +0100
Change: 2026-08-28 12:30:09.278084385 +0100
Birth: 2026-08-28 12:30:01.768181423 +0100

(kali@kali)~/ICDFA_Lab01/01_Evidence_Original
$ md5sum Ch01InChap01.dd
a117773bcf1fc88ec0ab8e0a349fbcb  Ch01InChap01.dd

(kali@kali)~/ICDFA_Lab01/01_Evidence_Original
$ md5sum Ch01InChap01.dd | tee ~/ICDFA_Lab01/06_Hashes/original_md5.txt
a117773bcf1fc88ec0ab8e0a349fbcb  Ch01InChap01.dd

(kali@kali)~/ICDFA_Lab01/01_Evidence_Original
$ sha256sum Ch01InChap01.dd
3ce8053e4f3d9c8ab98b3aadb2480685efb8e4980d34297b83bd5a09b1a7b122  Ch01InChap01.dd

(kali@kali)~/ICDFA_Lab01/01_Evidence_Original
$
```

Fig 1.9 Calculating the SHA-256 hash.

Explanation

I also calculated the SHA-256 hash of the original evidence image.

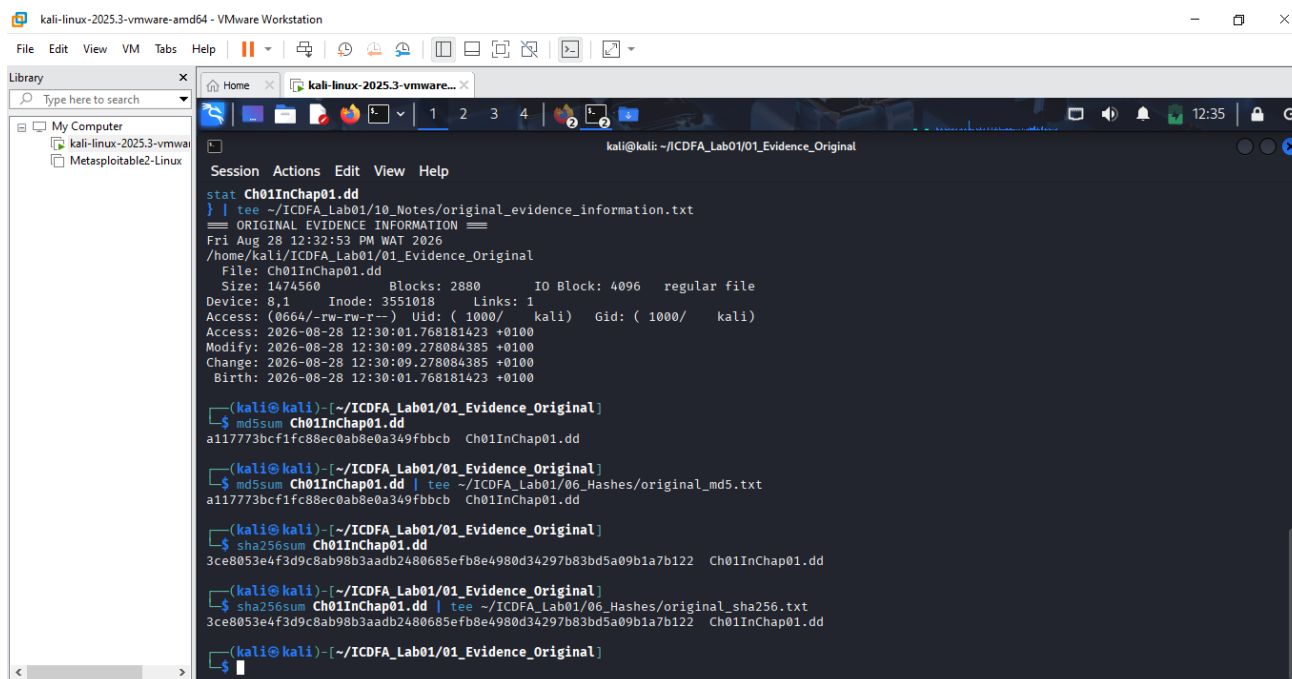


Fig 1.10 Saving the result of SHA-256 hash.

Explanation

The SHA-256 result was saved as part of the evidence record.

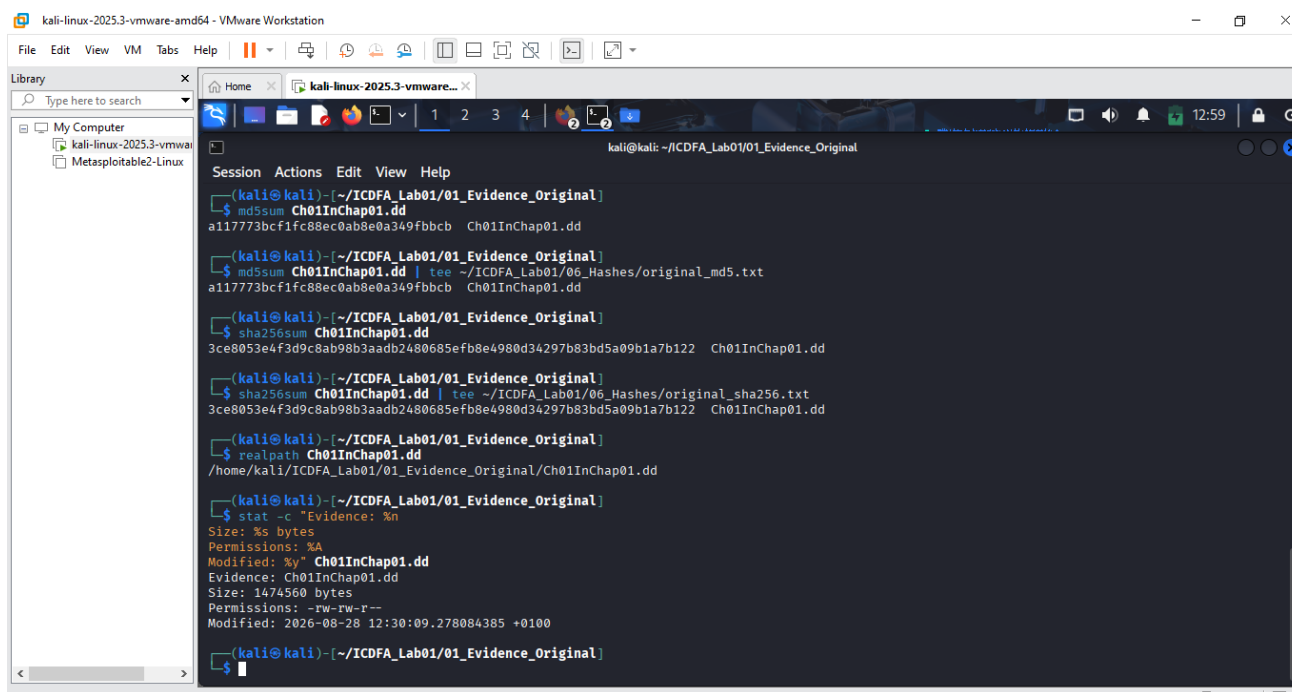


Fig 1.11 Evidence file size and location.

Explanation

I confirmed the exact file size and location of the evidence image.

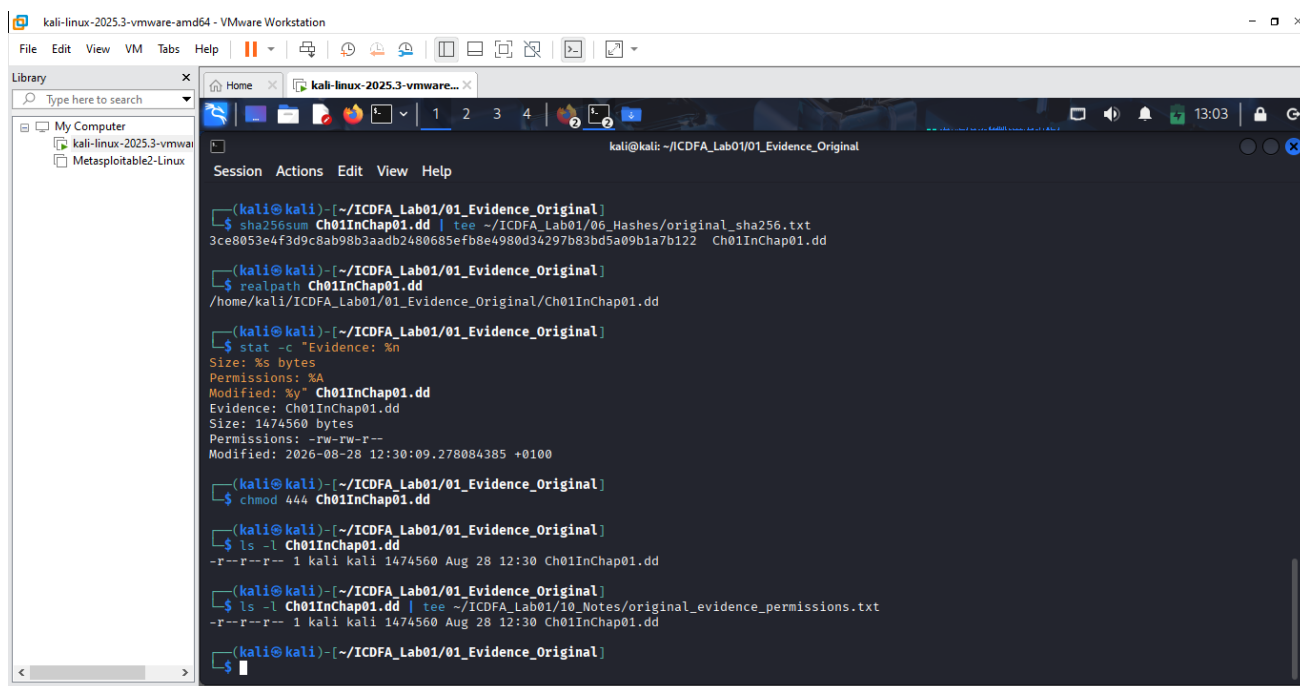


Fig 1.12 Make the original image read-only and sending it to original_evidence_permissions.txt.

Explanation

I made the original image read-only to protect it from accidental changes.

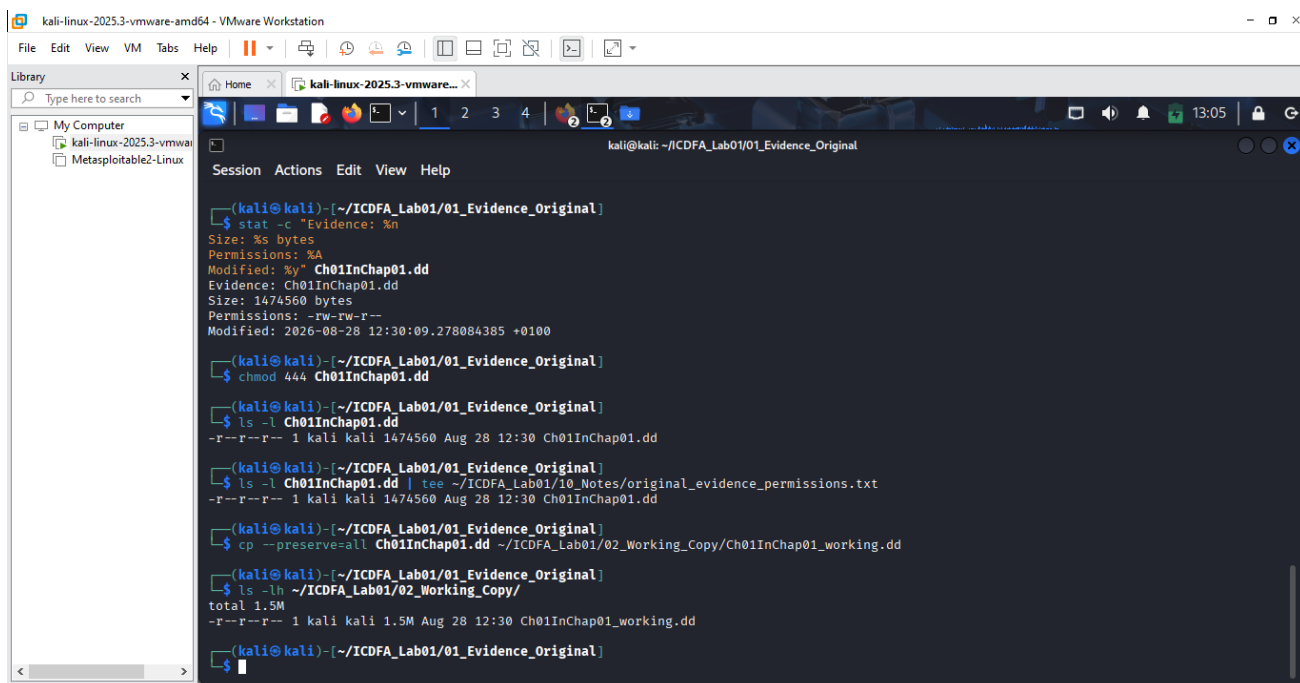
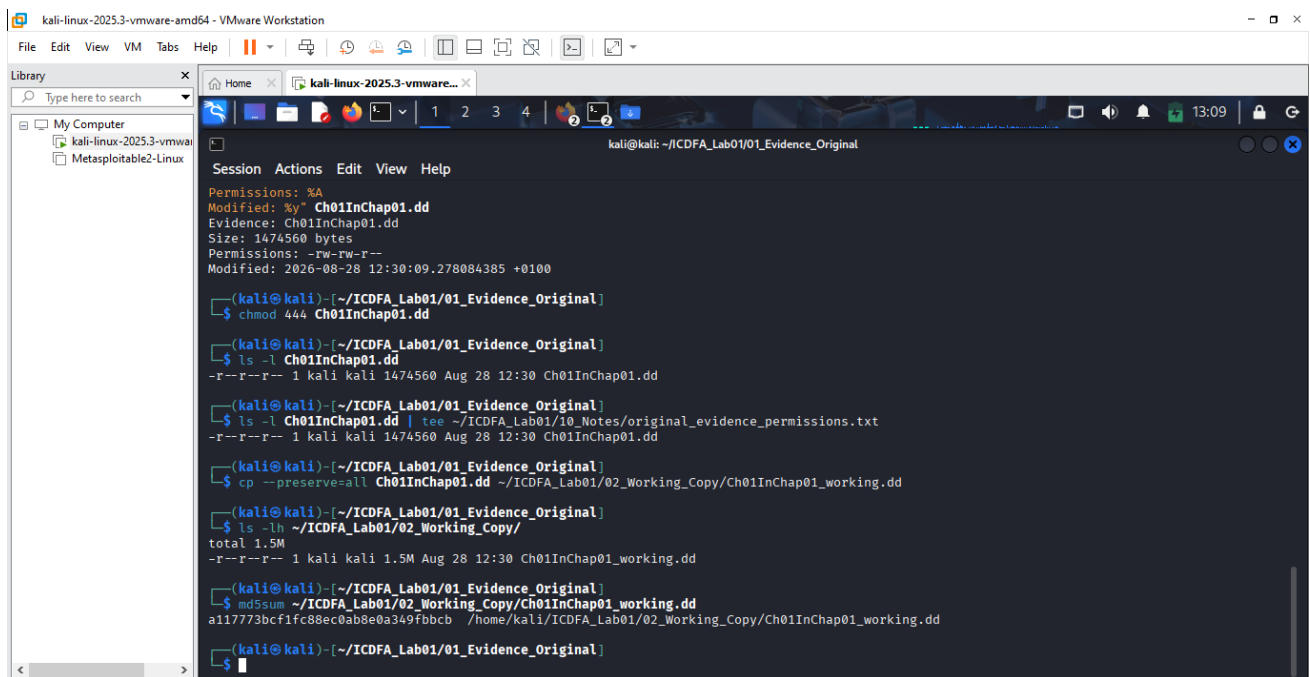


Fig 1.13 Creating the forensic working copy.

Explanation

I created a separate working copy so the original evidence would remain untouched.



```
kali@kali: ~/ICDFA_Lab01/01_Evidence_Original
Permissions: -rw-rw-r--
Modified: 2026-08-28 12:30:09.278084385 +0100

(kali@kali)-[~/ICDFA_Lab01/01_Evidence_Original]
$ chmod 444 Ch01InChap01.dd

(kali@kali)-[~/ICDFA_Lab01/01_Evidence_Original]
$ ls -l Ch01InChap01.dd
-r--r--r-- 1 kali kali 1474560 Aug 28 12:30 Ch01InChap01.dd

(kali@kali)-[~/ICDFA_Lab01/01_Evidence_Original]
$ ls -l Ch01InChap01.dd | tee ~/ICDFA_Lab01/10_Notes/original_evidence_permissions.txt
-r--r--r-- 1 kali kali 1474560 Aug 28 12:30 Ch01InChap01.dd

(kali@kali)-[~/ICDFA_Lab01/01_Evidence_Original]
$ cp --preserve=all Ch01InChap01.dd ~/ICDFA_Lab01/02_Working_Copy/Ch01InChap01_working.dd

(kali@kali)-[~/ICDFA_Lab01/02_Working_Copy/]
$ ls -lh ~/ICDFA_Lab01/02_Working_Copy/
total 1.5M
-r--r--r-- 1 kali kali 1.5M Aug 28 12:30 Ch01InChap01_working.dd

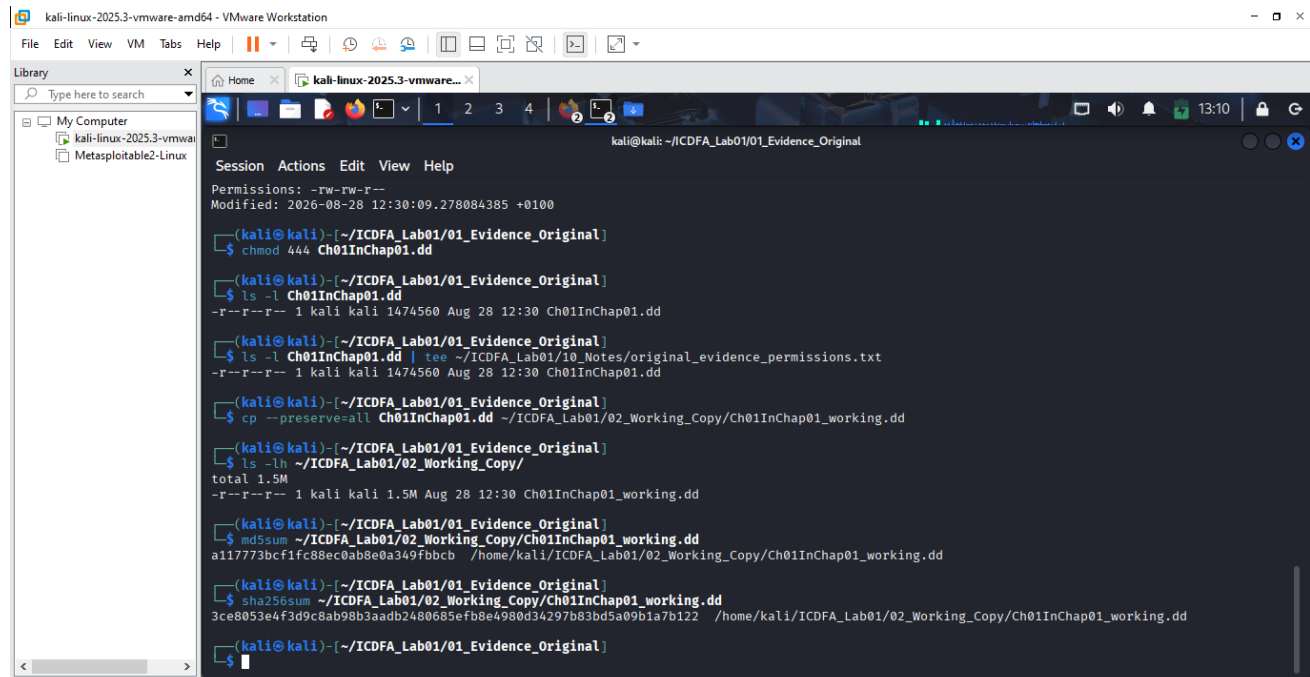
(kali@kali)-[~/ICDFA_Lab01/01_Evidence_Original]
$ md5sum ~/ICDFA_Lab01/02_Working_Copy/Ch01InChap01_working.dd
a11773bcf1fc88ec0ab8e0a349fbbcb /home/kali/ICDFA_Lab01/02_Working_Copy/Ch01InChap01_working.dd

(kali@kali)-[~/ICDFA_Lab01/01_Evidence_Original]
$
```

Fig 1.14 Hashing the working copy using md5sum.

Explanation

I calculated the MD5 hash of the working copy for comparison with the original.



```
kali@kali: ~/ICDFA_Lab01/01_Evidence_Original
Permissions: -rw-rw-r--
Modified: 2026-08-28 12:30:09.278084385 +0100

(kali@kali)-[~/ICDFA_Lab01/01_Evidence_Original]
$ chmod 444 Ch01InChap01.dd

(kali@kali)-[~/ICDFA_Lab01/01_Evidence_Original]
$ ls -l Ch01InChap01.dd
-r--r--r-- 1 kali kali 1474560 Aug 28 12:30 Ch01InChap01.dd

(kali@kali)-[~/ICDFA_Lab01/01_Evidence_Original]
$ ls -l Ch01InChap01.dd | tee ~/ICDFA_Lab01/10_Notes/original_evidence_permissions.txt
-r--r--r-- 1 kali kali 1474560 Aug 28 12:30 Ch01InChap01.dd

(kali@kali)-[~/ICDFA_Lab01/01_Evidence_Original]
$ cp --preserve=all Ch01InChap01.dd ~/ICDFA_Lab01/02_Working_Copy/Ch01InChap01_working.dd

(kali@kali)-[~/ICDFA_Lab01/02_Working_Copy/]
$ ls -lh ~/ICDFA_Lab01/02_Working_Copy/
total 1.5M
-r--r--r-- 1 kali kali 1.5M Aug 28 12:30 Ch01InChap01_working.dd

(kali@kali)-[~/ICDFA_Lab01/01_Evidence_Original]
$ md5sum ~/ICDFA_Lab01/02_Working_Copy/Ch01InChap01_working.dd
a11773bcf1fc88ec0ab8e0a349fbbcb /home/kali/ICDFA_Lab01/02_Working_Copy/Ch01InChap01_working.dd

(kali@kali)-[~/ICDFA_Lab01/01_Evidence_Original]
$ sha256sum ~/ICDFA_Lab01/02_Working_Copy/Ch01InChap01_working.dd
3ce8053e4f3d9c8ab98b3aadb2480685efb8e4980d34297b83bd5a09b1a7b122 /home/kali/ICDFA_Lab01/02_Working_Copy/Ch01InChap01_working.dd

(kali@kali)-[~/ICDFA_Lab01/01_Evidence_Original]
$
```

Fig 1.15 Hashing the working copy using sha256sum.

Explanation

I calculated the SHA-256 hash of the working copy as an additional integrity check.

```
(kali@kali)-[~/ICDFA_Lab01/01_Evidence_Original]
$ ls -l Ch01InChap01.dd
-r--r--r-- 1 kali kali 1474560 Aug 28 12:30 Ch01InChap01.dd

(kali@kali)-[~/ICDFA_Lab01/01_Evidence_Original]
$ ls -l Ch01InChap01.dd | tee ~/ICDFA_Lab01/10_Notes/original_evidence_permissions.txt
-r--r--r-- 1 kali kali 1474560 Aug 28 12:30 Ch01InChap01.dd

(kali@kali)-[~/ICDFA_Lab01/01_Evidence_Original]
$ cp --preserve=all Ch01InChap01.dd ~/ICDFA_Lab01/02_Working_Copy/Ch01InChap01_working.dd

(kali@kali)-[~/ICDFA_Lab01/01_Evidence_Original]
$ ls -lh ~/ICDFA_Lab01/02_Working_Copy/
total 1.5M
-r--r--r-- 1 kali kali 1.5M Aug 28 12:30 Ch01InChap01_working.dd

(kali@kali)-[~/ICDFA_Lab01/01_Evidence_Original]
$ md5sum ~/ICDFA_Lab01/02_Working_Copy/Ch01InChap01_working.dd
a117773bcf1fc88ec0ab8e0a349fbbcb /home/kali/ICDFA_Lab01/02_Working_Copy/Ch01InChap01_working.dd

(kali@kali)-[~/ICDFA_Lab01/01_Evidence_Original]
$ sha256sum ~/ICDFA_Lab01/02_Working_Copy/Ch01InChap01_working.dd
3ce8053e4f3d9c8ab98b3aadb2480685efb8e4980d34297b83bd5a09b1a7b122 /home/kali/ICDFA_Lab01/02_Working_Copy/Ch01InChap01_working.dd

(kali@kali)-[~/ICDFA_Lab01/01_Evidence_Original]
$ md5sum ~/ICDFA_Lab01/02_Working_Copy/Ch01InChap01_working.dd \
| tee ~/ICDFA_Lab01/06_Hashes/working_copy_md5.txt
a117773bcf1fc88ec0ab8e0a349fbbcb /home/kali/ICDFA_Lab01/02_Working_Copy/Ch01InChap01_working.dd

(kali@kali)-[~/ICDFA_Lab01/01_Evidence_Original]
$
```

Fig 1.16 Saving the working of md5sum.

Explanation

The MD5 result for the working copy was saved for documentation.

```
(kali@kali)-[~/ICDFA_Lab01/01_Evidence_Original]
$ ls -l Ch01InChap01.dd | tee ~/ICDFA_Lab01/10_Notes/original_evidence_permissions.txt
-r--r--r-- 1 kali kali 1474560 Aug 28 12:30 Ch01InChap01.dd

(kali@kali)-[~/ICDFA_Lab01/01_Evidence_Original]
$ cp --preserve=all Ch01InChap01.dd ~/ICDFA_Lab01/02_Working_Copy/Ch01InChap01_working.dd

(kali@kali)-[~/ICDFA_Lab01/01_Evidence_Original]
$ ls -lh ~/ICDFA_Lab01/02_Working_Copy/
total 1.5M
-r--r--r-- 1 kali kali 1.5M Aug 28 12:30 Ch01InChap01_working.dd

(kali@kali)-[~/ICDFA_Lab01/01_Evidence_Original]
$ md5sum ~/ICDFA_Lab01/02_Working_Copy/Ch01InChap01_working.dd
a117773bcf1fc88ec0ab8e0a349fbbcb /home/kali/ICDFA_Lab01/02_Working_Copy/Ch01InChap01_working.dd

(kali@kali)-[~/ICDFA_Lab01/01_Evidence_Original]
$ sha256sum ~/ICDFA_Lab01/02_Working_Copy/Ch01InChap01_working.dd
3ce8053e4f3d9c8ab98b3aadb2480685efb8e4980d34297b83bd5a09b1a7b122 /home/kali/ICDFA_Lab01/02_Working_Copy/Ch01InChap01_working.dd

(kali@kali)-[~/ICDFA_Lab01/01_Evidence_Original]
$ md5sum ~/ICDFA_Lab01/02_Working_Copy/Ch01InChap01_working.dd \
| tee ~/ICDFA_Lab01/06_Hashes/working_copy_md5.txt
a117773bcf1fc88ec0ab8e0a349fbbcb /home/kali/ICDFA_Lab01/02_Working_Copy/Ch01InChap01_working.dd

(kali@kali)-[~/ICDFA_Lab01/01_Evidence_Original]
$ sha256sum ~/ICDFA_Lab01/02_Working_Copy/Ch01InChap01_working.dd \
| tee ~/ICDFA_Lab01/06_Hashes/working_copy_sha256.txt
3ce8053e4f3d9c8ab98b3aadb2480685efb8e4980d34297b83bd5a09b1a7b122 /home/kali/ICDFA_Lab01/02_Working_Copy/Ch01InChap01_working.dd

(kali@kali)-[~/ICDFA_Lab01/01_Evidence_Original]
$
```

Fig 1.17 Saving the working of sha256sum.

Explanation

The SHA-256 result for the working copy was also saved.

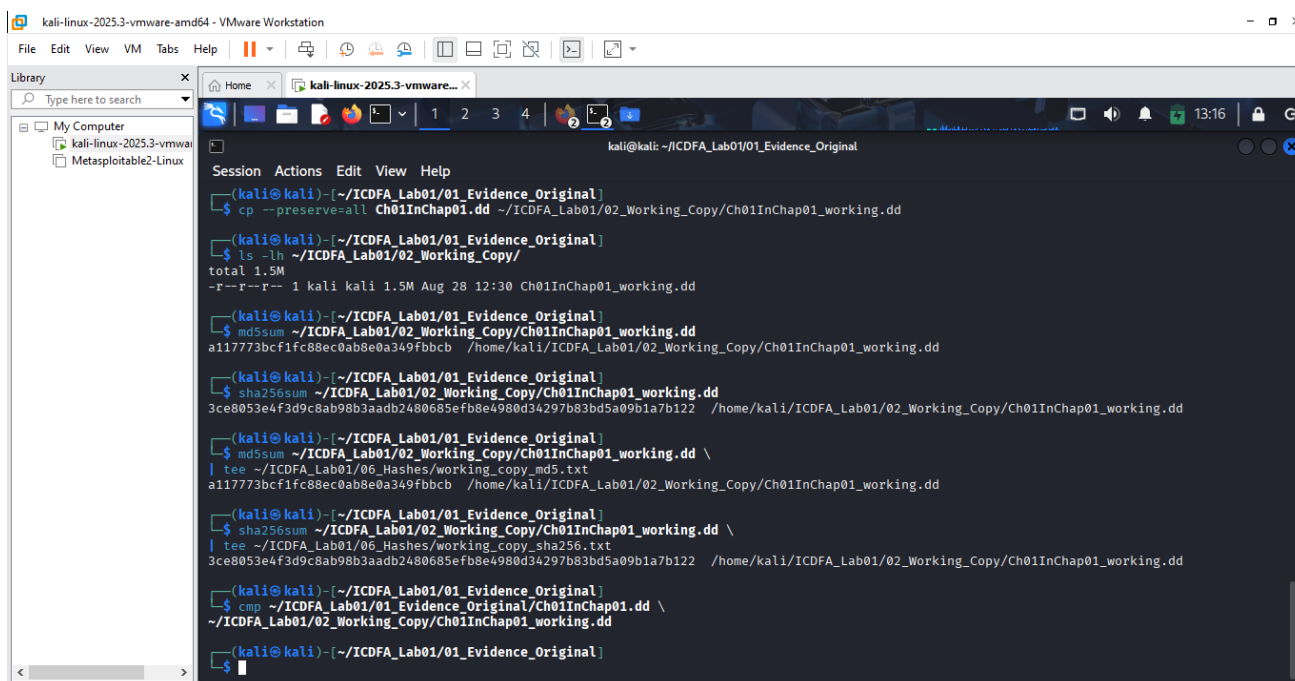


Fig 1.18 Directly verify that both files are identical.

Explanation

I directly compared the original image with the working copy.

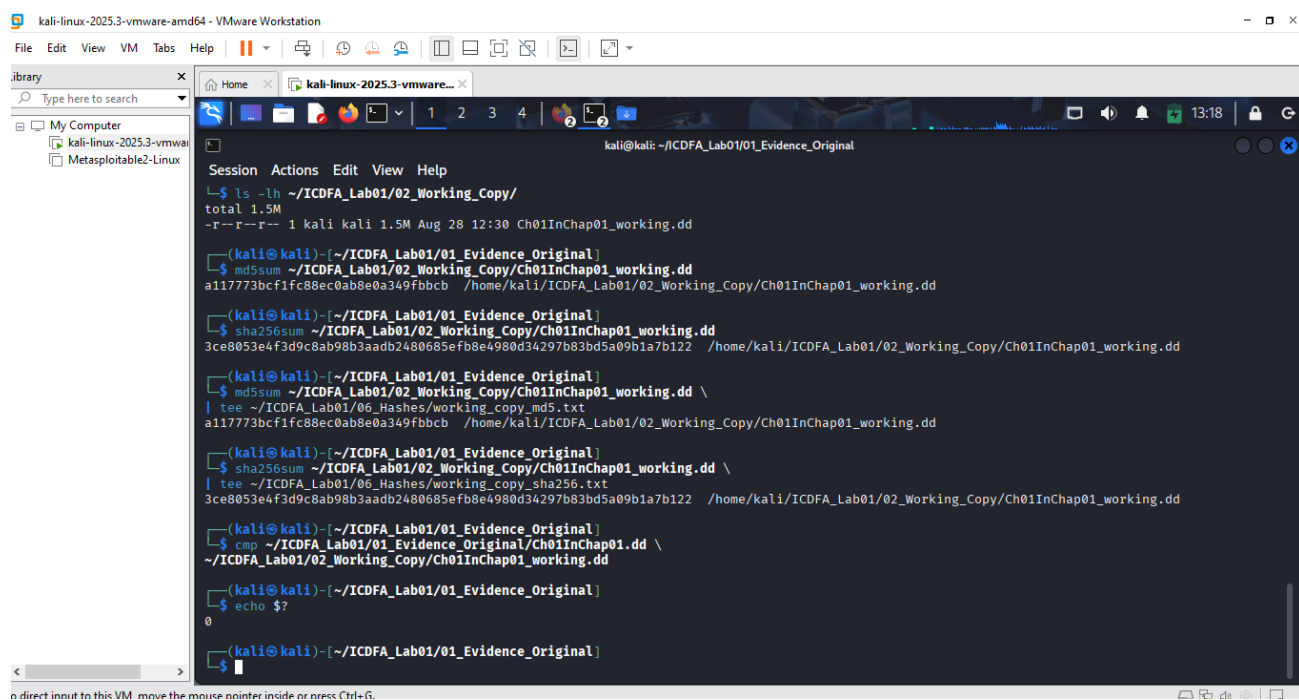


Fig 1.19 Verify that both files are identical 1.

Explanation

This shows the result of the file comparison process.

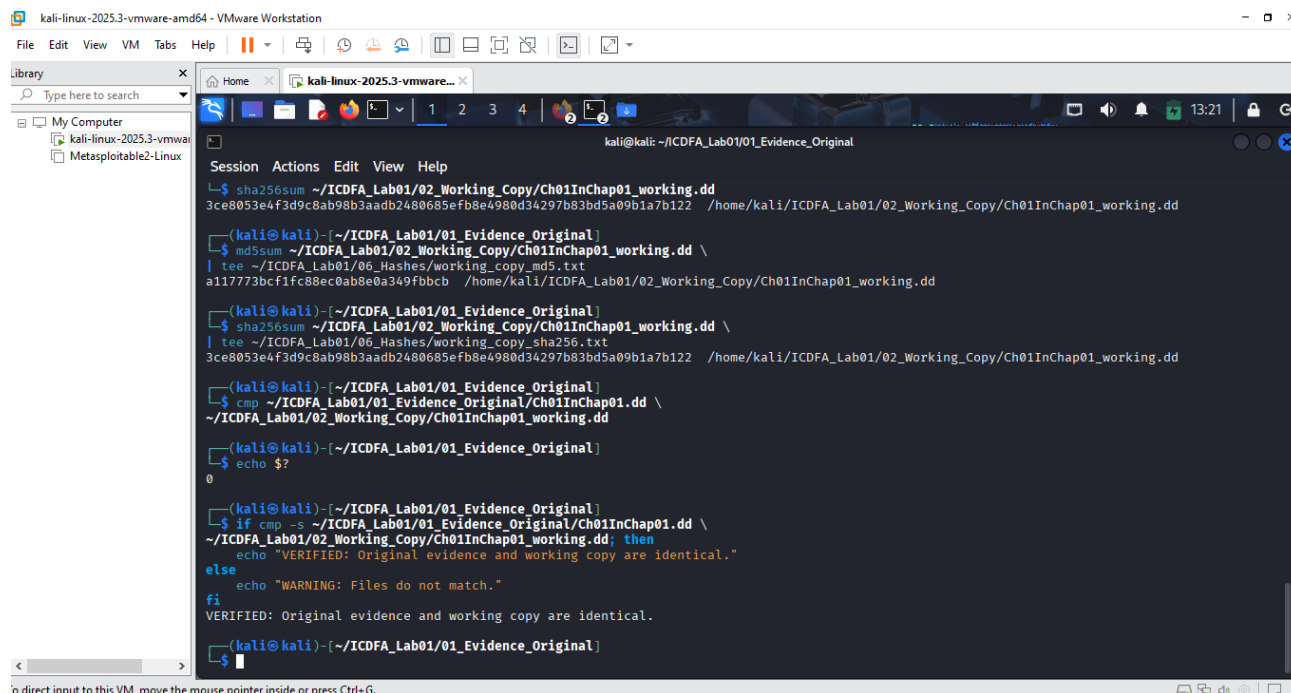


Fig 1.20 Verify that both files are identical 2.

Explanation

The comparison confirmed that the original and working copy were identical.

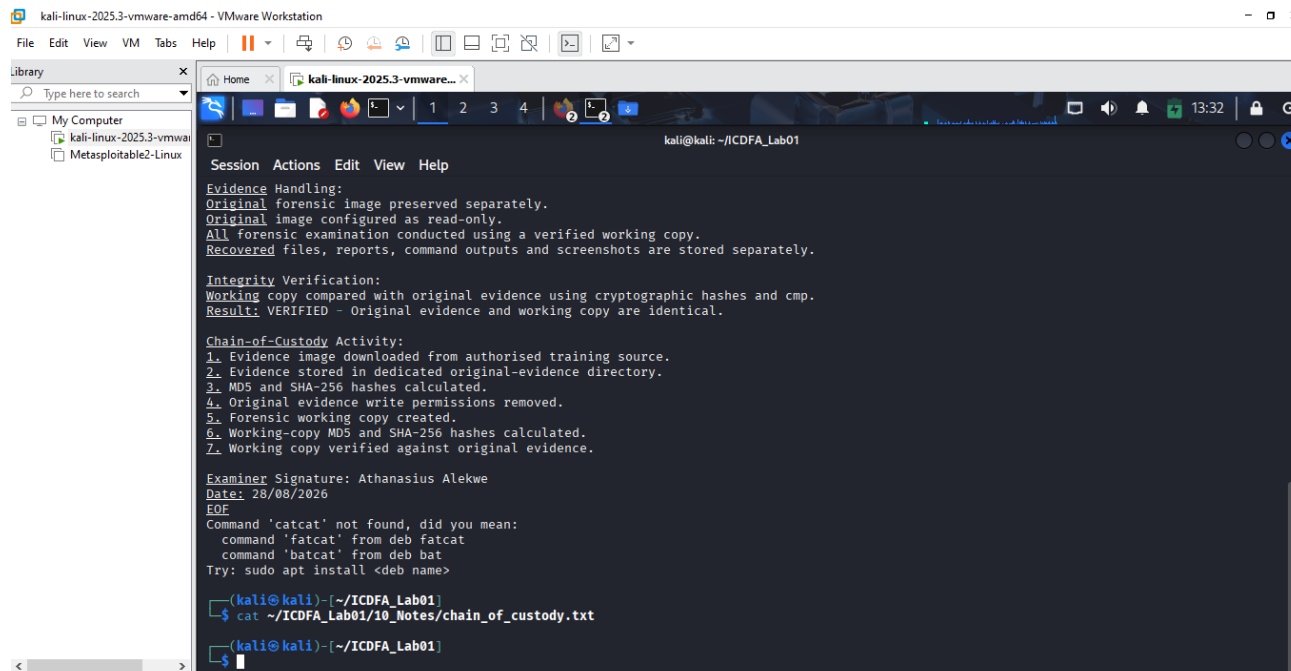


Fig 2. Creating a Mini Chain-of-Custody / Evidence Worksheet.

Explanation

I prepared a mini chain-of-custody record to document how the evidence was handled.

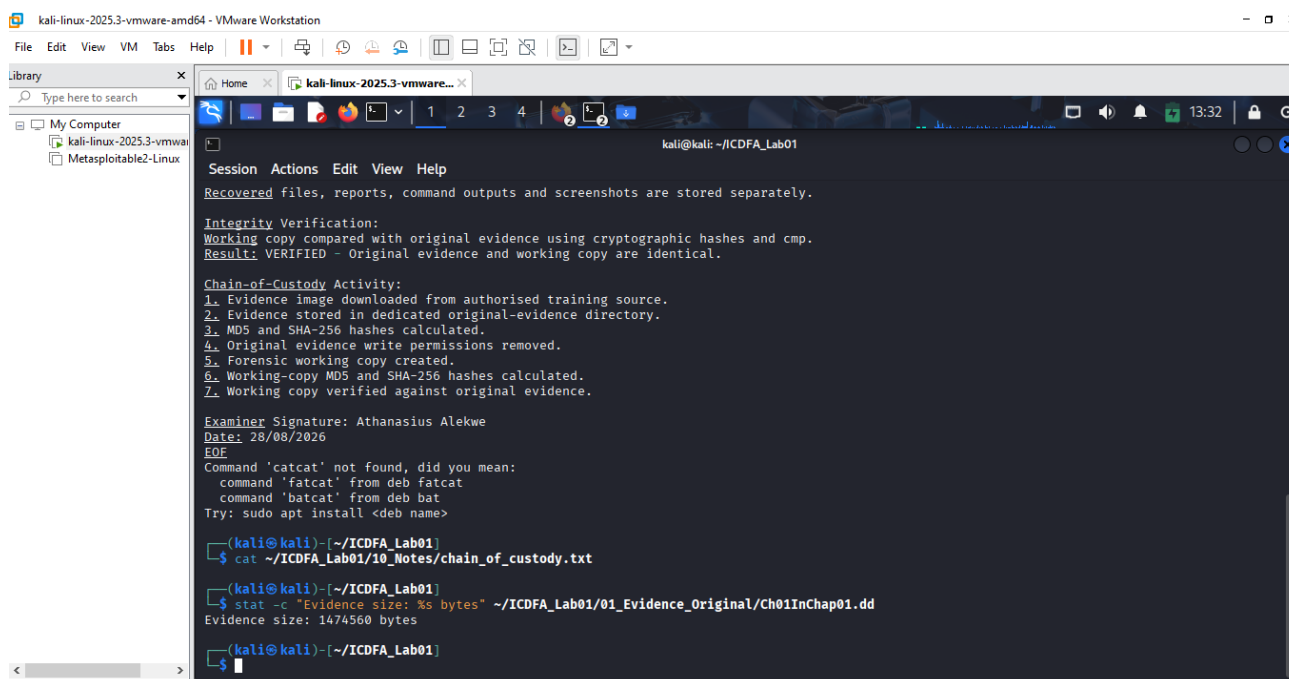


Fig 2.1 Getting the actual evidence size.

Explanation

I checked the evidence image to obtain its exact size in bytes.

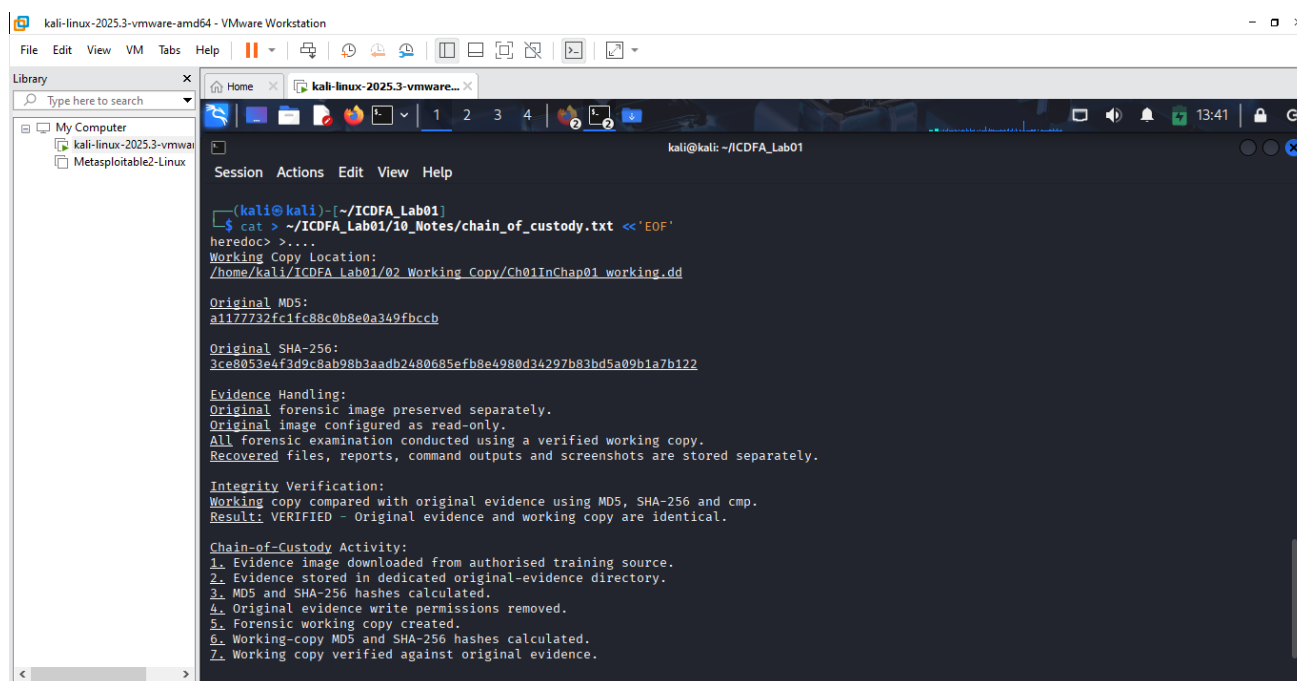


Fig 2.2 Creating a Mini Chain-of-Custody / Evidence Worksheet.

Explanation

This shows the main details entered into the chain-of-custody worksheet.

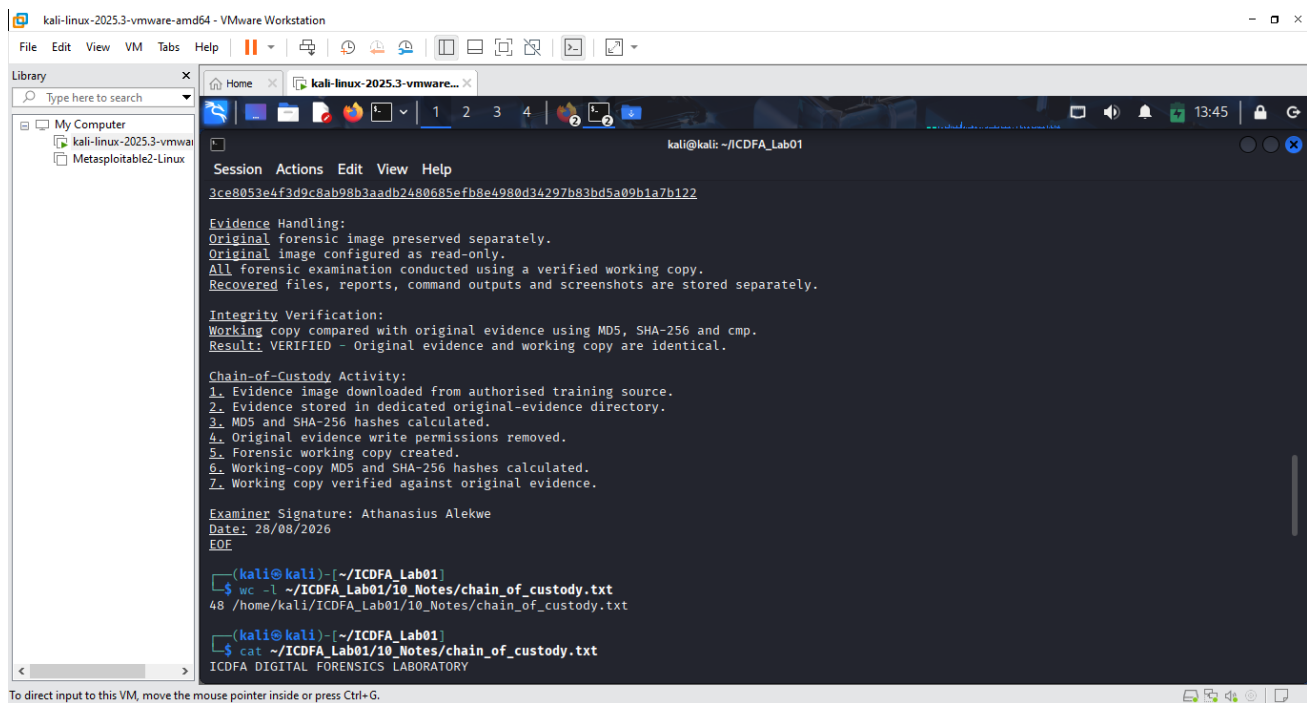


Fig 2.2.1 Creating a Mini Chain-of-Custody / Evidence Worksheet page 1.

Explanation

This continues the evidence-handling and integrity information in the worksheet.

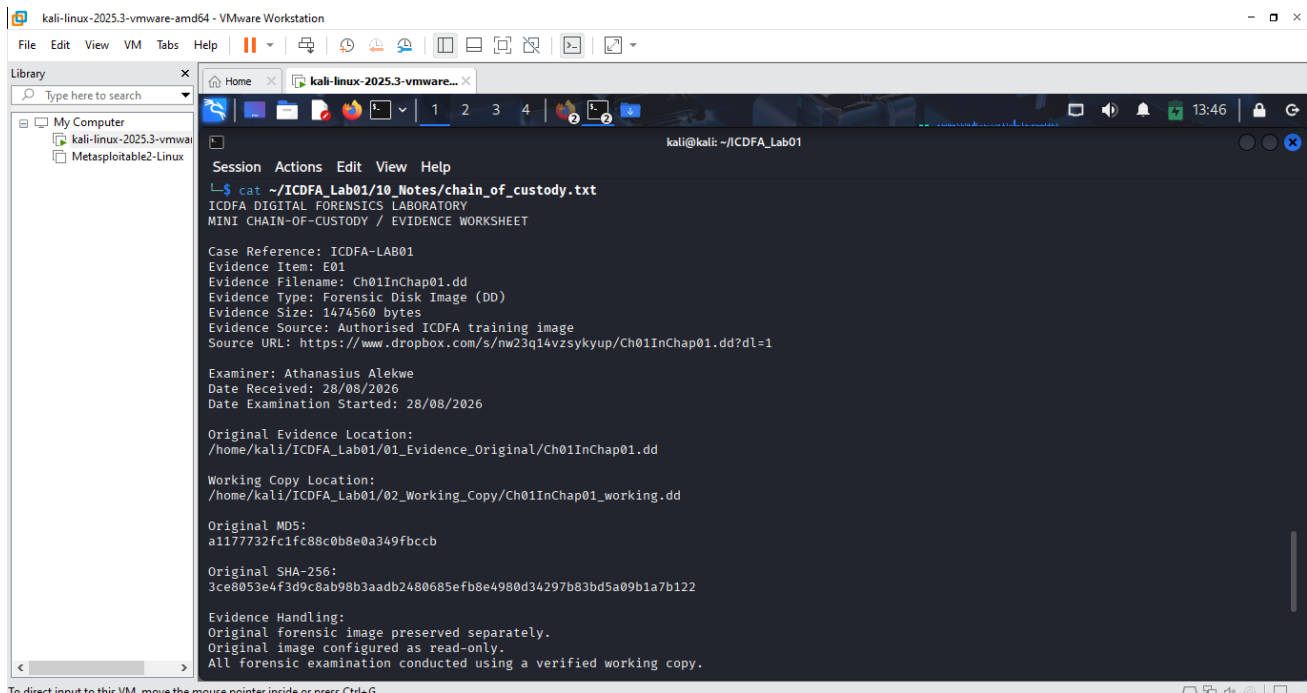


Fig 2.3 Verifying the chain of custody.

Explanation

I checked the saved chain-of-custody file to make sure the information was recorded correctly.

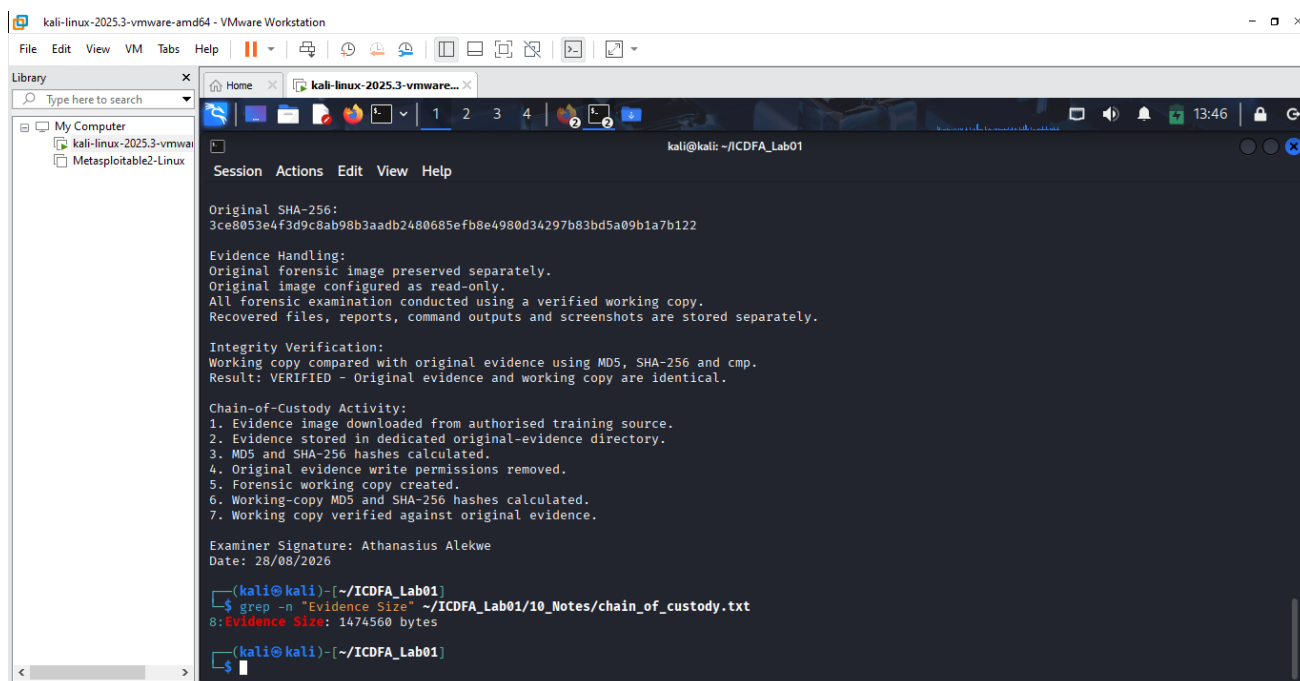


Fig 2.4 Verifying the evidence size.

Explanation

This confirms the evidence size as **1,474,560 bytes**.

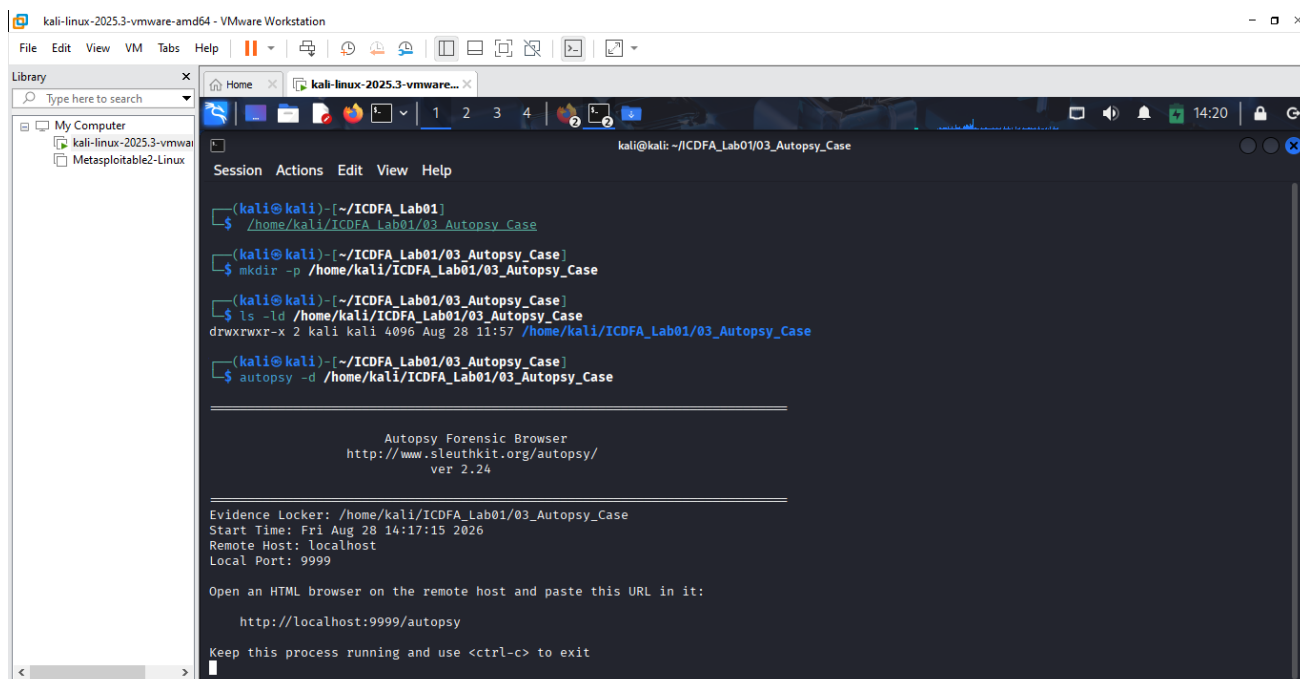


Fig 2.5 Autopsy server started.

Explanation

I successfully started the Autopsy forensic server for the examination.

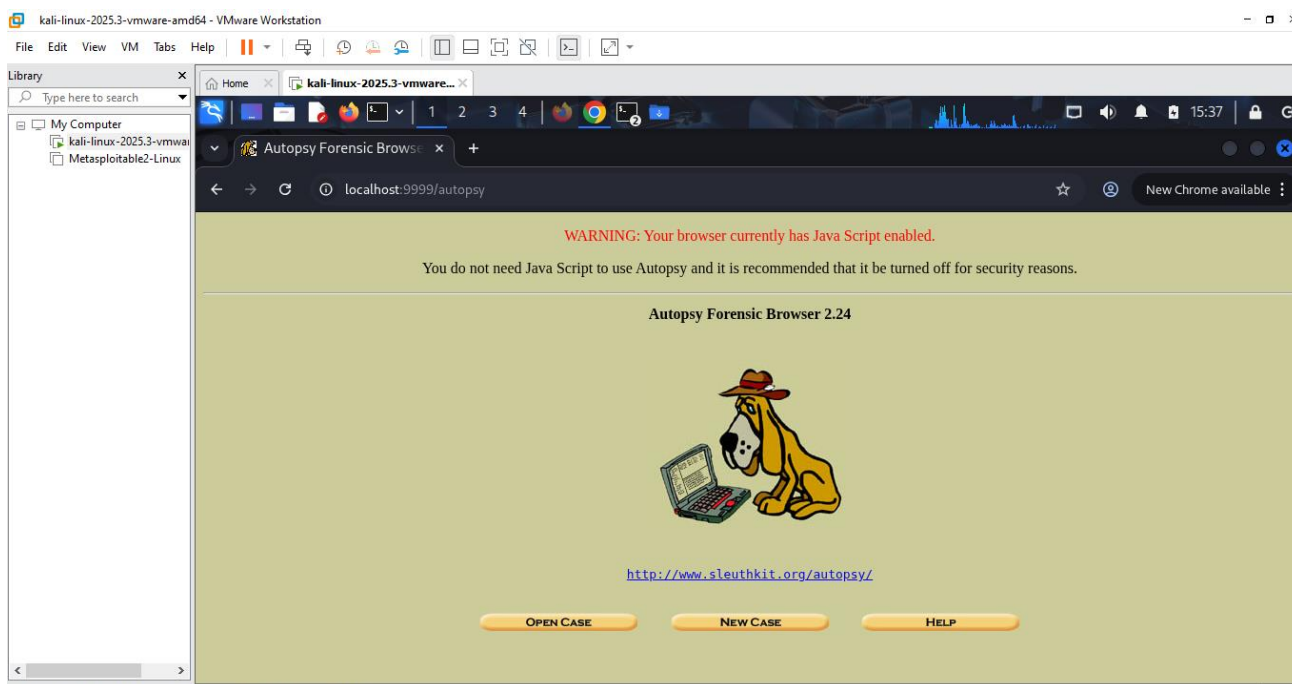


Fig 2.6 Autopsy home page.

Explanation

This is the Autopsy home page used to begin the case setup.

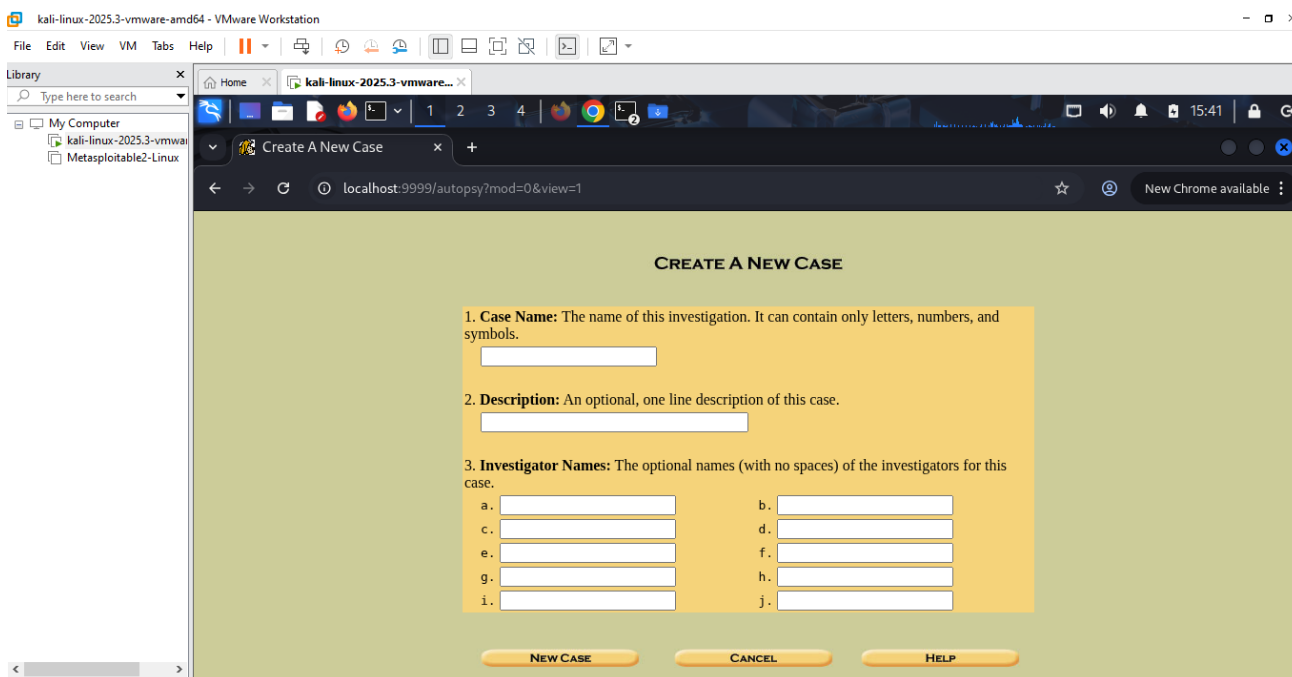


Fig 2.7 Autopsy new case details.

Explanation

I opened the form for creating a new forensic case.

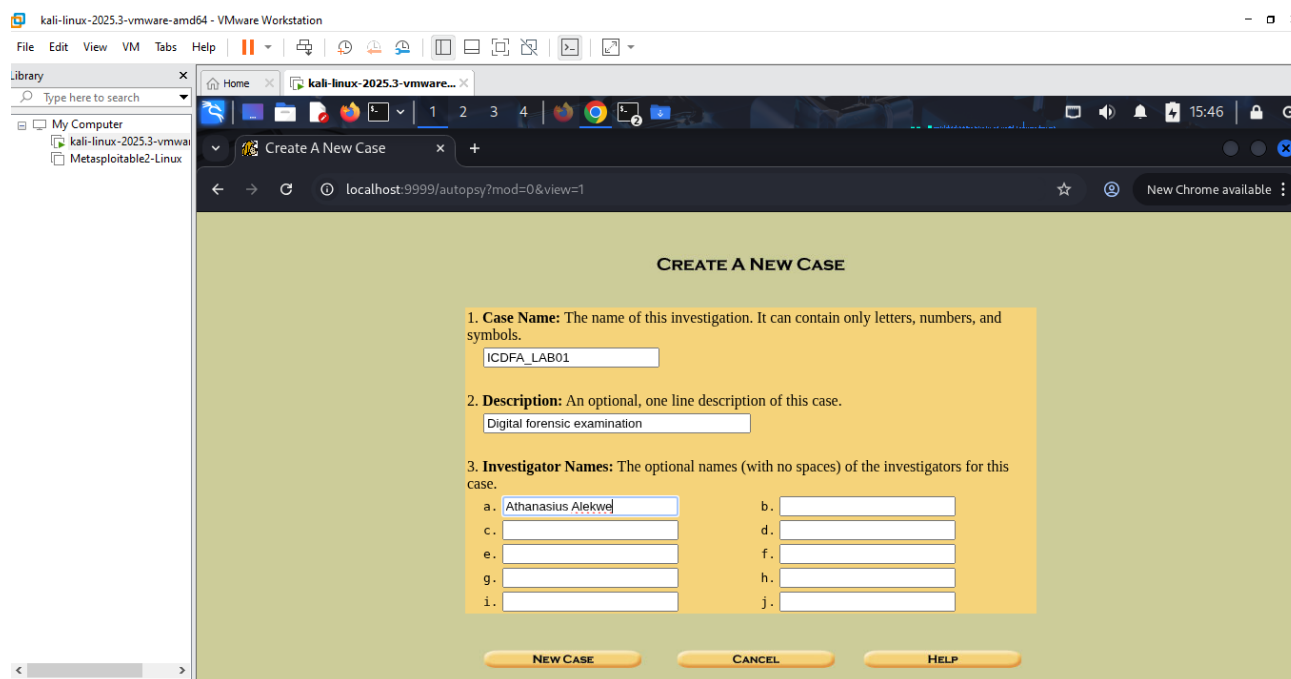


Fig 2.8 Autopsy new case complete details.

Explanation

This shows the completed case information before the case was created.

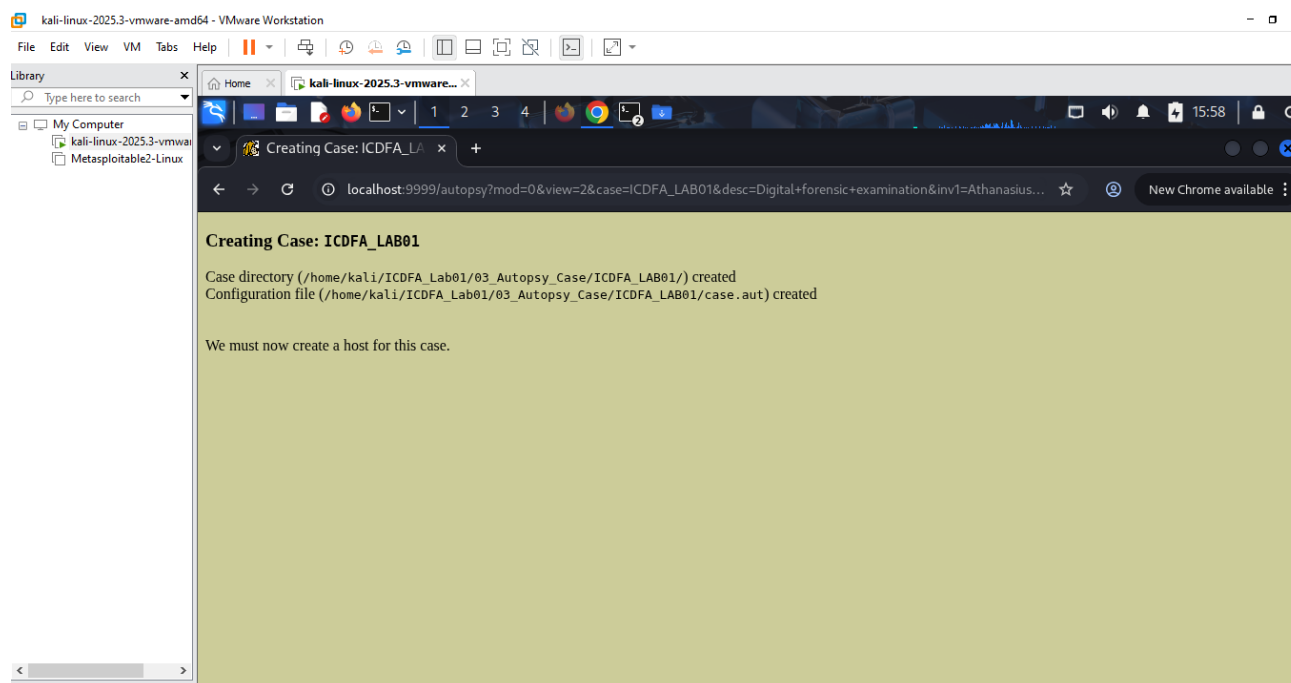


Fig 2.9 Autopsy case created.

Explanation

The Autopsy case was created successfully.

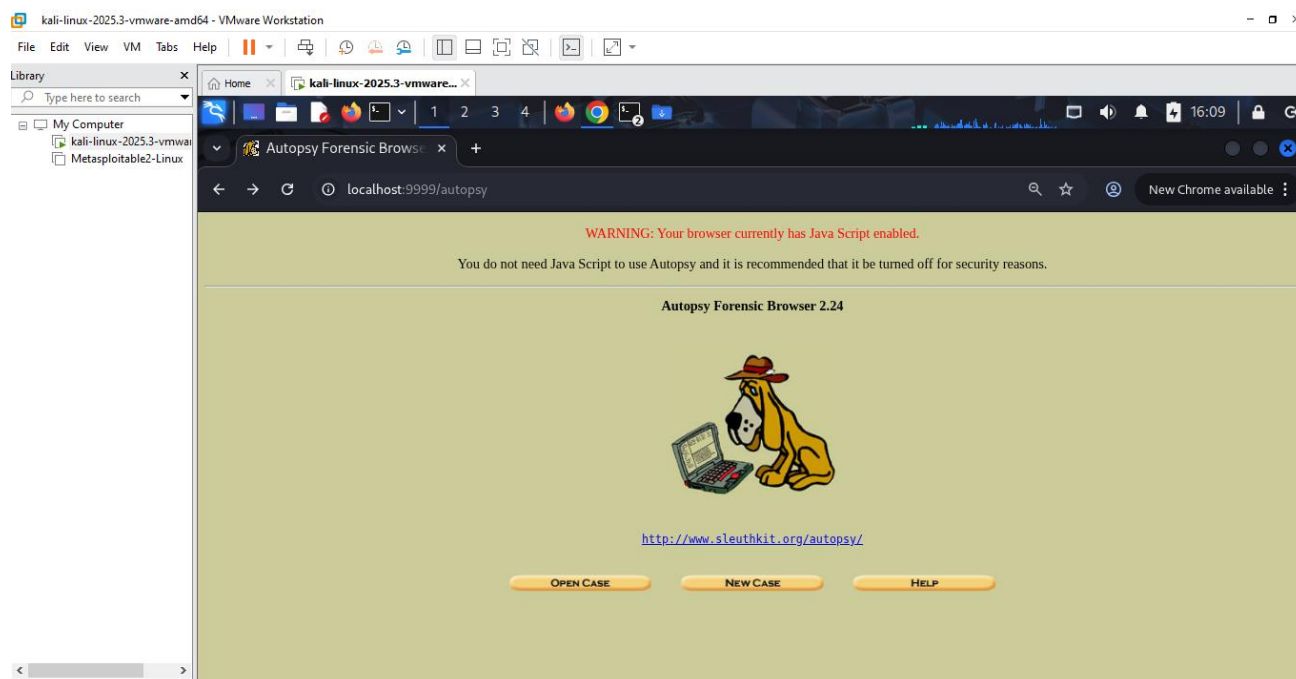


Fig 3. Autopsy home page.

Explanation

I returned to the Autopsy home page to continue the investigation.

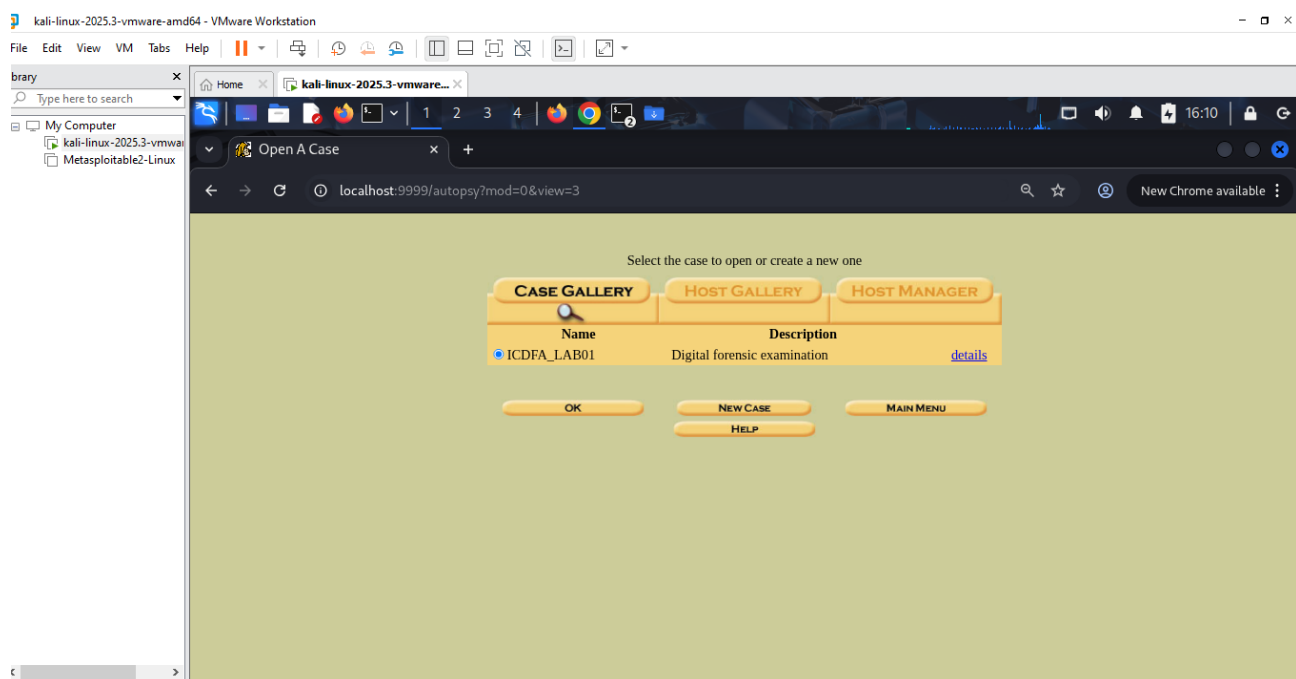


Fig 3.1 Autopsy open case.

Explanation

I selected and opened the existing ICDFA_LAB01 case.

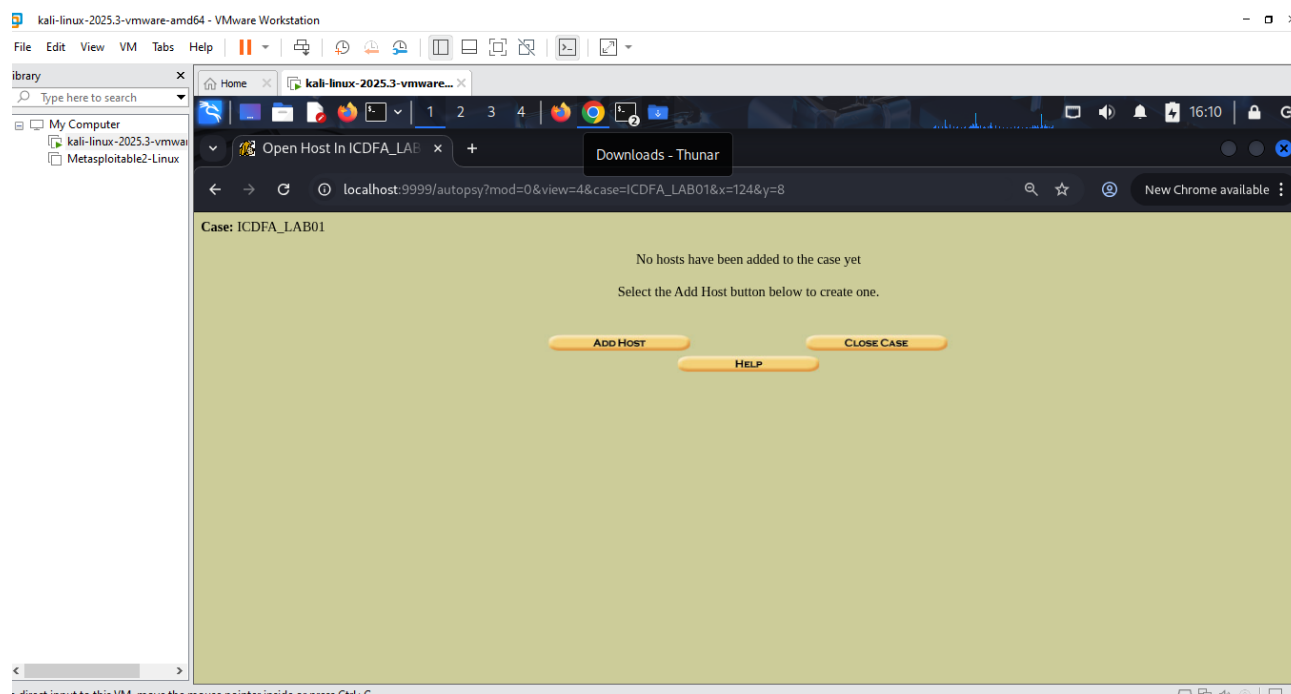


Fig 3.2 Adding a host to the ICDFA_LAB01 Autopsy.

Explanation

I moved to the stage where a host was added to the case.

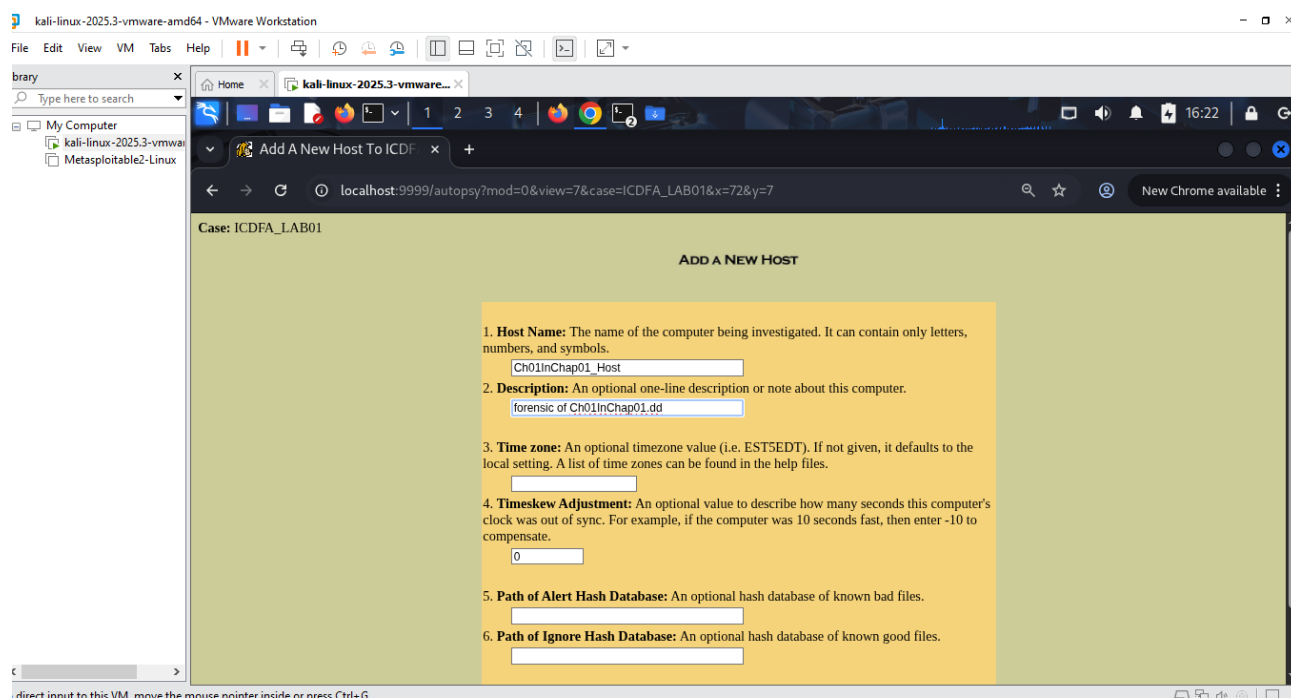


Fig 3.3 Autopsy host configuration.

Explanation

I entered the required information for the forensic host.

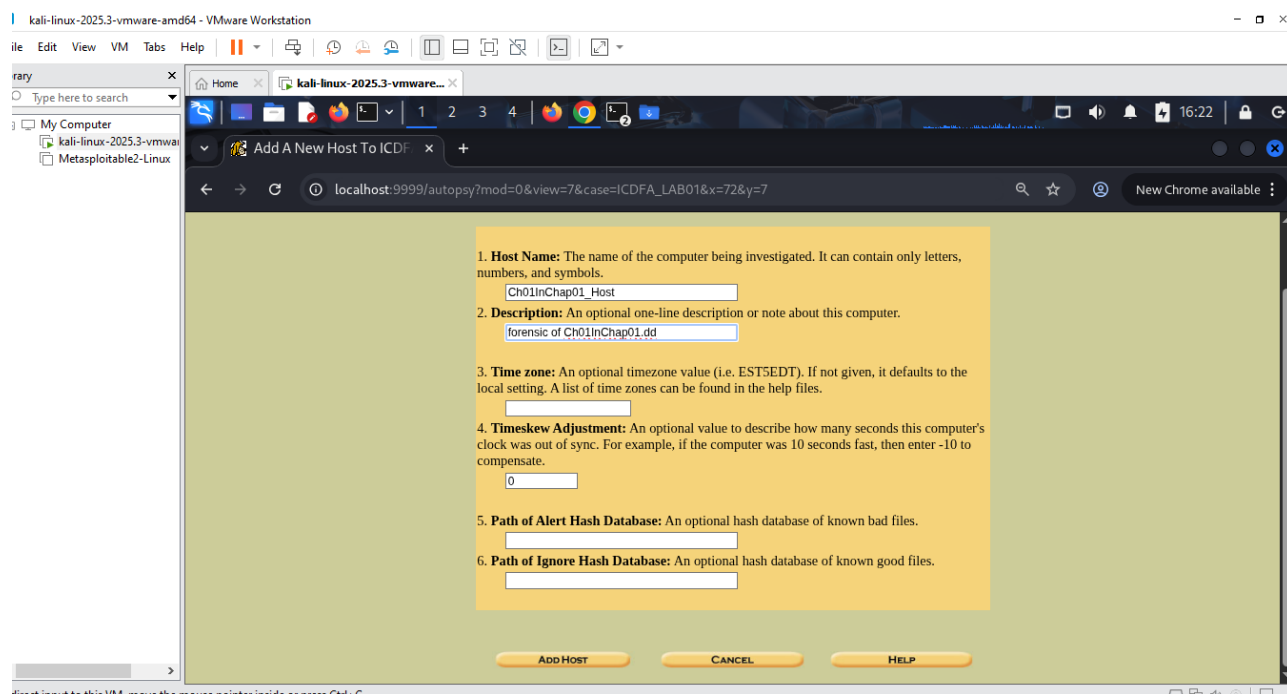


Fig 3.4 Autopsy host configuration page 1.

Explanation

This shows the completed host settings before it was added.

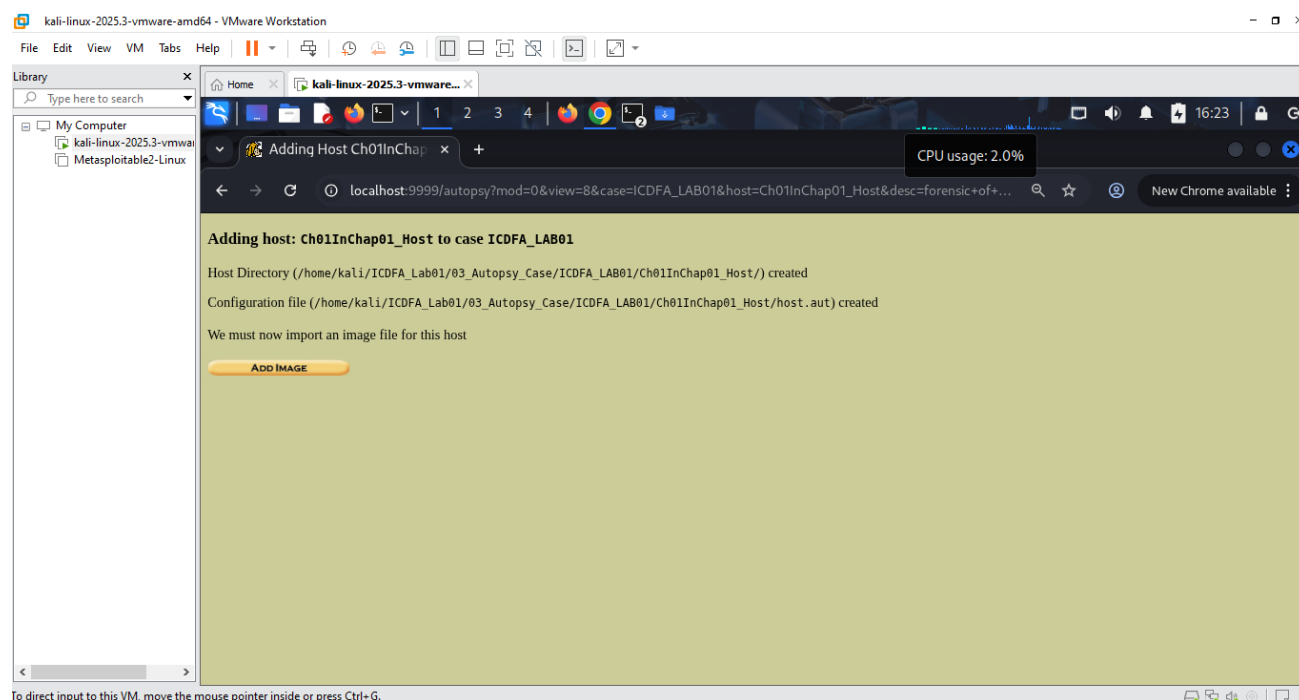


Fig 3.5 Autopsy host created.

Explanation

The host was created successfully and was ready for the image.

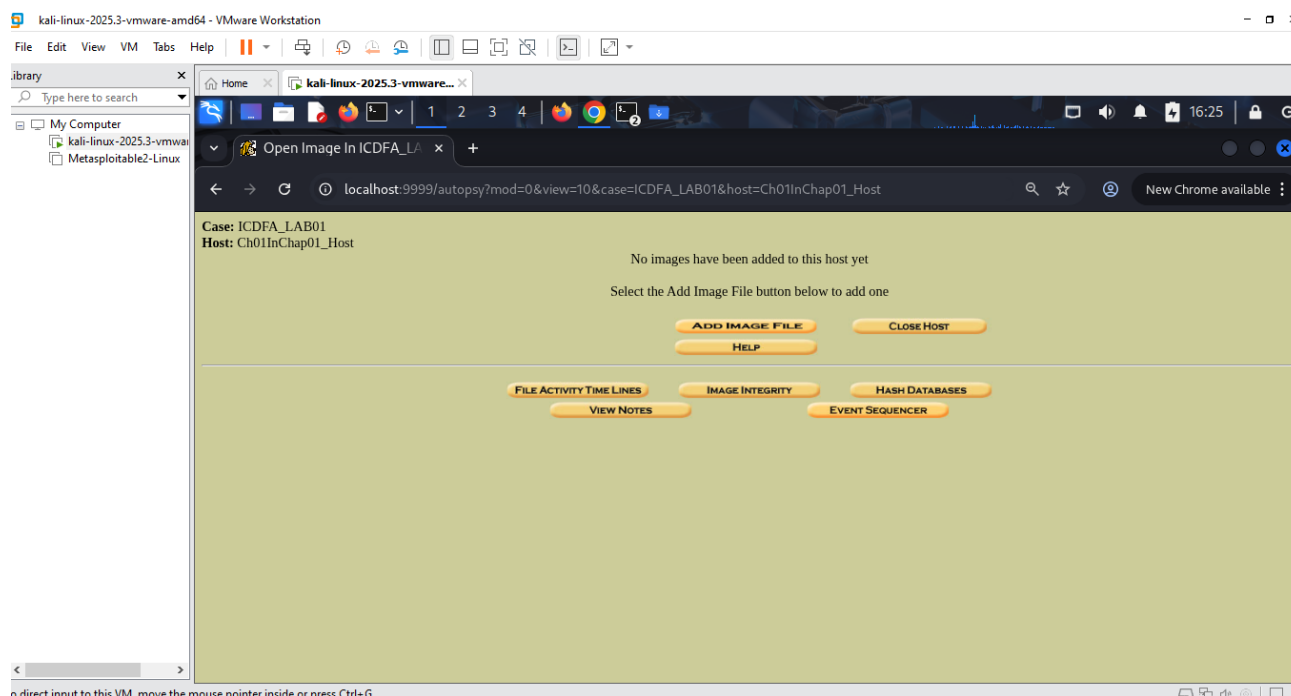


Fig 3.6 Autopsy add image configuration.

Explanation

I selected the option to add the forensic image to the host.

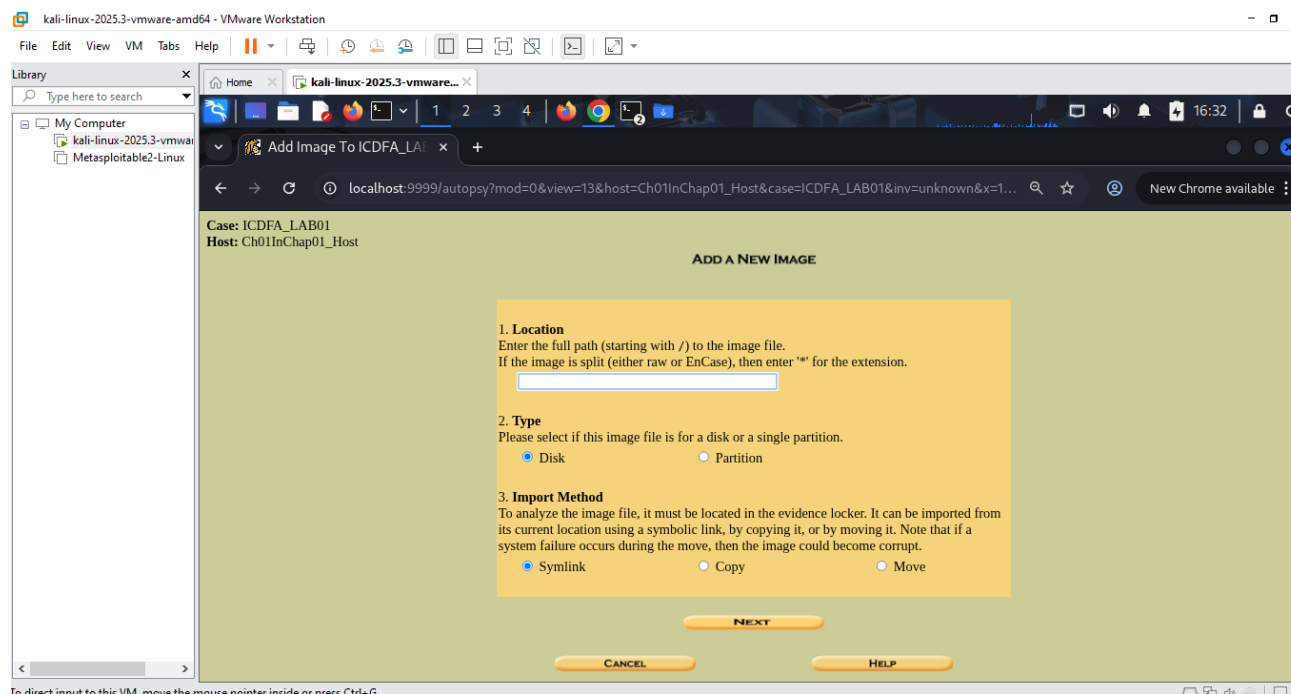


Fig 3.7 Autopsy add image configuration page 1.

Explanation

I entered the working-copy location and selected the appropriate import settings.

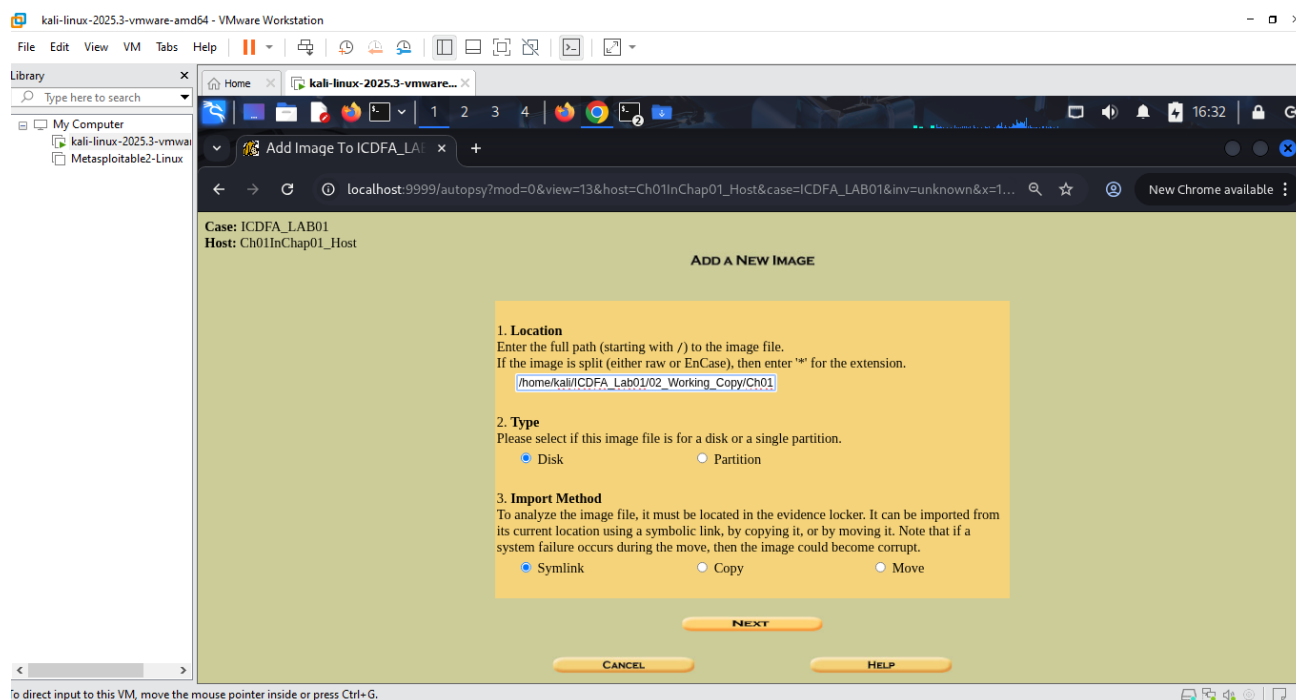


Fig 3.8 Autopsy add image configuration page 2.

Explanation

This shows the completed image configuration before moving forward.

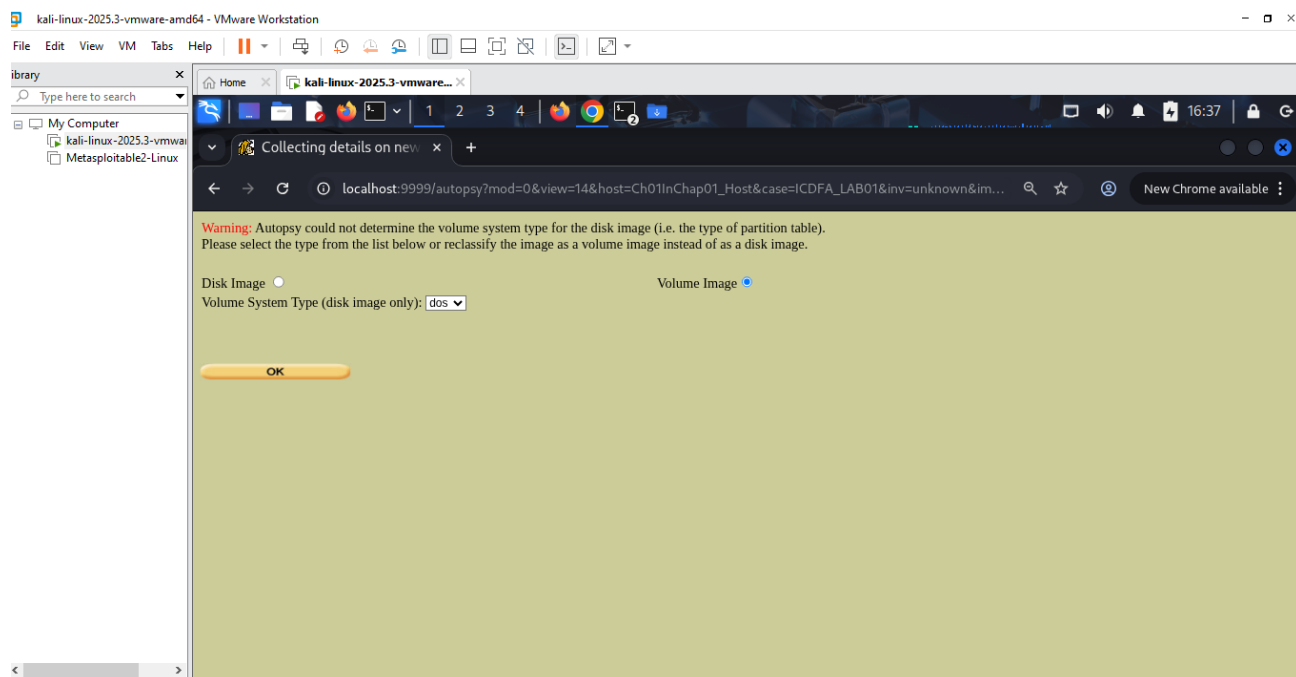


Fig 3.9 Autopsy volume system warning.

Explanation

Autopsy could not detect a partition table, so the image was treated as a volume image.

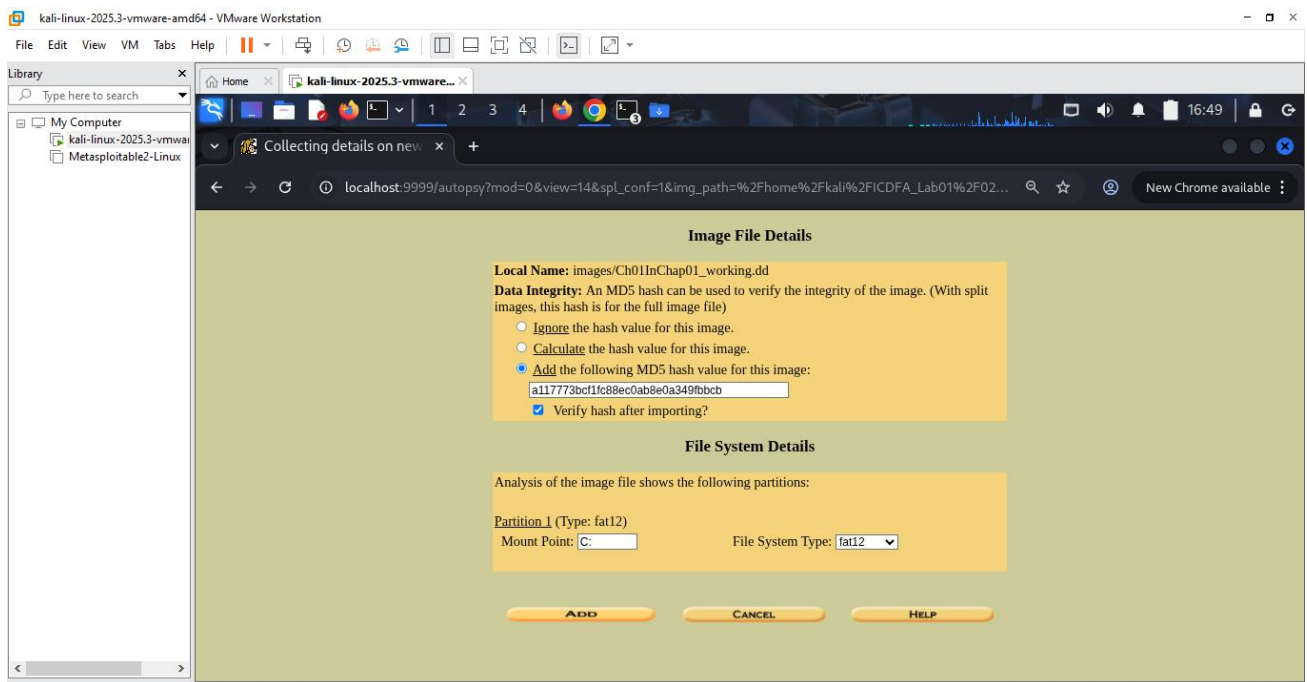


Fig 4.0 Autopsy image hash and filesystem.

Explanation

I supplied the MD5 value and confirmed the FAT12 filesystem before adding the image.

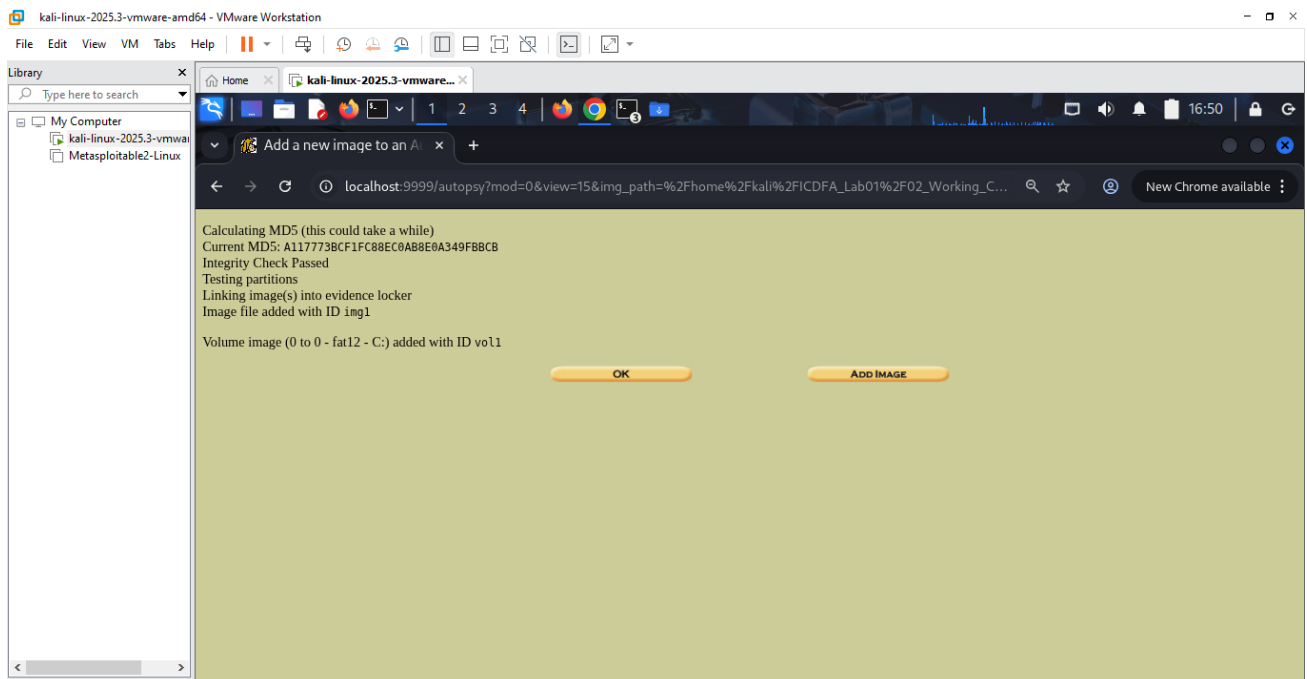
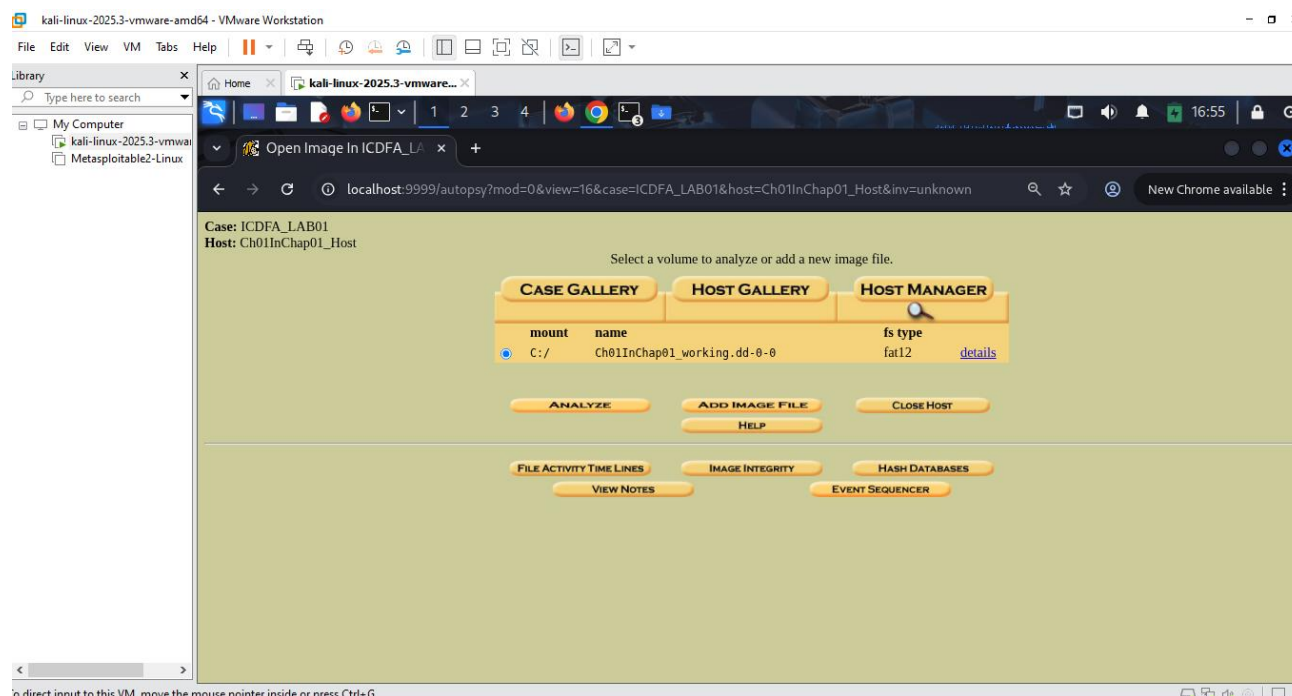


Fig 4.1 Autopsy md5 integrity check passed.

Explanation

Autopsy verified the MD5 successfully, confirming that the image had not changed.

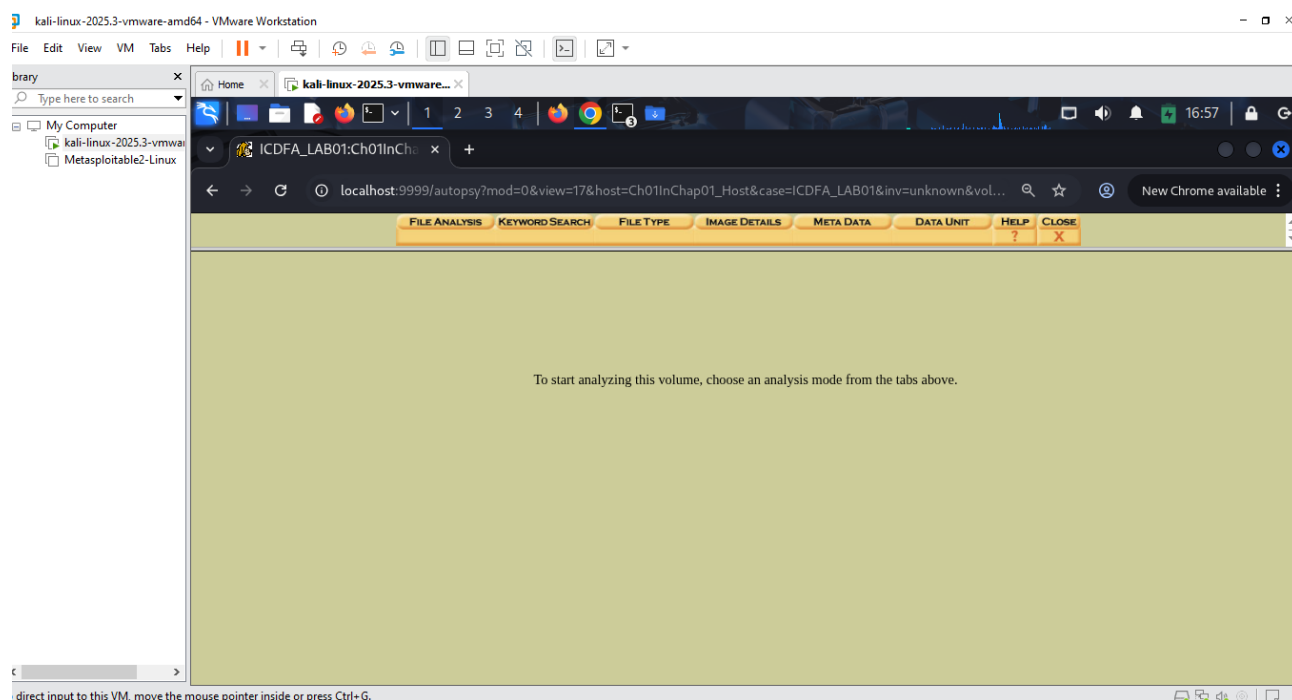


direct input to this VM, move the mouse pointer inside or press Ctrl+G.

Fig 4.2 Selecting the case to examine.

Explanation

I selected the attached FAT12 volume so that analysis could begin.



direct input to this VM, move the mouse pointer inside or press Ctrl+G.

Fig 4.3 File Analyzing.

Explanation

This shows the start of the Autopsy file-analysis process.

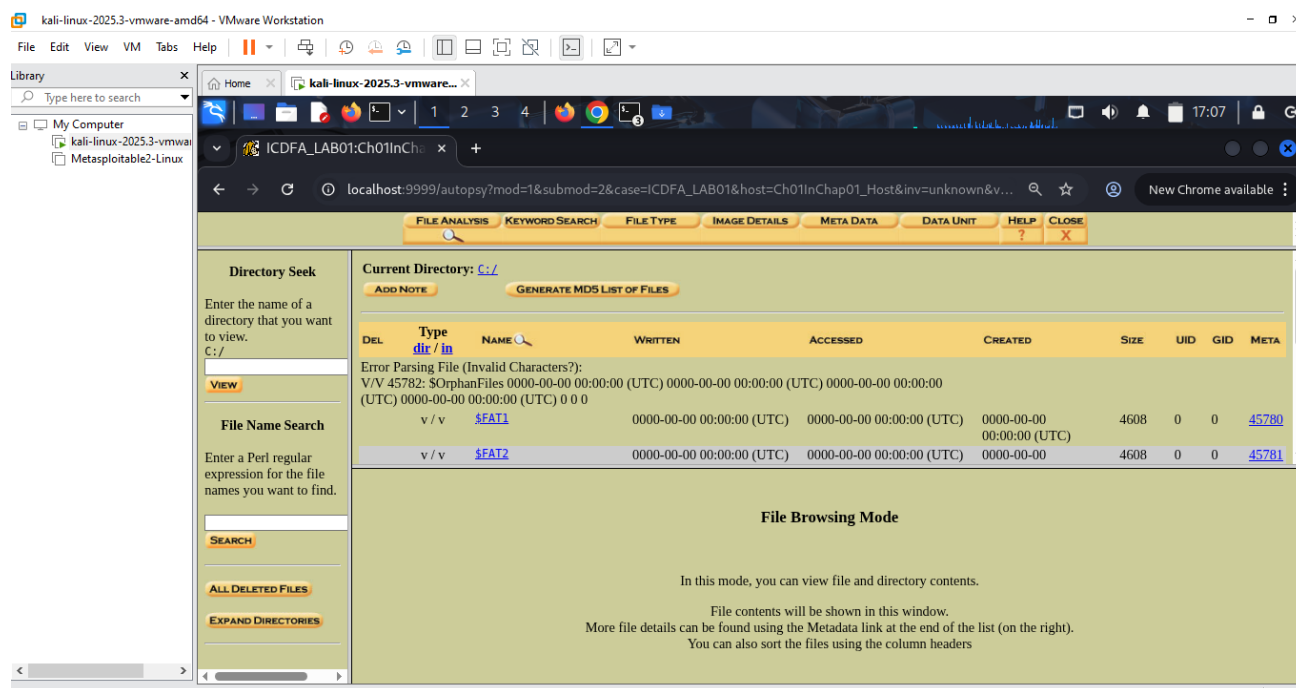


Fig 4.3.1 File Analyzing Page 1.

Explanation

The filesystem contents became available for further examination.

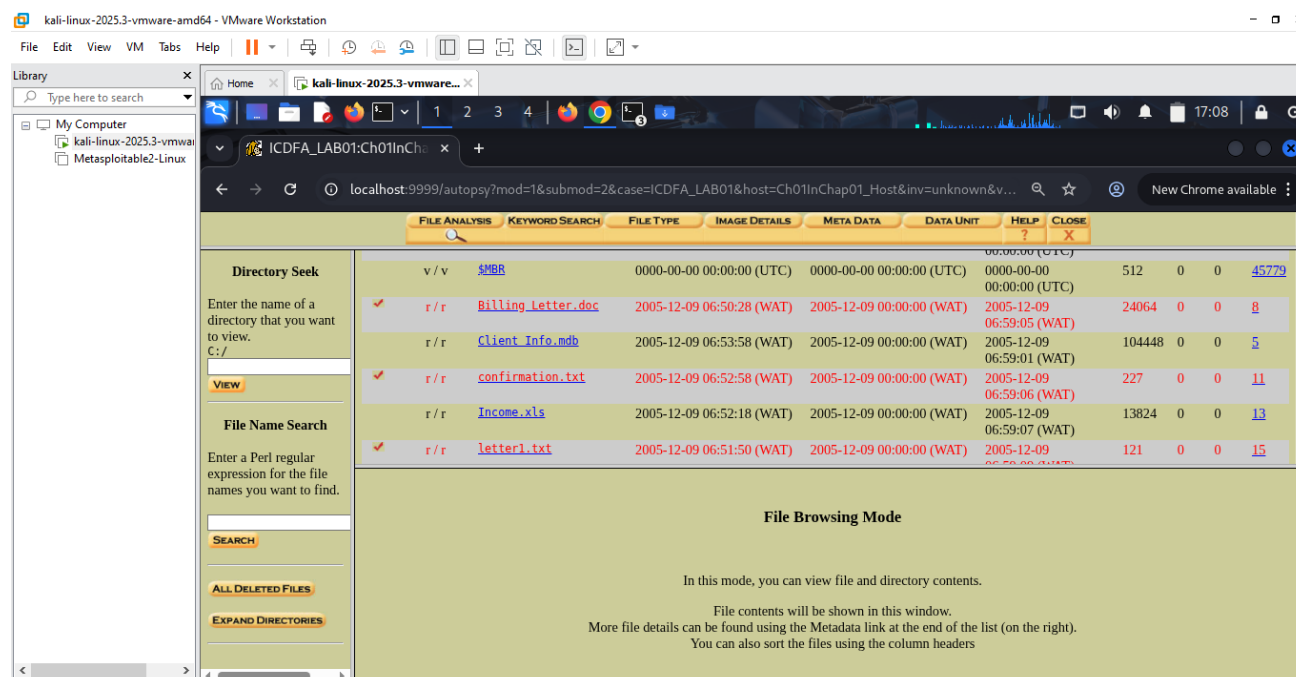


Fig 4.4 File Analyzing and all deleted files.

Explanation

Autopsy displayed both normal and deleted files found in the image.

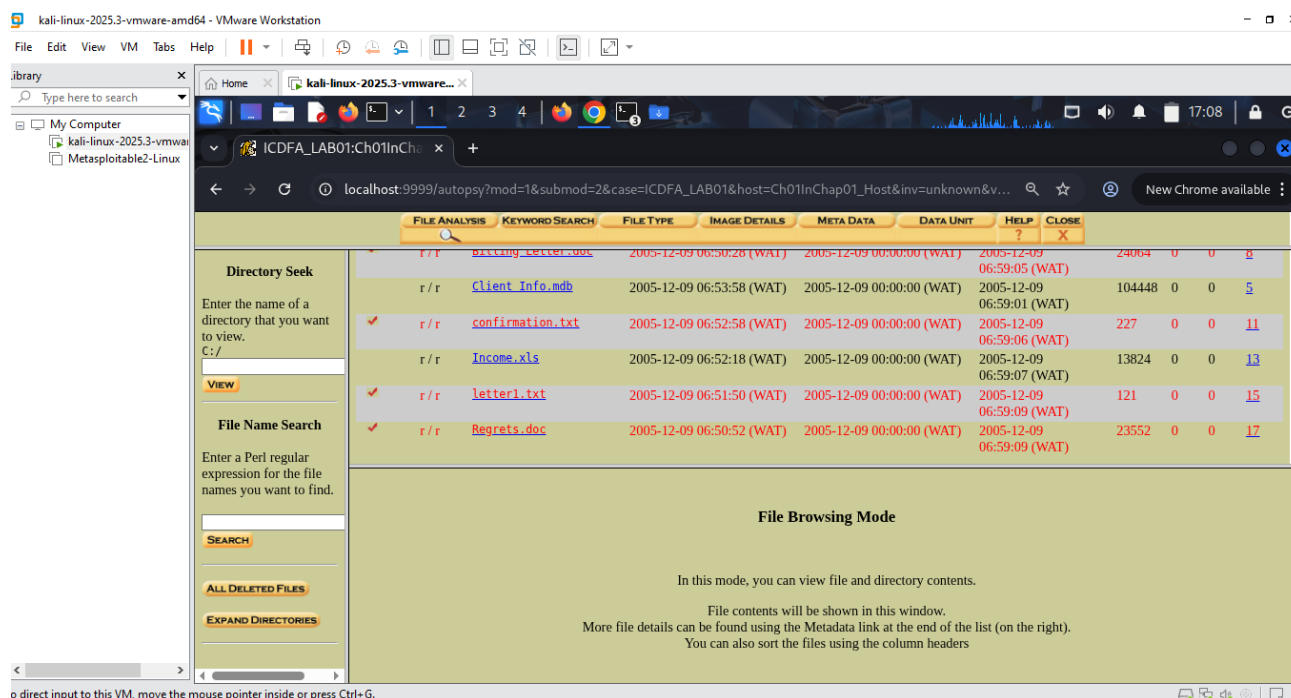


Fig 4.4.1 File Analyzing and all deleted files page 1.

Explanation

This continues the file listing and shows more deleted entries.

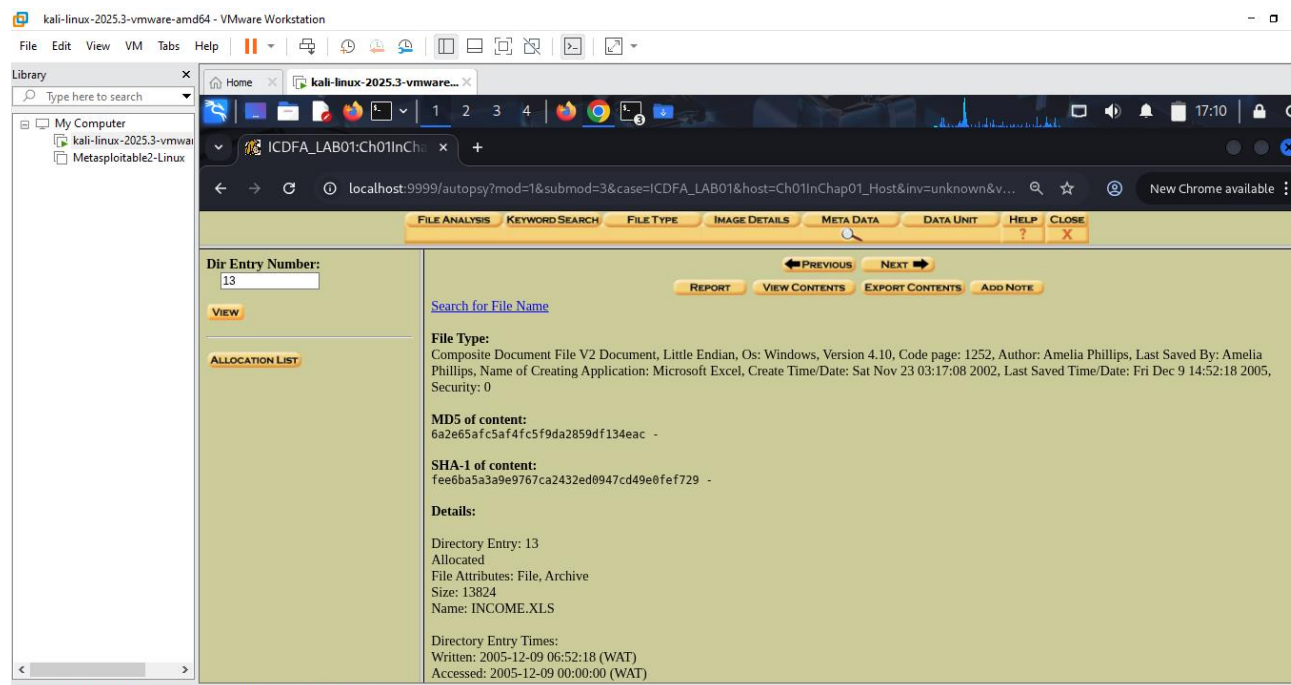


Fig 4.5 Selecting the Income.xls file.

Explanation

I selected Income.xls to examine its forensic metadata.

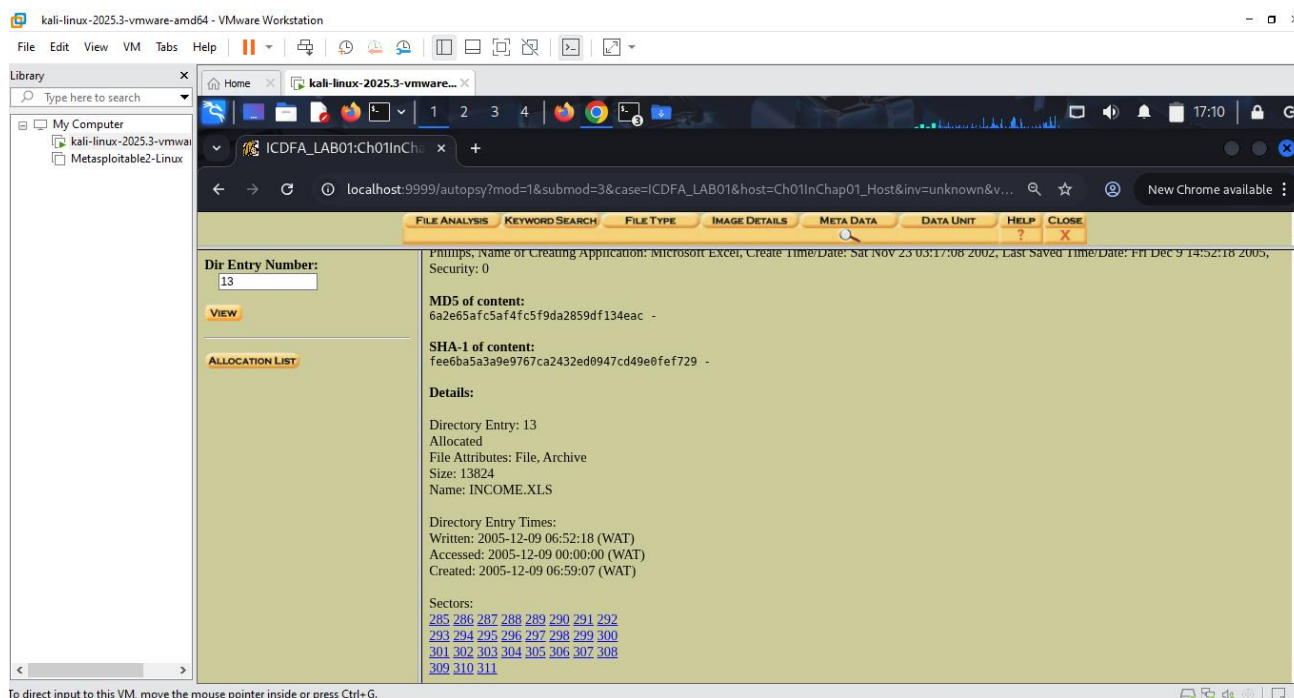


Fig 4.5.1 Selecting the Income.xls file page 1.

Explanation

This shows more details about INCOME.XLS, including its size and data sectors.

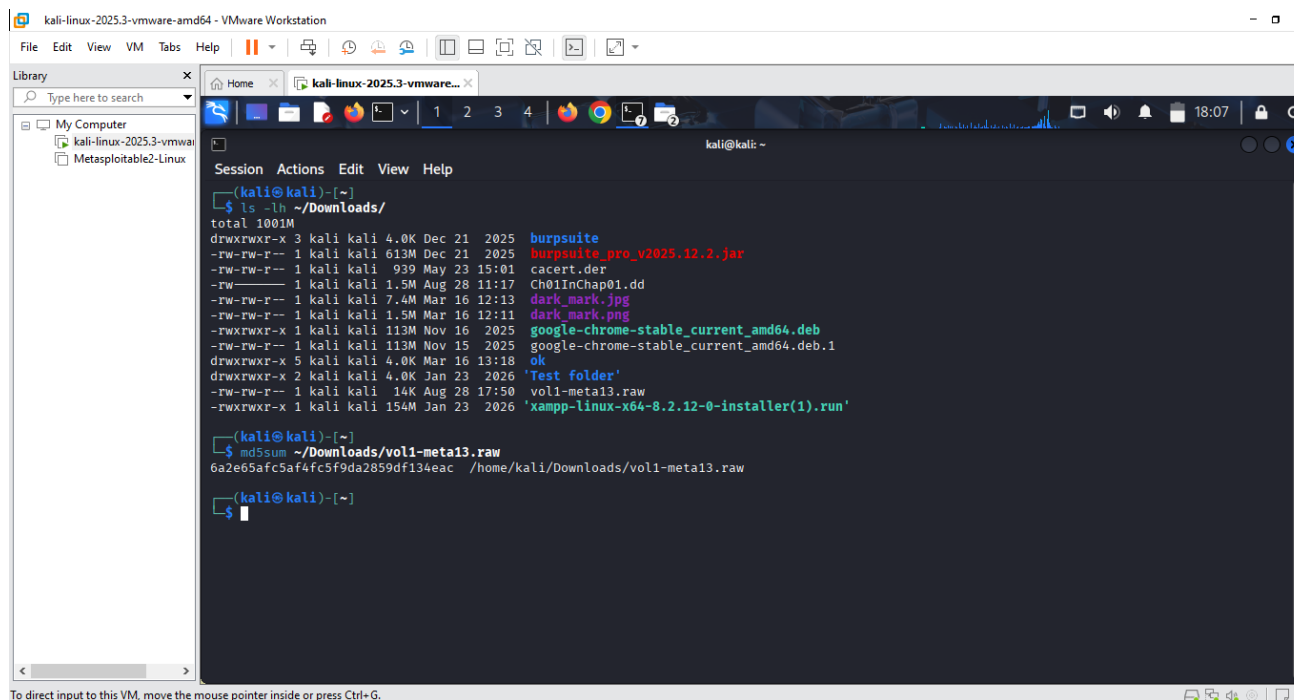


Fig 4.6 Verifying the md5sum hash after download.

Explanation

I checked the MD5 hash of the INCOME.XLS file exported from Autopsy.

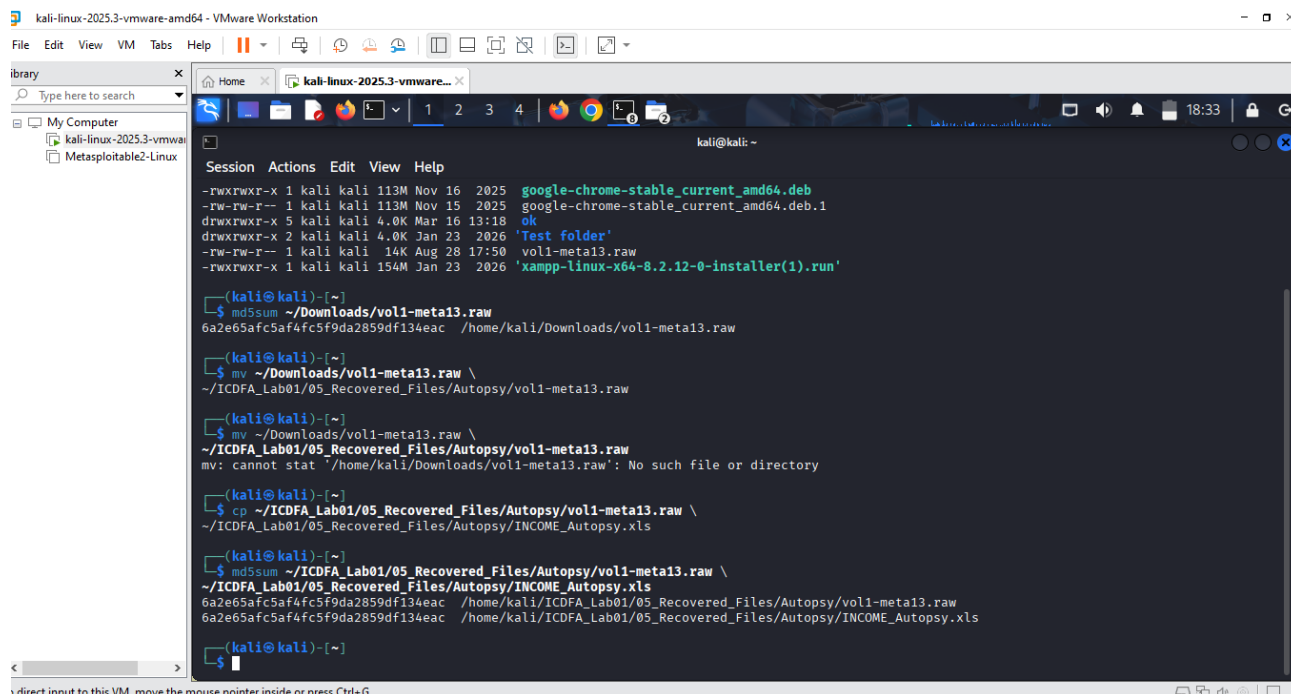


Fig 4.7 Autopsy income xls export hash verified.

Explanation

The exported INCOME.XLS copy was verified using its hash value.

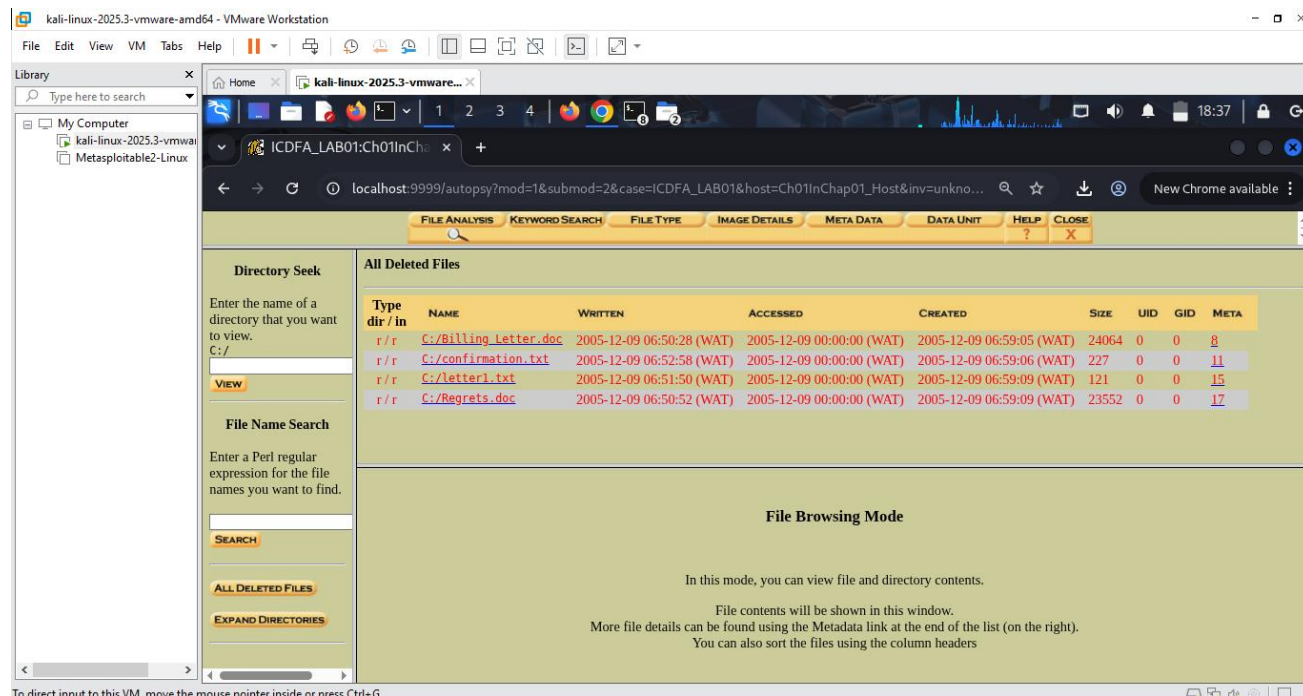


Fig 4.8 List of all deleted files.

Explanation

This shows the complete list of deleted files identified by Autopsy.

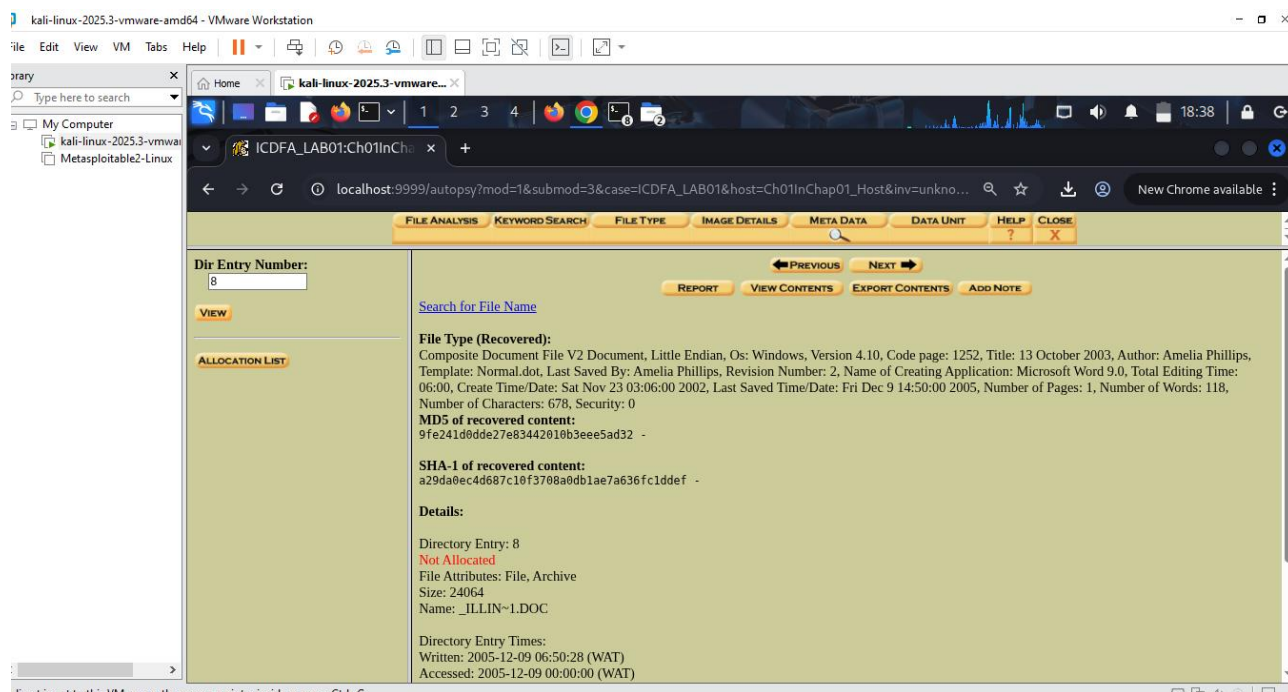


Fig 4.9 Autopsy deleted billing letter metadata.

Explanation

I examined the metadata of the deleted Billing Letter.doc file.

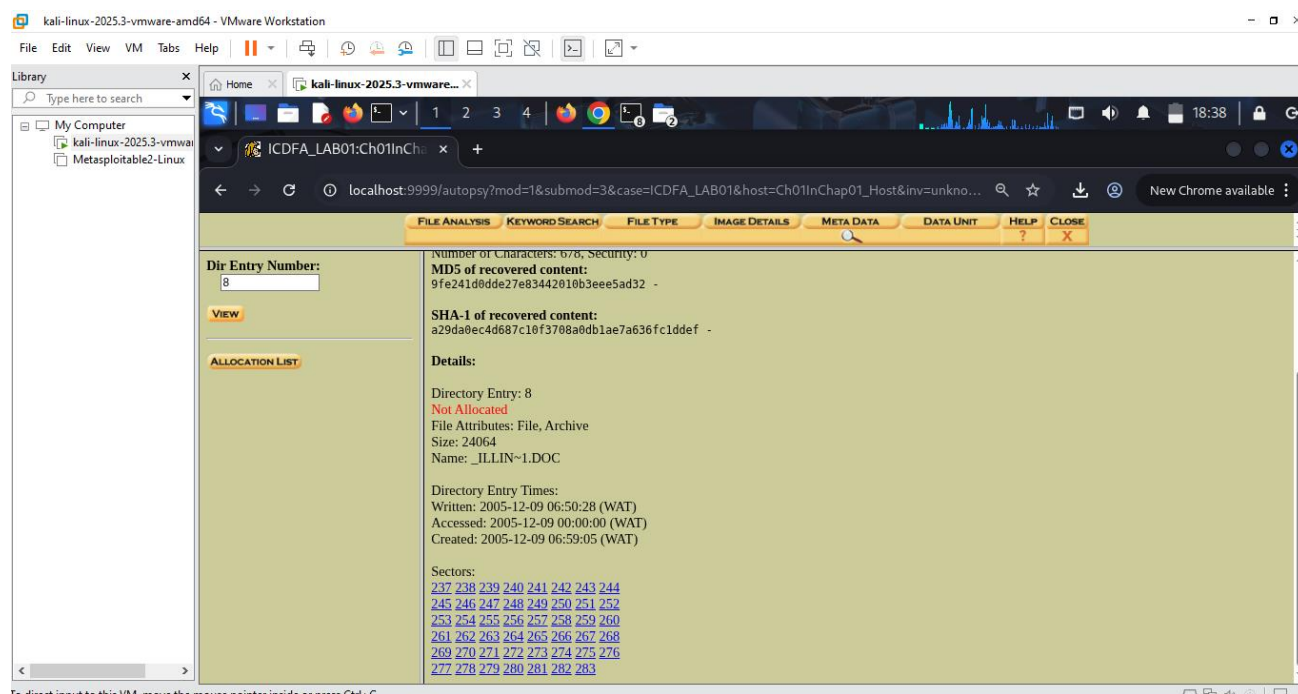


Fig 4.9.1 Autopsy deleted billing letter metadata page 1.

Explanation

This shows additional details and data sectors linked to the deleted Billing Letter.

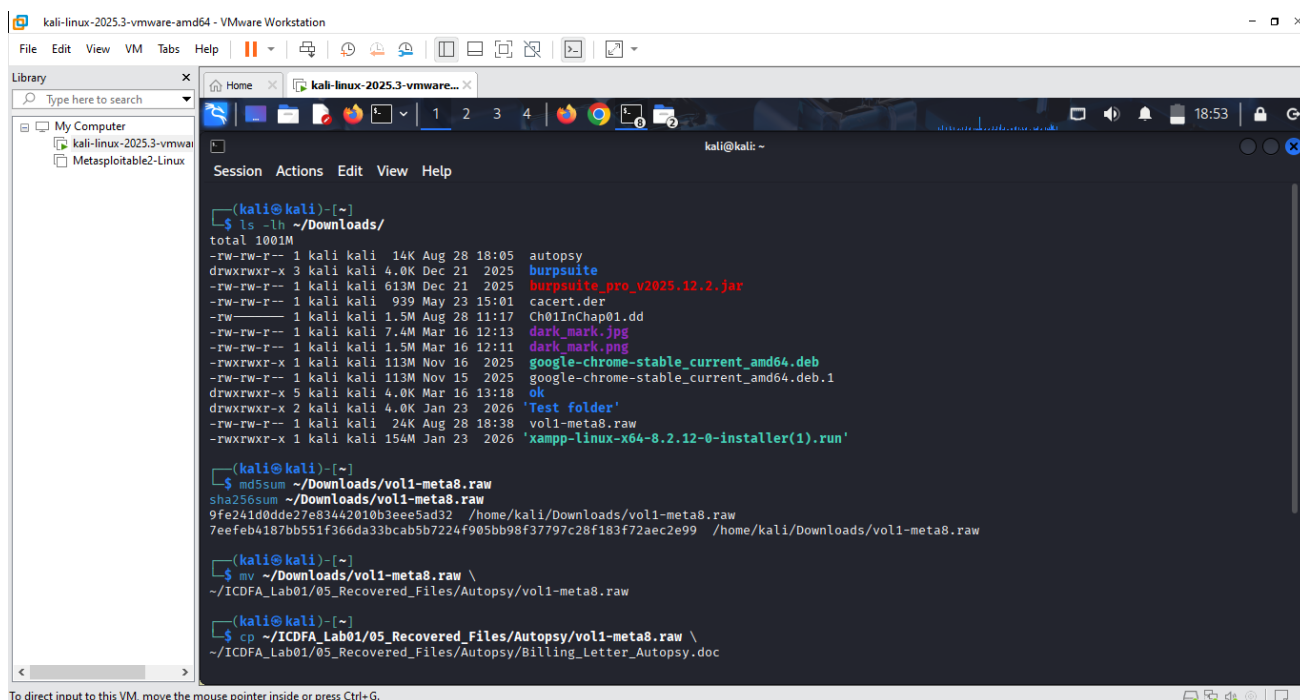


Fig 5. Autopsy billing letter recovery verified.

Explanation

I recovered the Billing Letter and checked its hash for integrity.

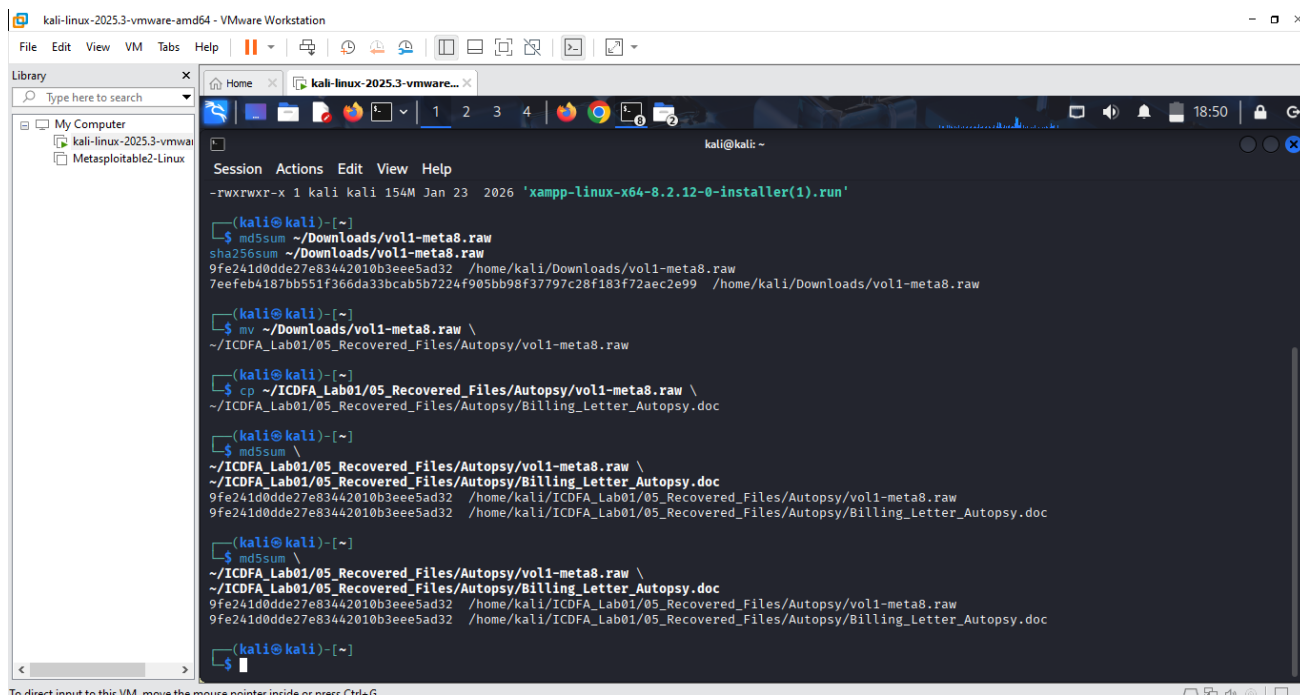


Fig 5.1 Autopsy billing letter recovery verified page 1.

Explanation

The recovered Billing Letter copy was verified against its recorded hash.

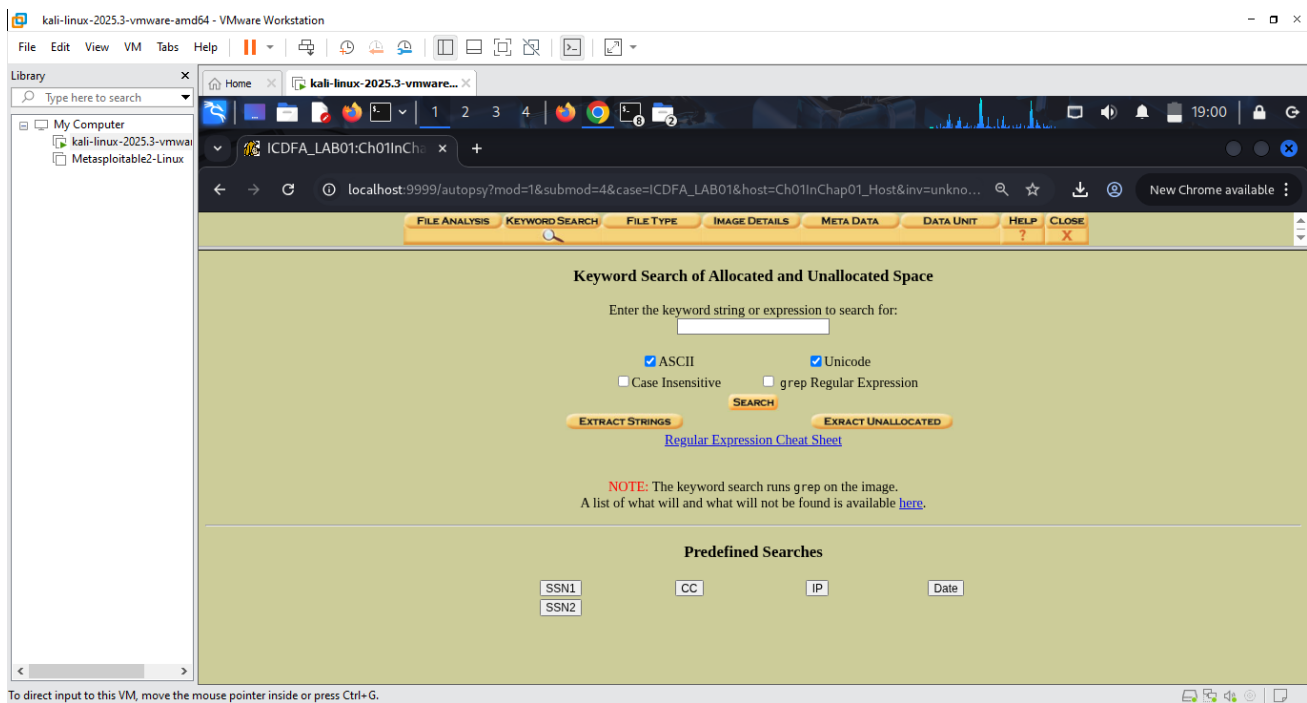


Fig 5.2 Keyword search.

Explanation

I opened Autopsy's keyword-search tool to search the evidence.

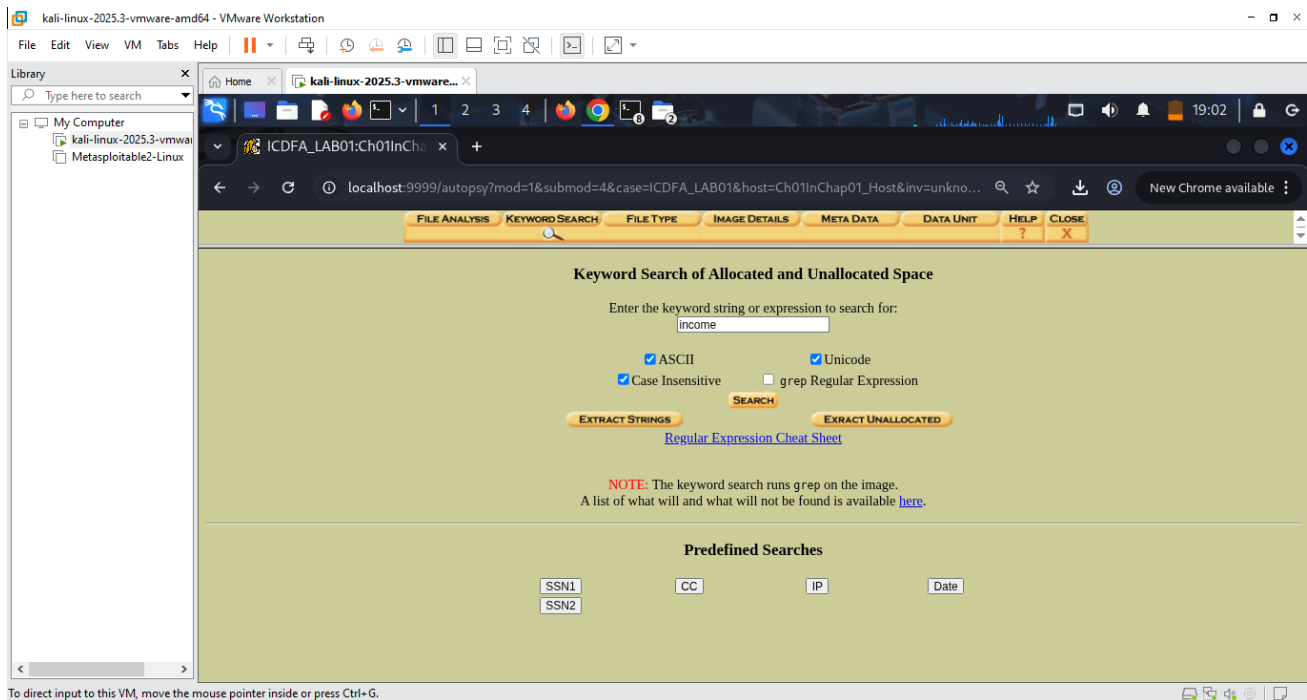


Fig 5.3 Keyword search for income.

Explanation

I searched the image for the keyword income.

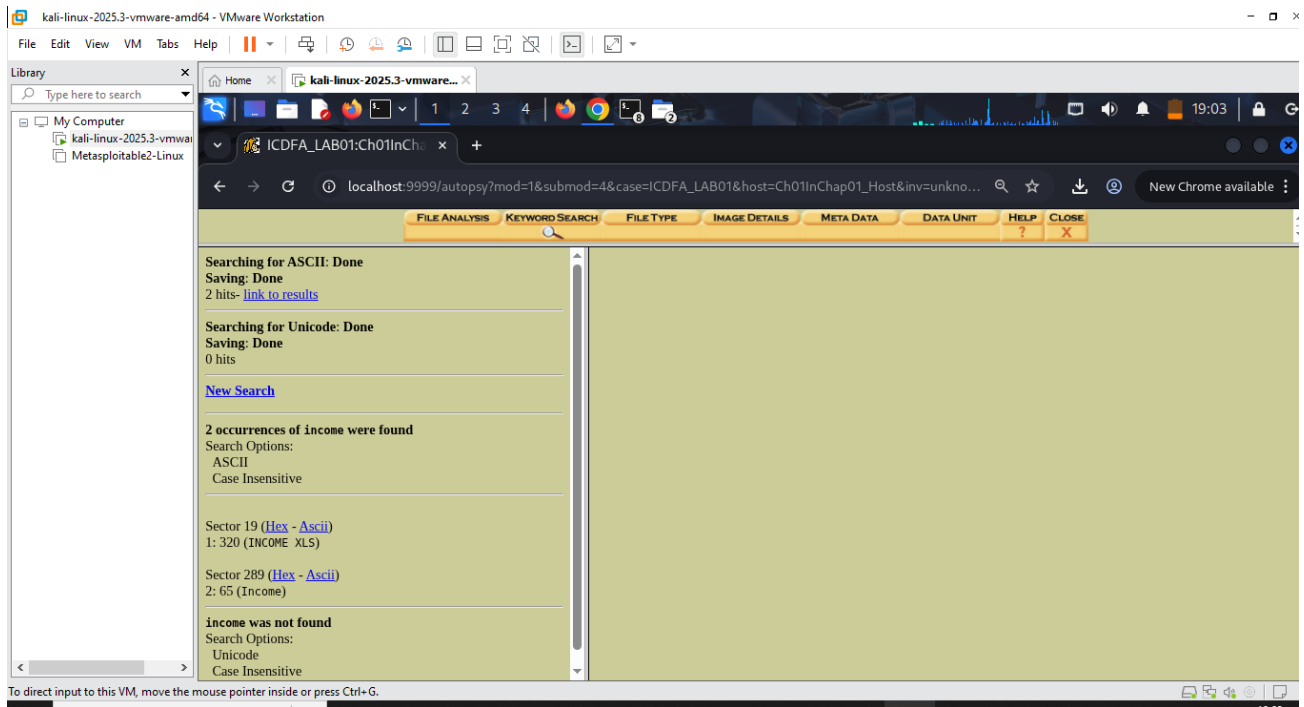


Fig 5.4 Autopsy keyword search income results.

Explanation

Autopsy returned matching results for the income search.

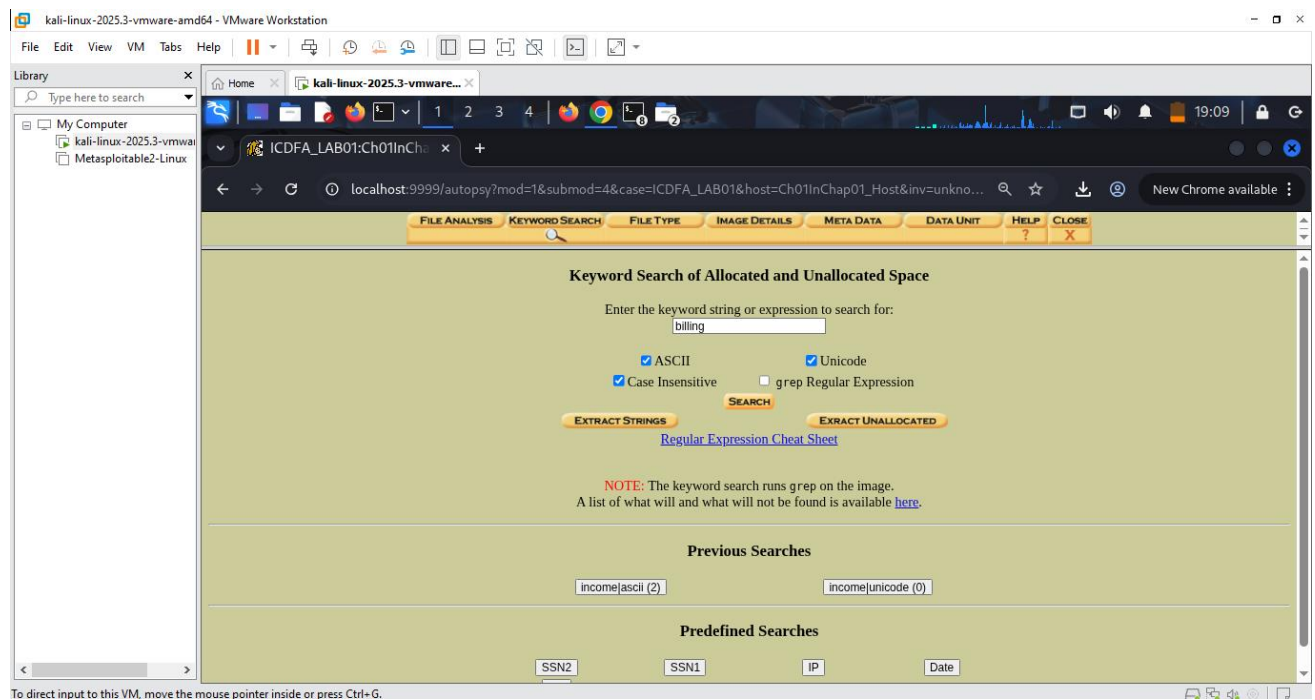


Fig 5.5 Keyword search for billing.

Explanation

I carried out another keyword search using billing.

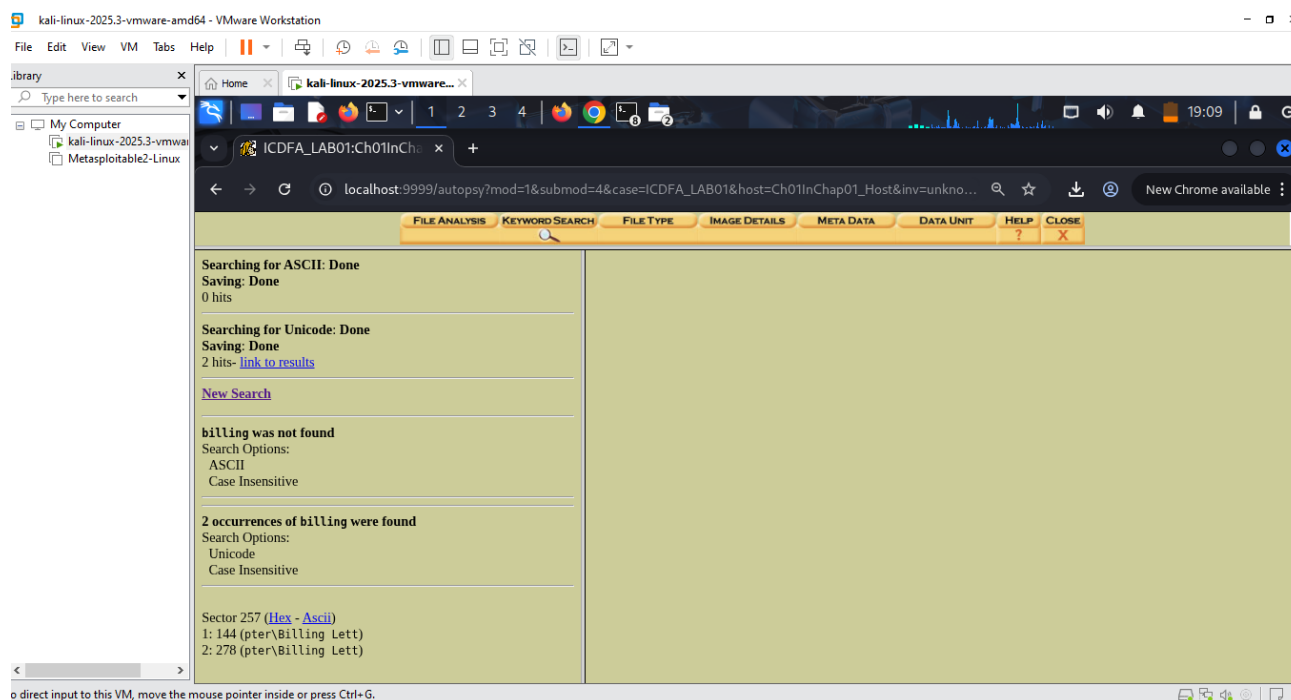


Fig 5.6 Autopsy keyword search billing results.

Explanation

The search returned occurrences of the word billing.

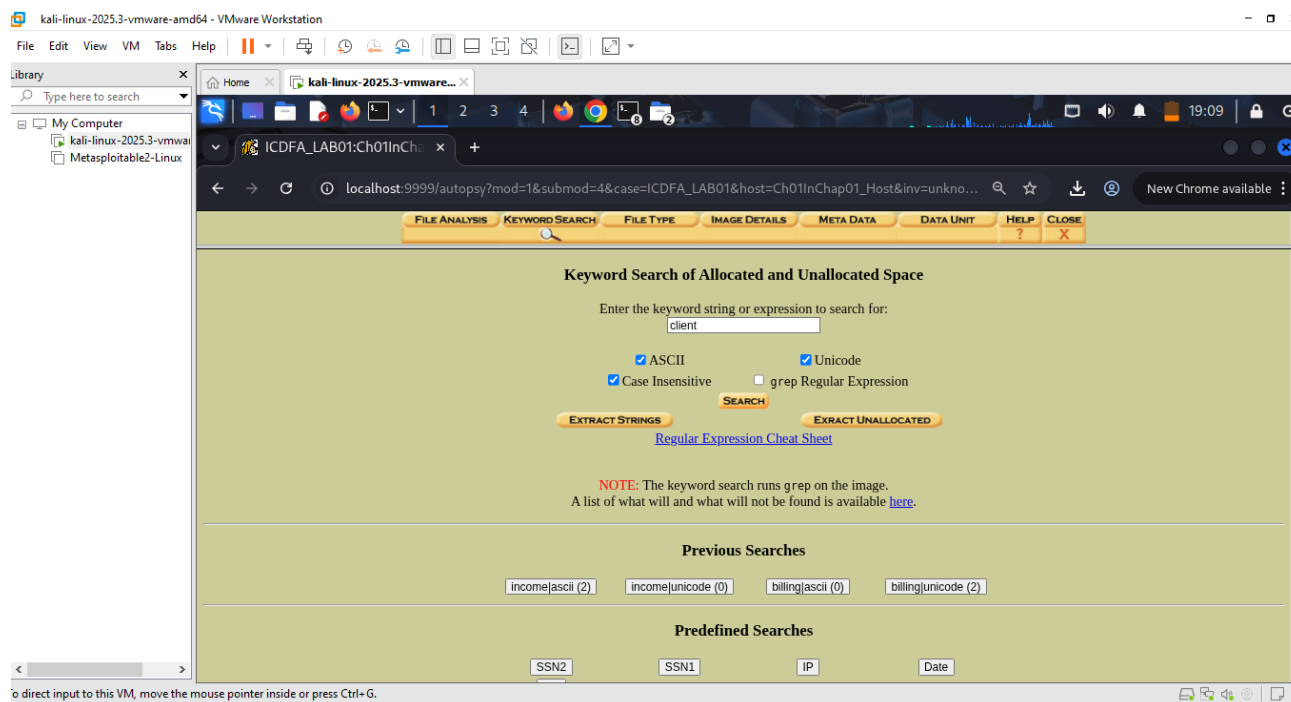


Fig 5.7 Keyword search for client.

Explanation

I then searched the evidence for the keyword client.

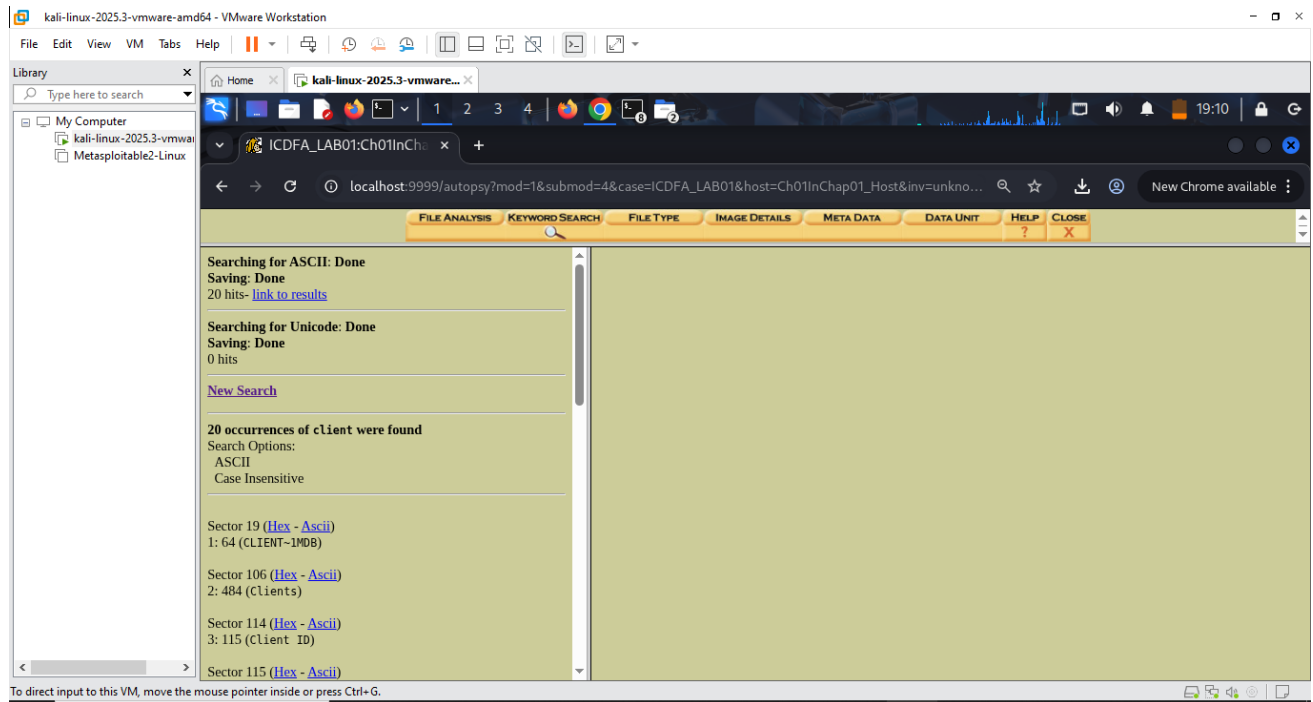


Fig 5.8. Autopsy keyword search client results.

Explanation

Autopsy found several matches for the word client.

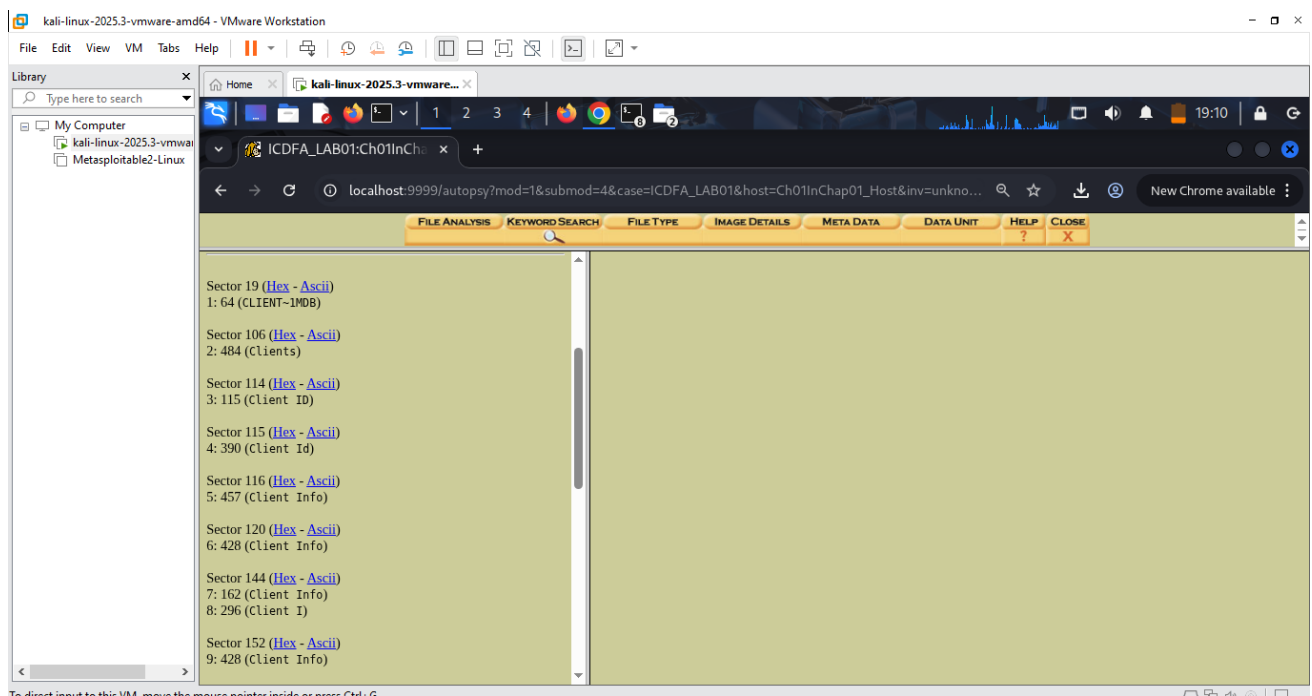


Fig 5.8.1 Autopsy keyword search client results page 1.

Explanation

This shows additional locations where client was found.

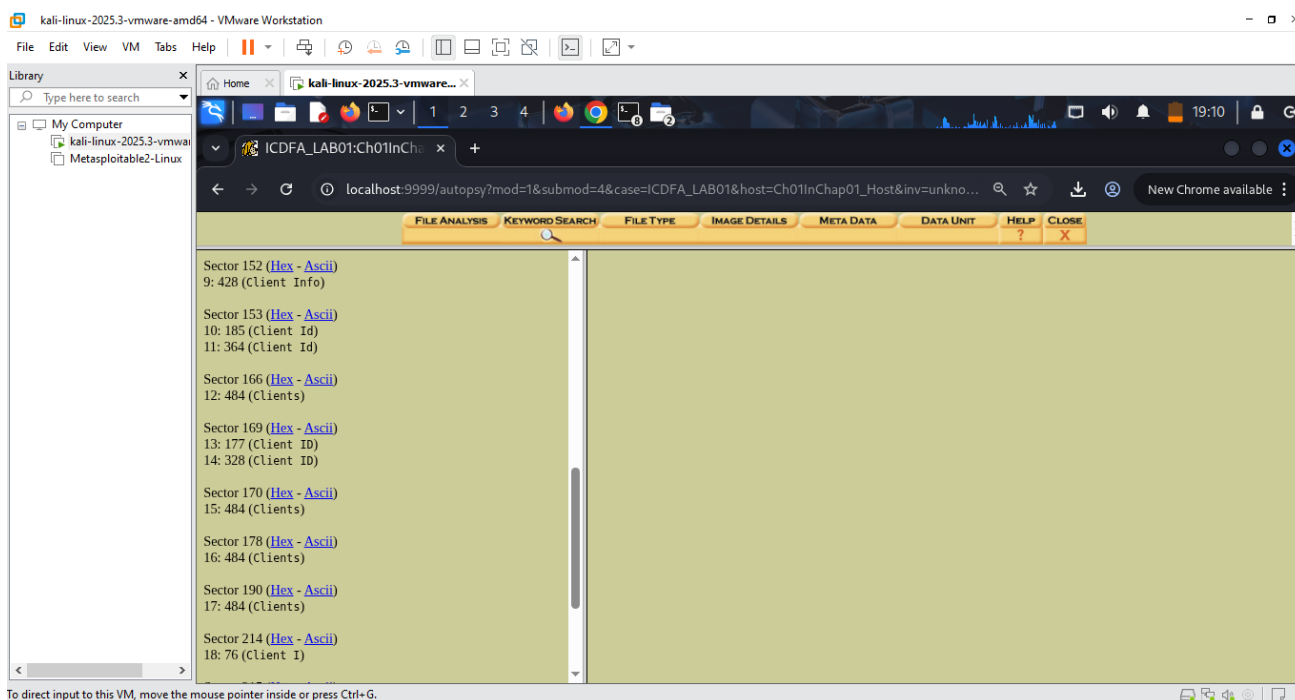


Fig 5.8.2 Autopsy keyword search client results page 2.

Explanation

More client keyword hits are displayed in this part of the results.

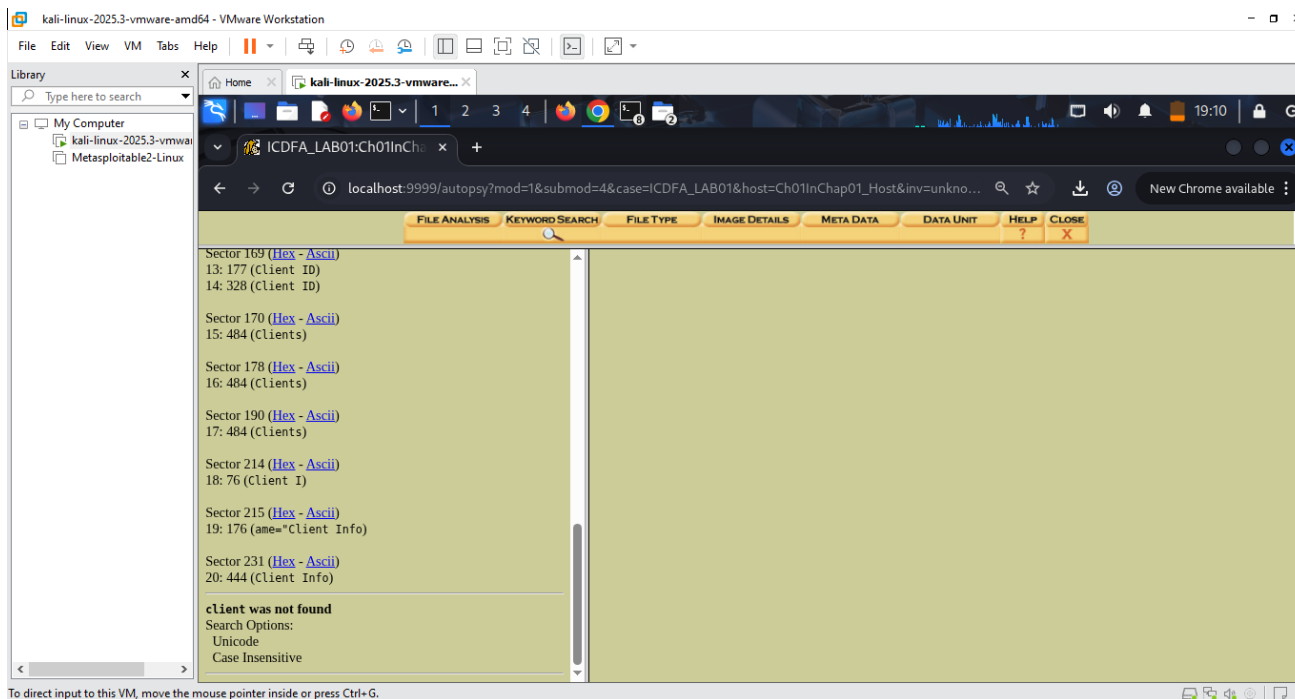


Fig 5.8.3 Autopsy keyword search client results page 3.

Explanation

This shows the remaining client keyword-search results.

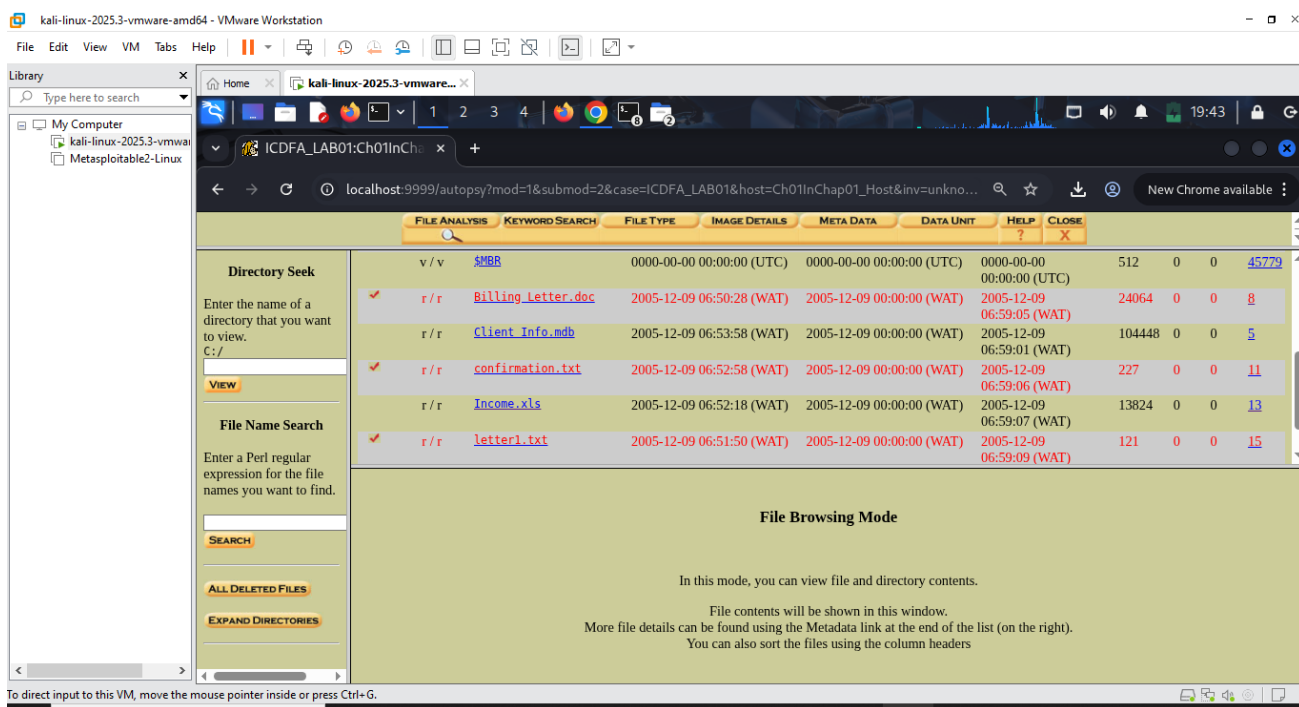


Fig 5.9 Tag relevant evidence for INCOME.XLS.

Explanation

I selected INCOME.XLS as relevant evidence for documentation.

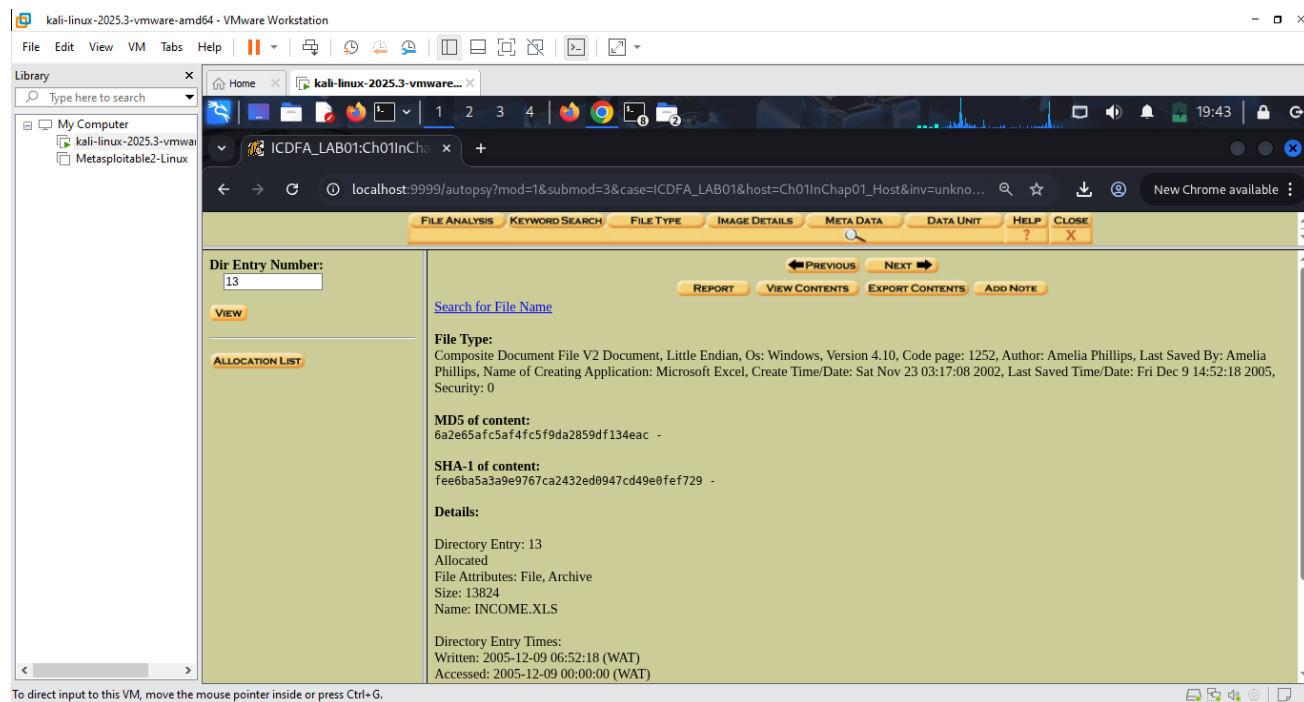


Fig 6. Adding note to tag relevant evidence for INCOME.XLS.

Explanation

I opened the option to add an evidence note to INCOME.XLS.

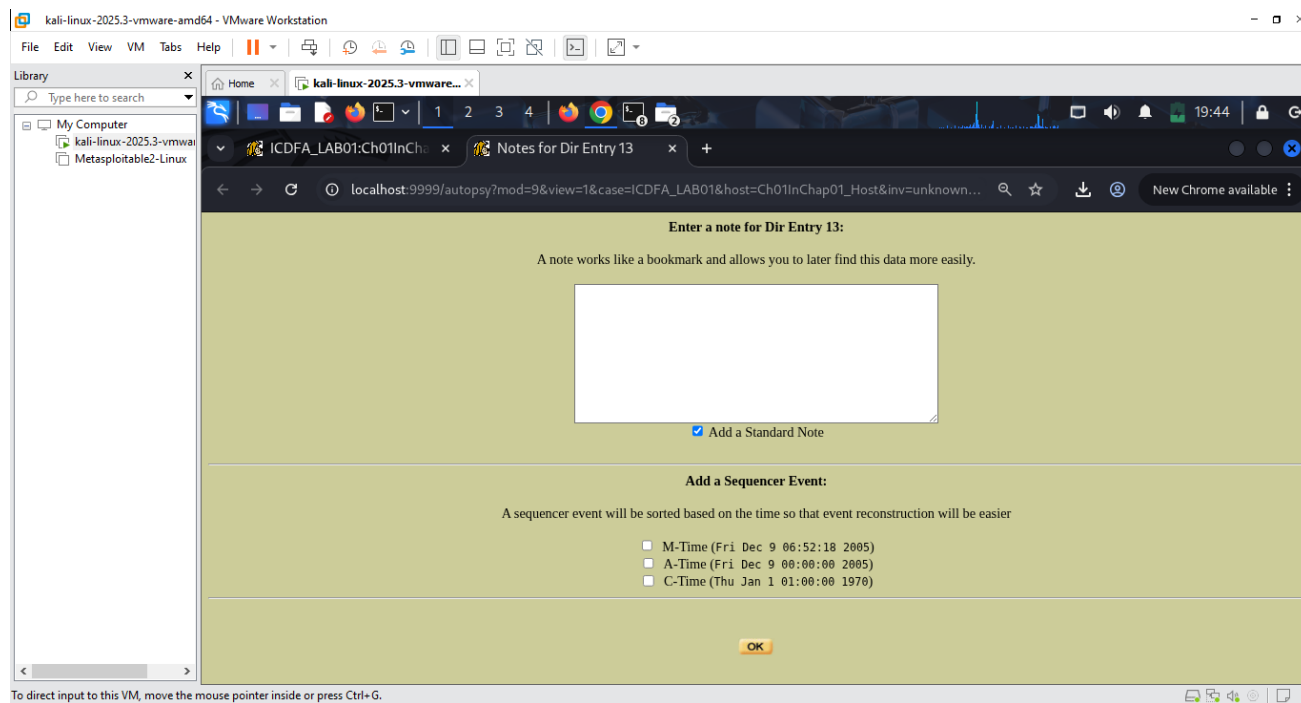


Fig 6.1 Adding note to tag relevant evidence for INCOME.XLS page 1.

Explanation

This shows the form used to enter a standard forensic note.

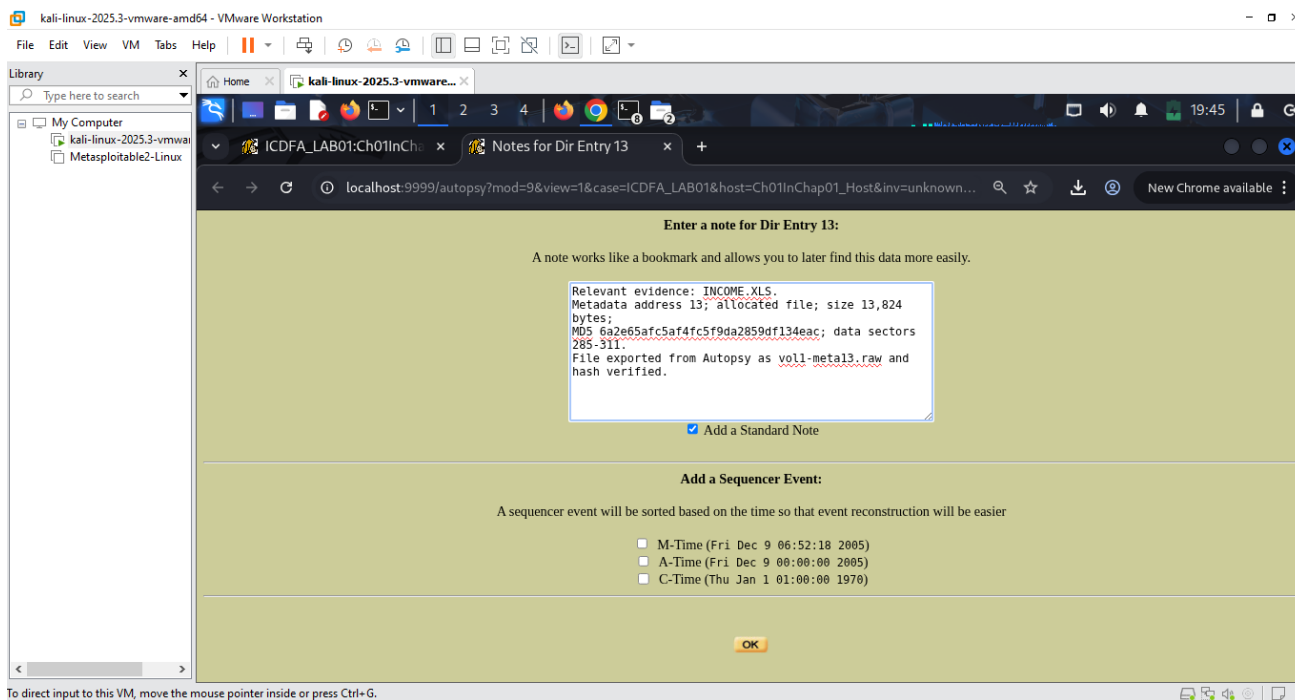


Fig 6.2 Autopsy income xls evidence note.

Explanation

I recorded the important details of INCOME.XLS in the evidence note.

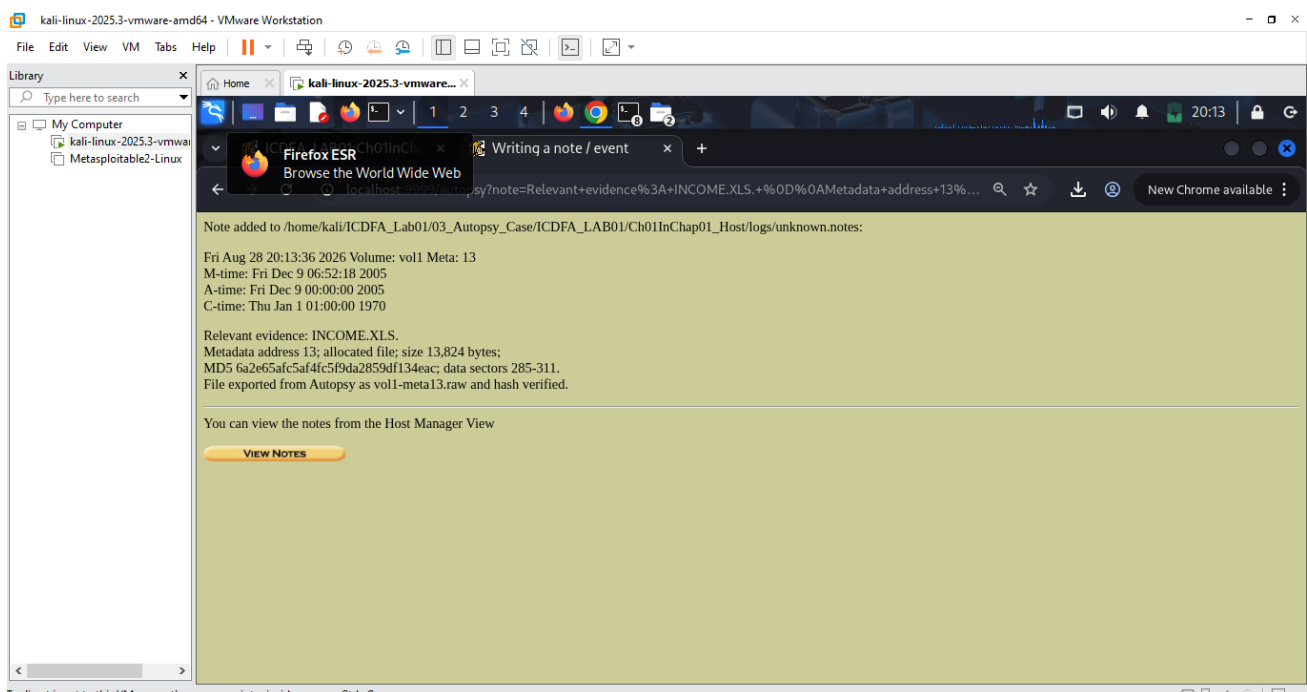


Fig 6.3 Autopsy income xls evidence note save.

Explanation

The INCOME.XLS evidence note was saved successfully.

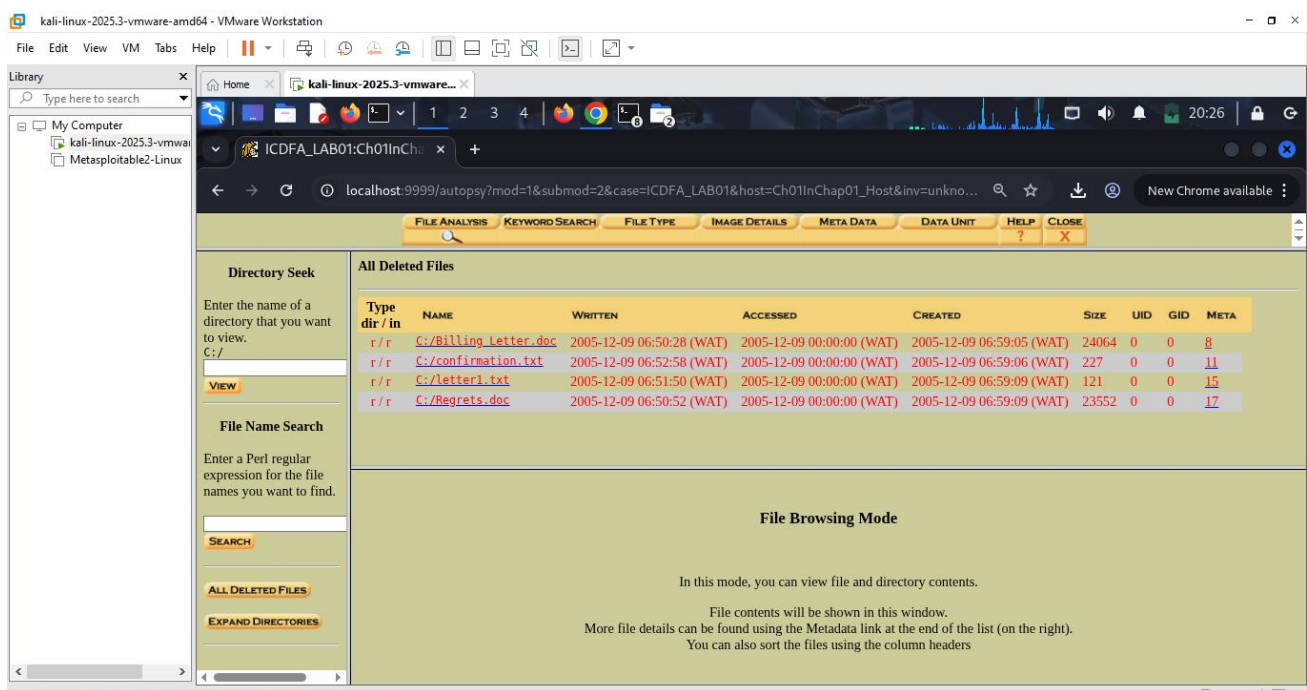


Fig 6.4 Tag relevant evidence for Billing_letter.doc.

Explanation

I also selected the deleted Billing Letter.doc as relevant evidence.

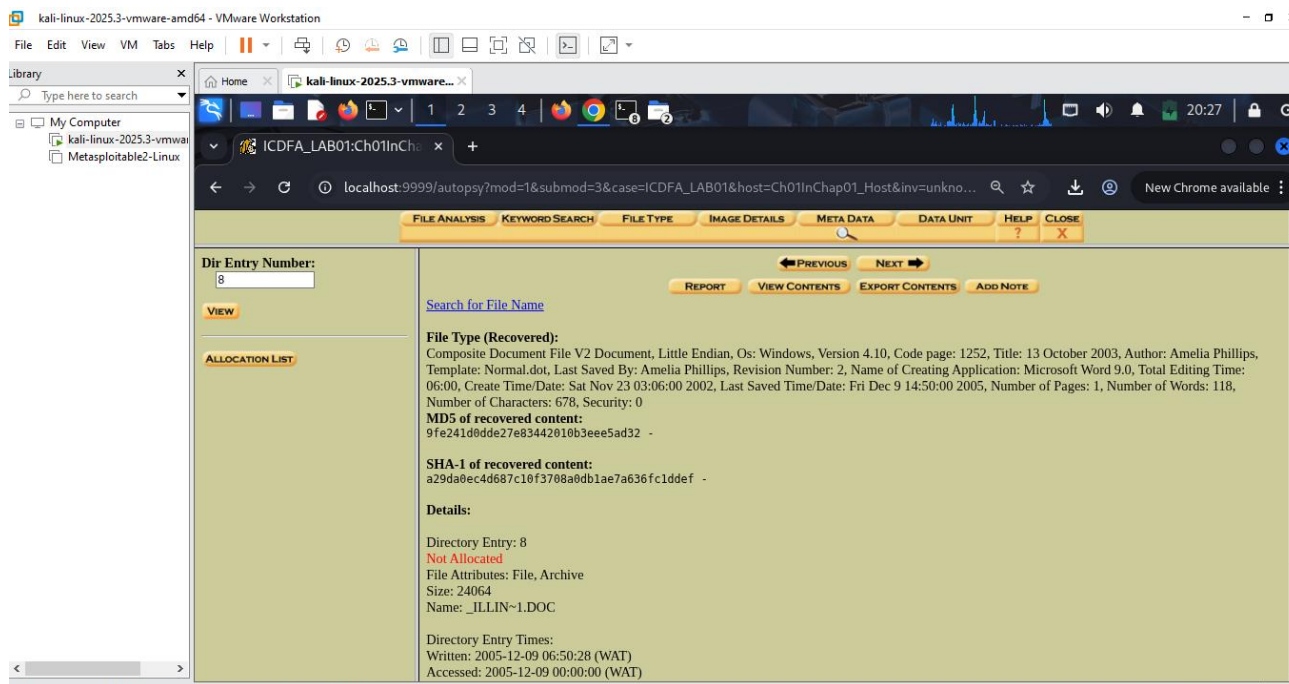


Fig 6.5 Adding note to tag relevant evidence for Billing_letter.doc.

Explanation

I reviewed the Billing Letter metadata before adding the note.

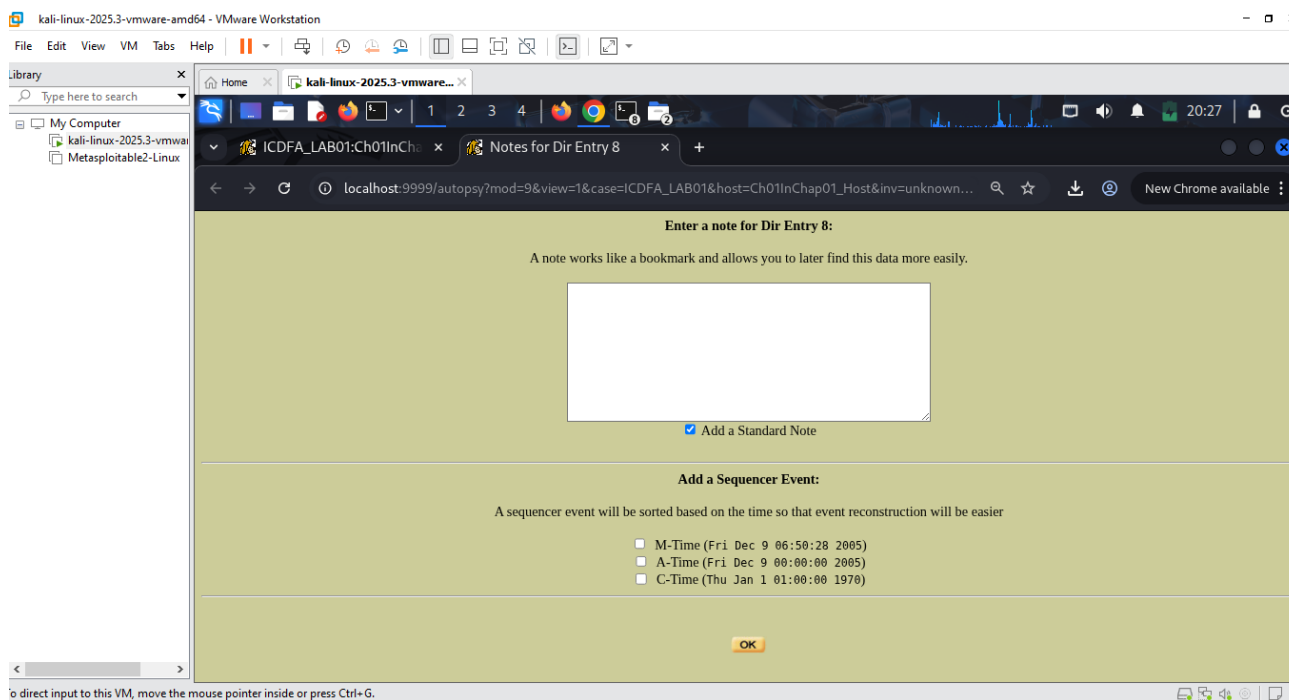


Fig 6.6 Autopsy Billing_letter.doc evidence note.

Explanation

This shows the note-entry screen for the Billing Letter.

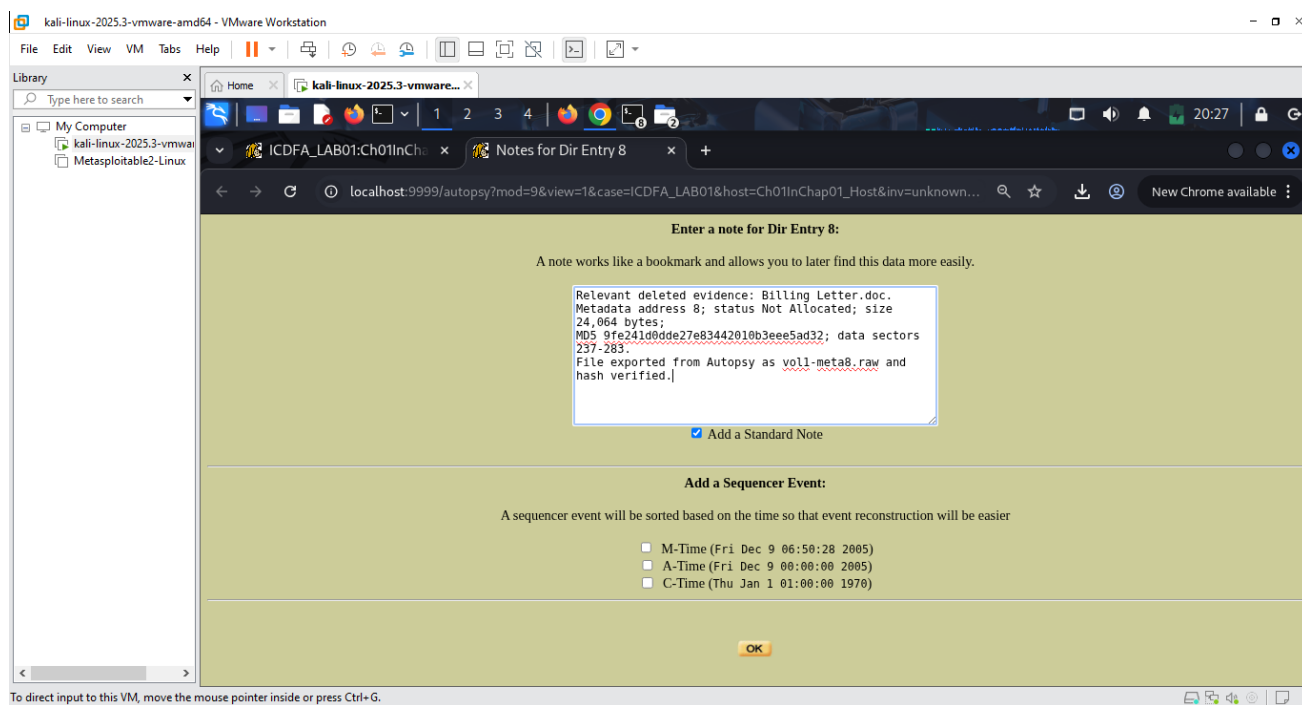


Fig 6.7 Autopsy Billing_letter.doc evidence note page 1.

Explanation

I entered the important recovery and metadata details for the Billing Letter.

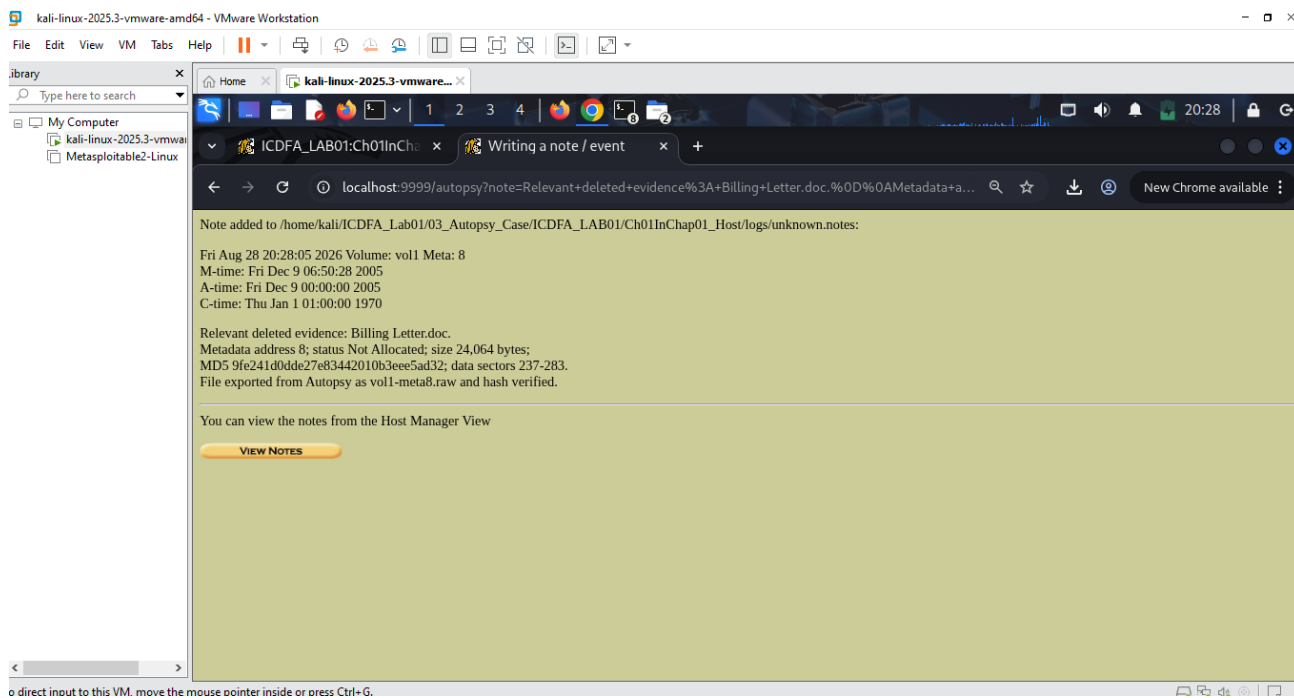


Fig 6.8 Autopsy Billing_letter.doc evidence note save.

Explanation

The Billing Letter evidence note was saved successfully.

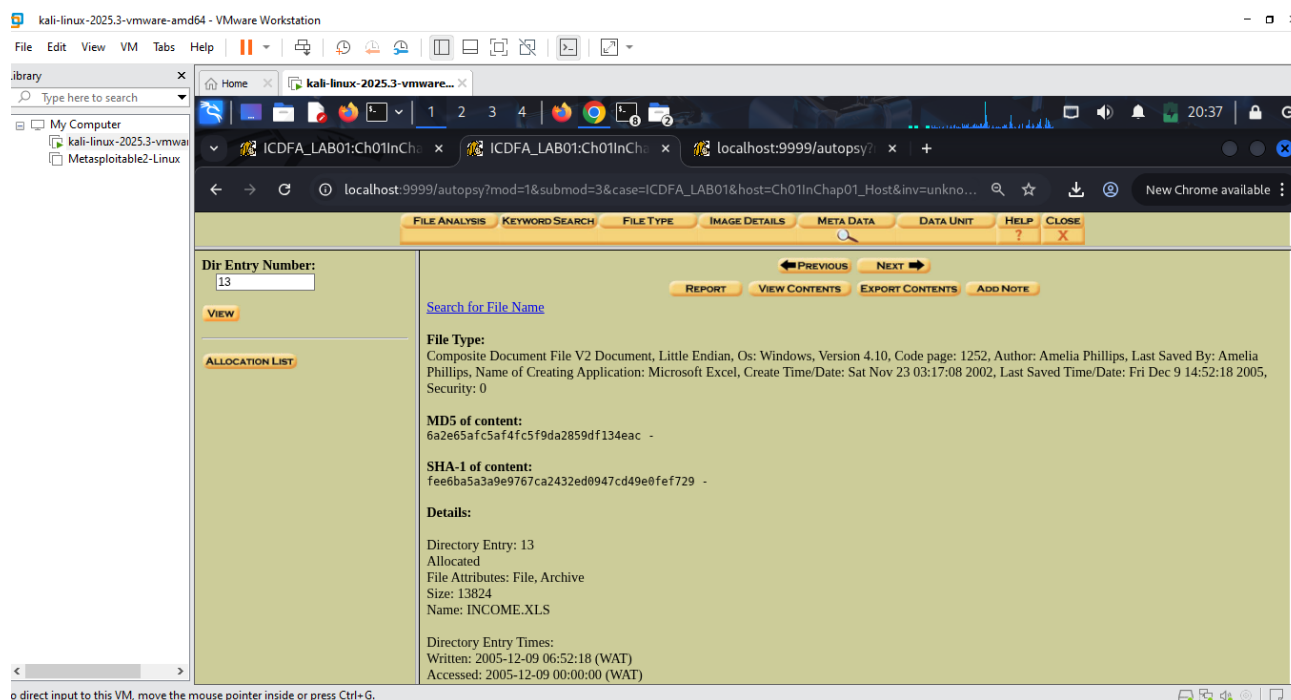


Fig 6.9 Generating the INCOME.XLS report.

Explanation

I selected the option to generate a forensic report for INCOME.XLS.

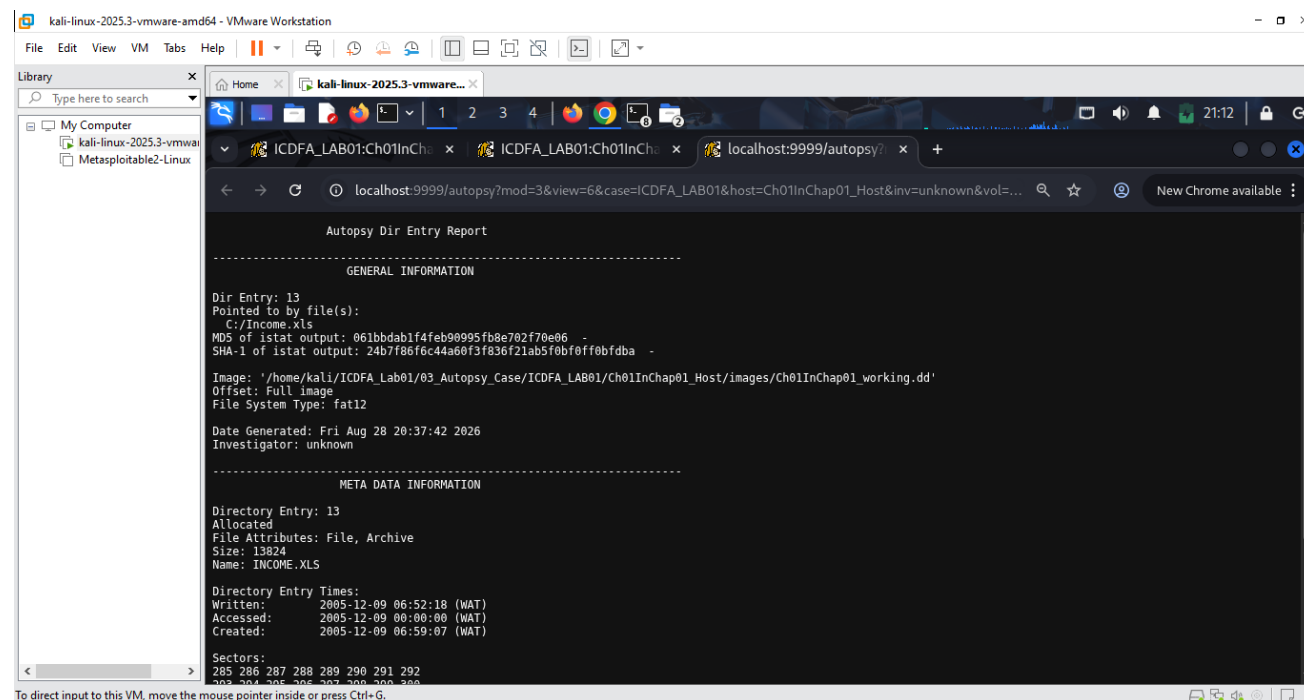


Fig 7. Generate report.

Explanation

Autopsy generated a report containing the file's forensic information.

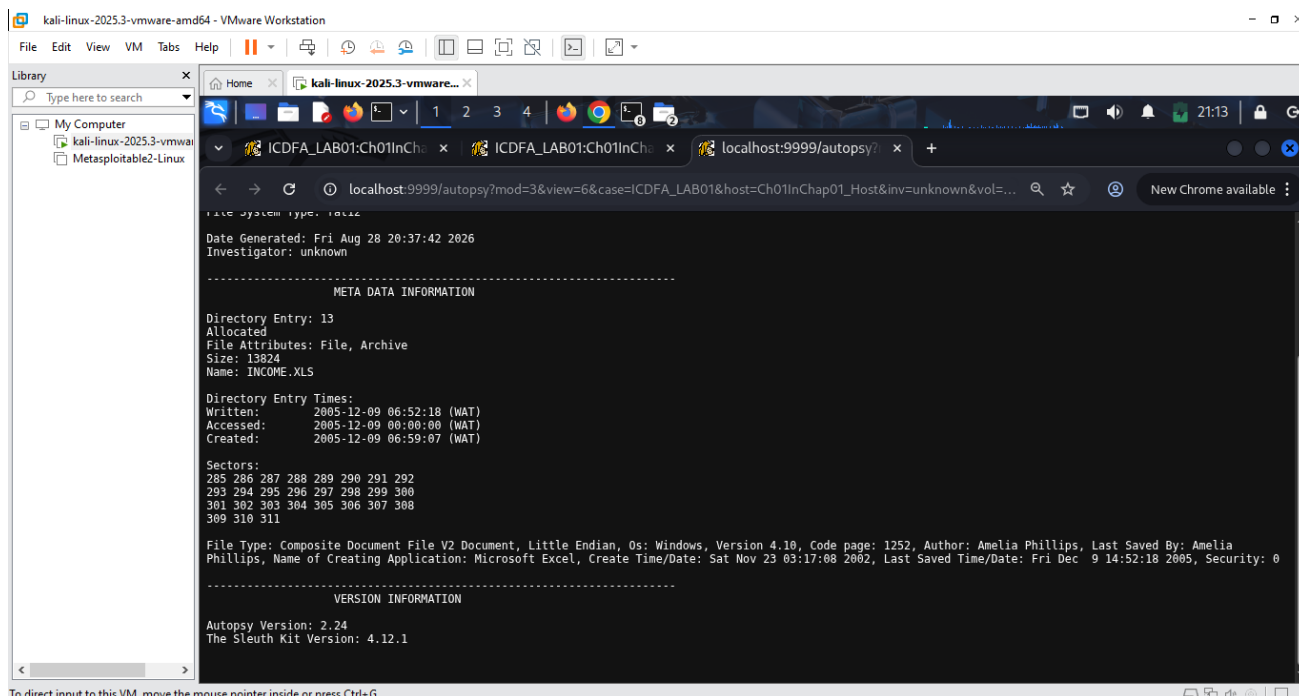


Fig 7.1 Generate report page 1.

Explanation

This part of the report shows additional metadata and sector information.

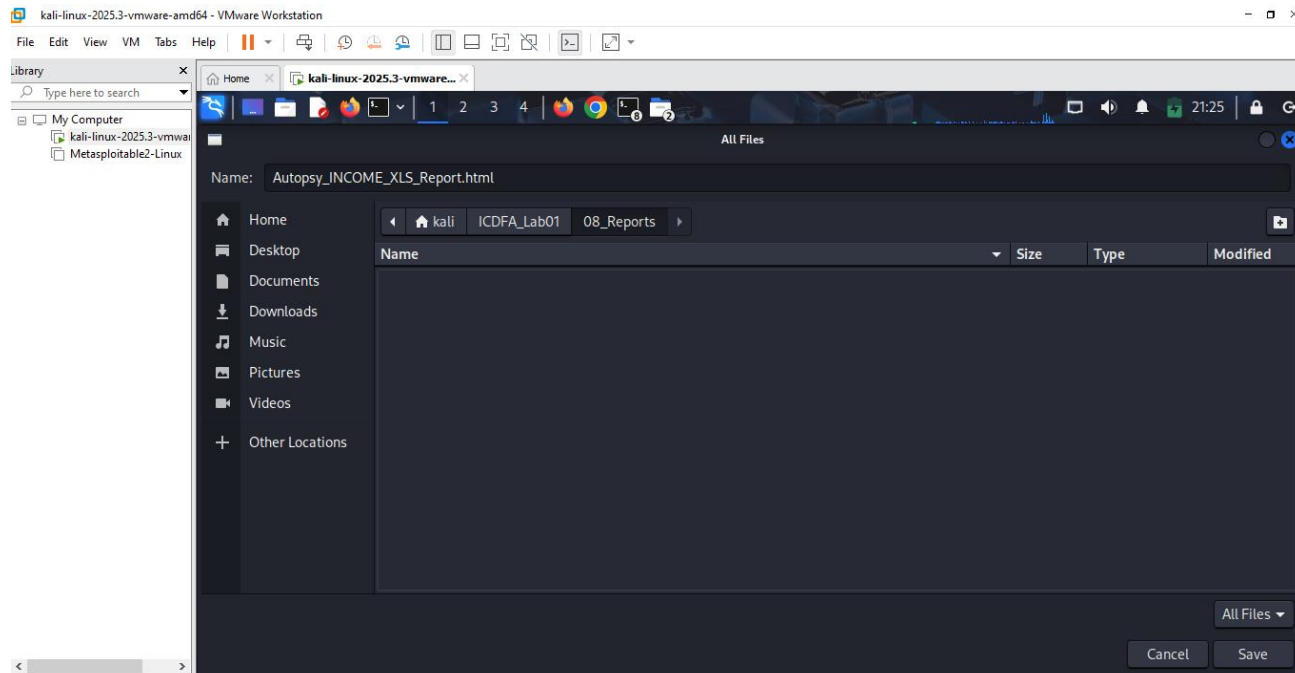


Fig 7.2 Saving the Report html file.

Explanation

I saved the generated Autopsy report as an HTML file.

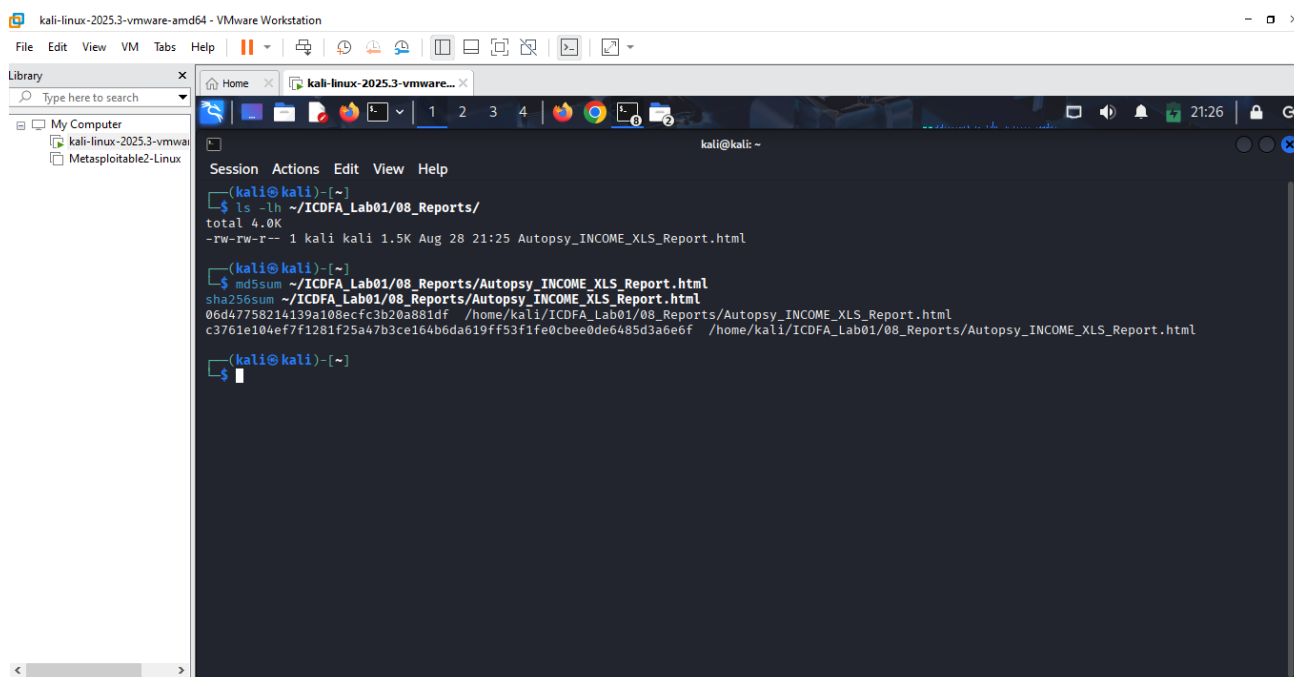


Fig 7.3 Verifying and hashing the Report html file.

Explanation

I verified and hashed the saved report to document its integrity.

Part B — Sleuth Kit Analysis

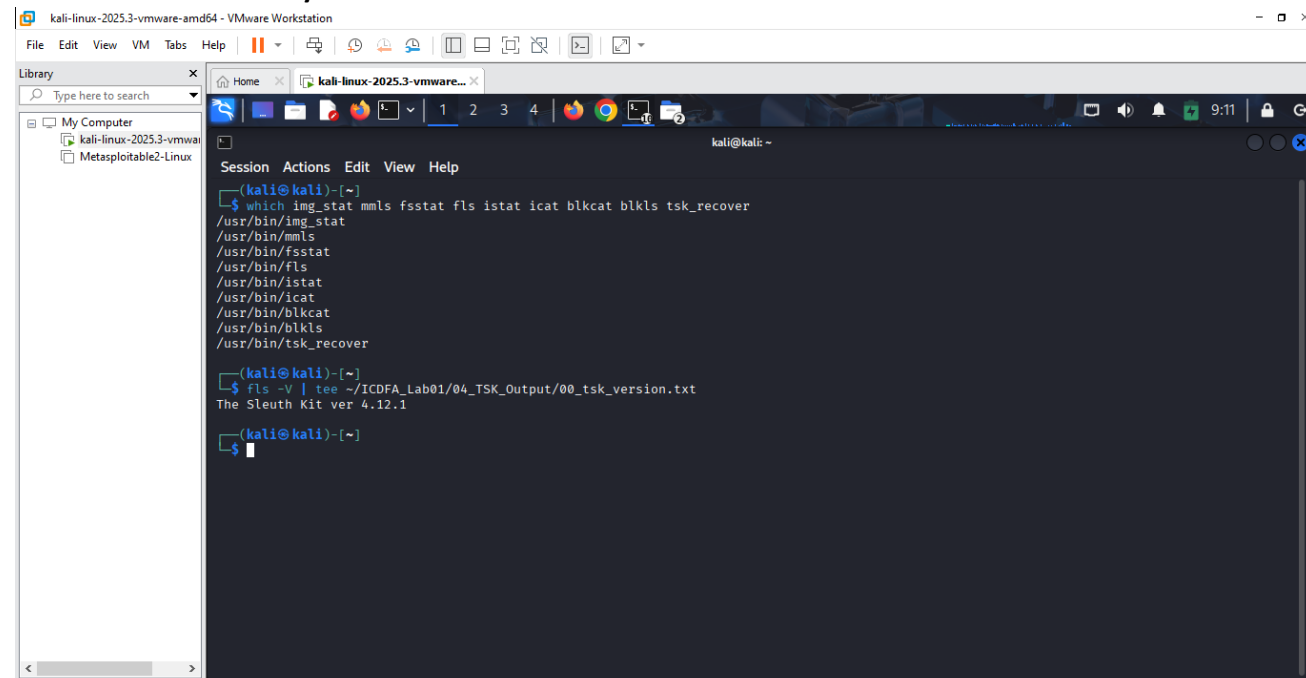


Fig 7.4 Confirm Sleuth Kit is installed and the version.

Explanation

I confirmed that the required Sleuth Kit tools were installed and available.

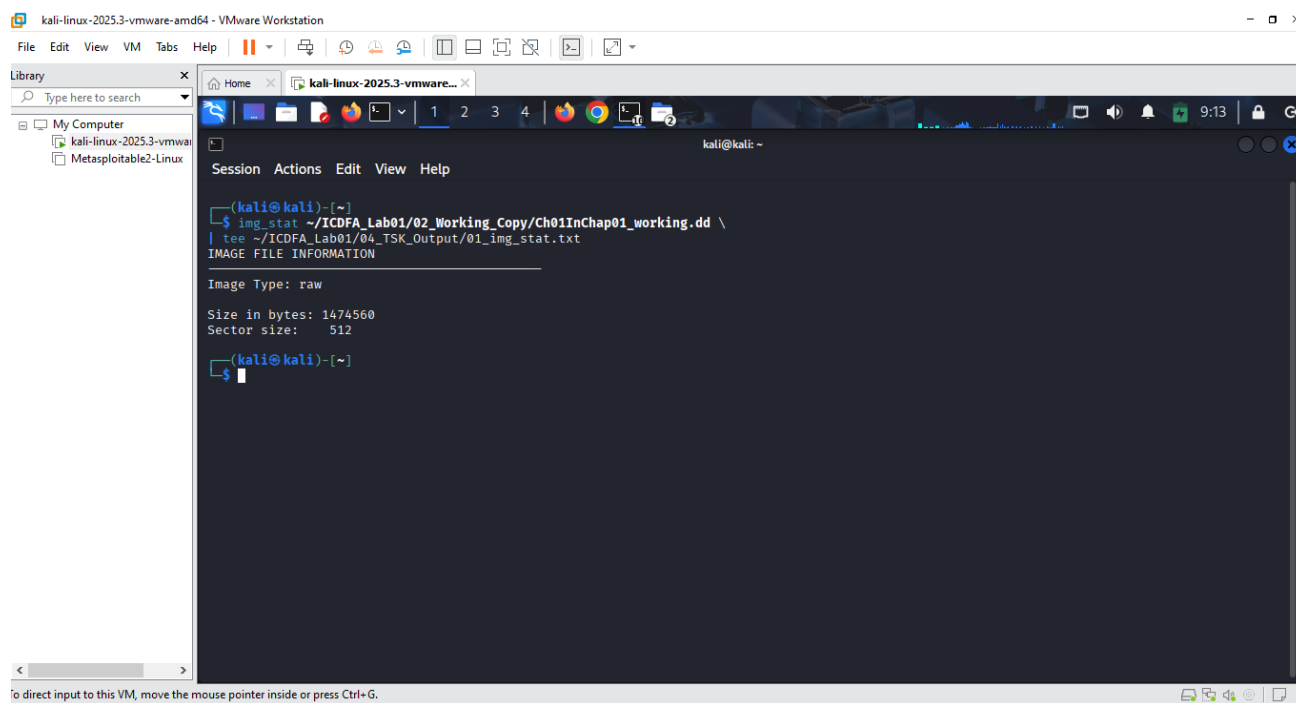


Fig 7.5 tsk img stat.

Explanation

img_stat confirmed that the evidence was a raw image and displayed its size.

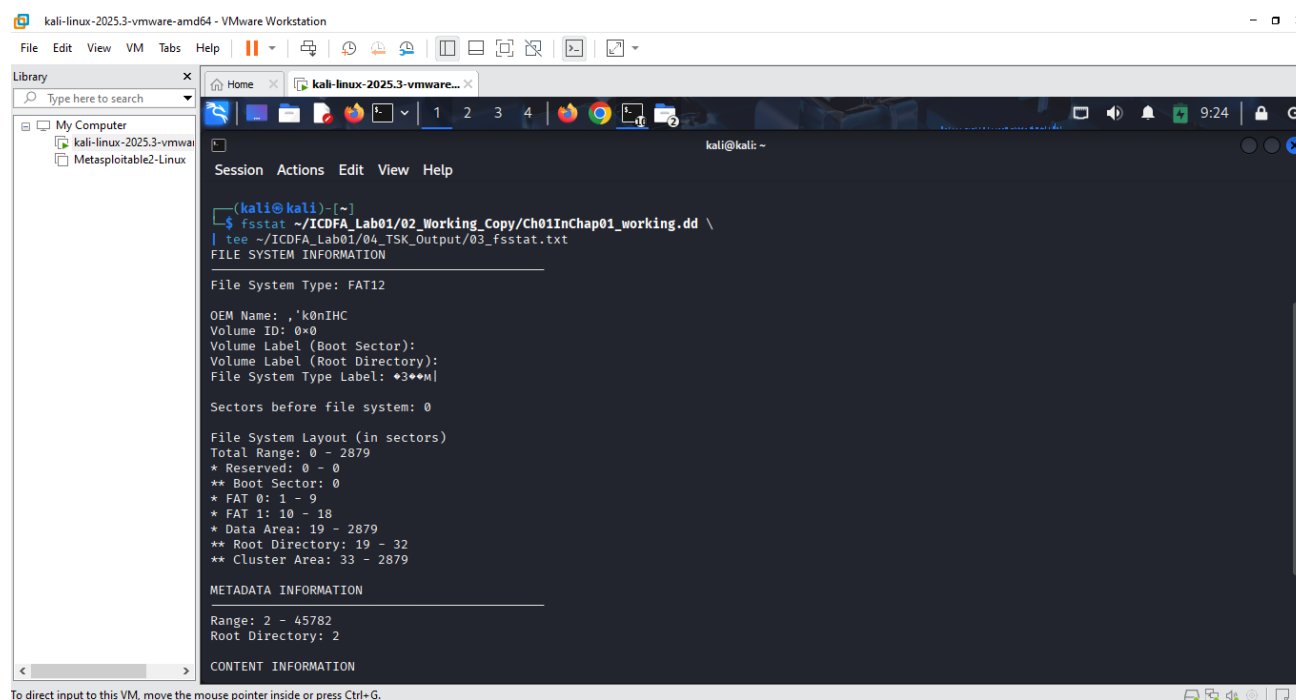


Fig 7.6 fsstat filesystem information.

Explanation

fsstat identified the filesystem as FAT12 and displayed its structure.

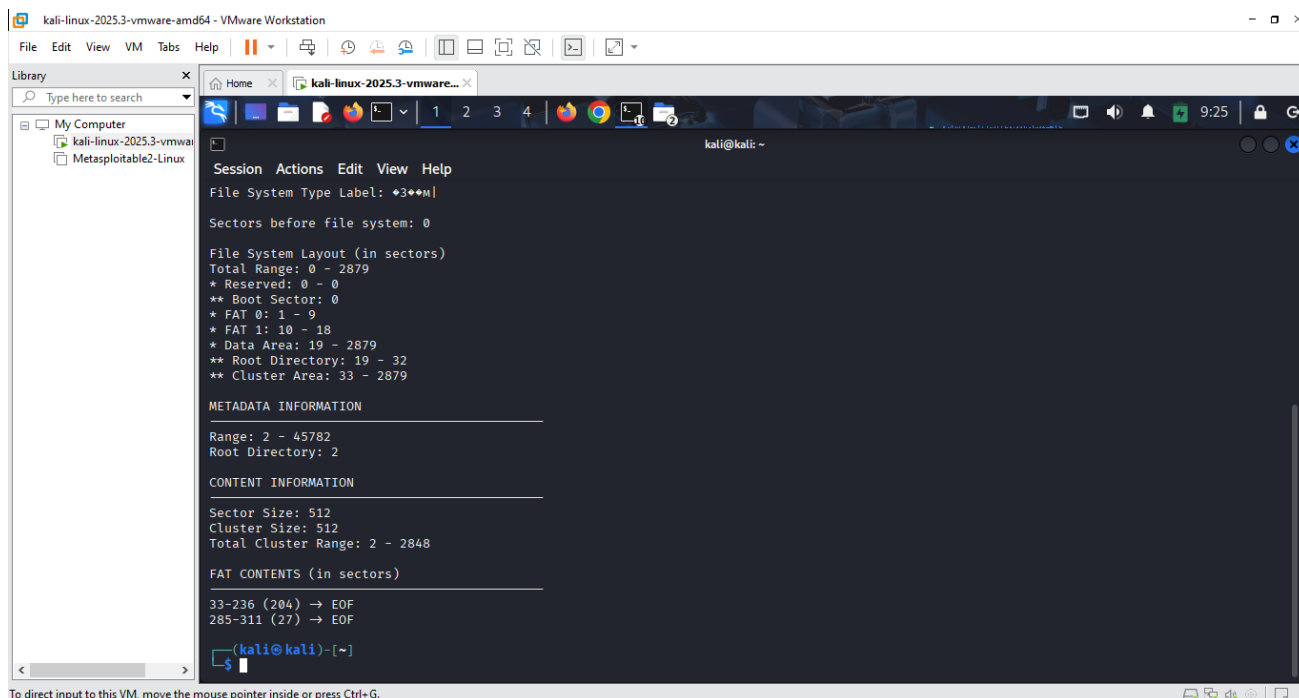


Fig 7.6.1 fsstat filesystem information page 1.

Explanation

This shows more details about the FAT12 filesystem layout and sectors.

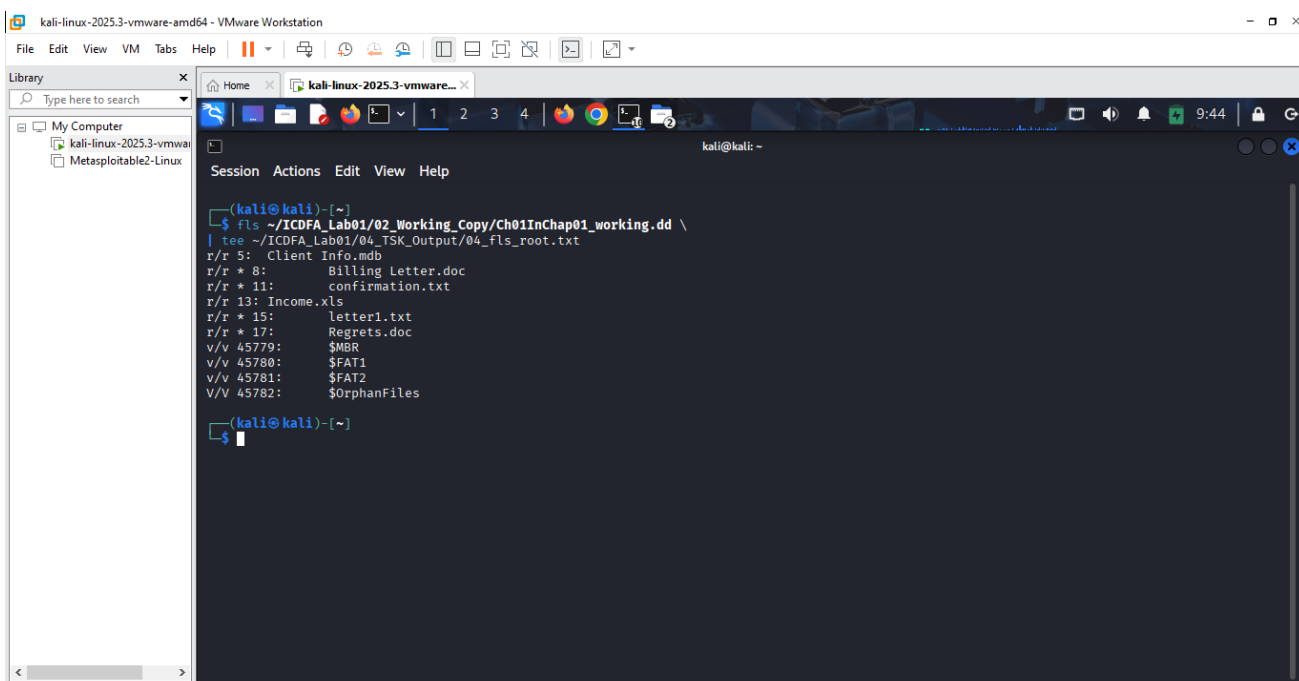


Fig 7.7 fls root listing.

Explanation

fls displayed the files and directory entries stored in the image.

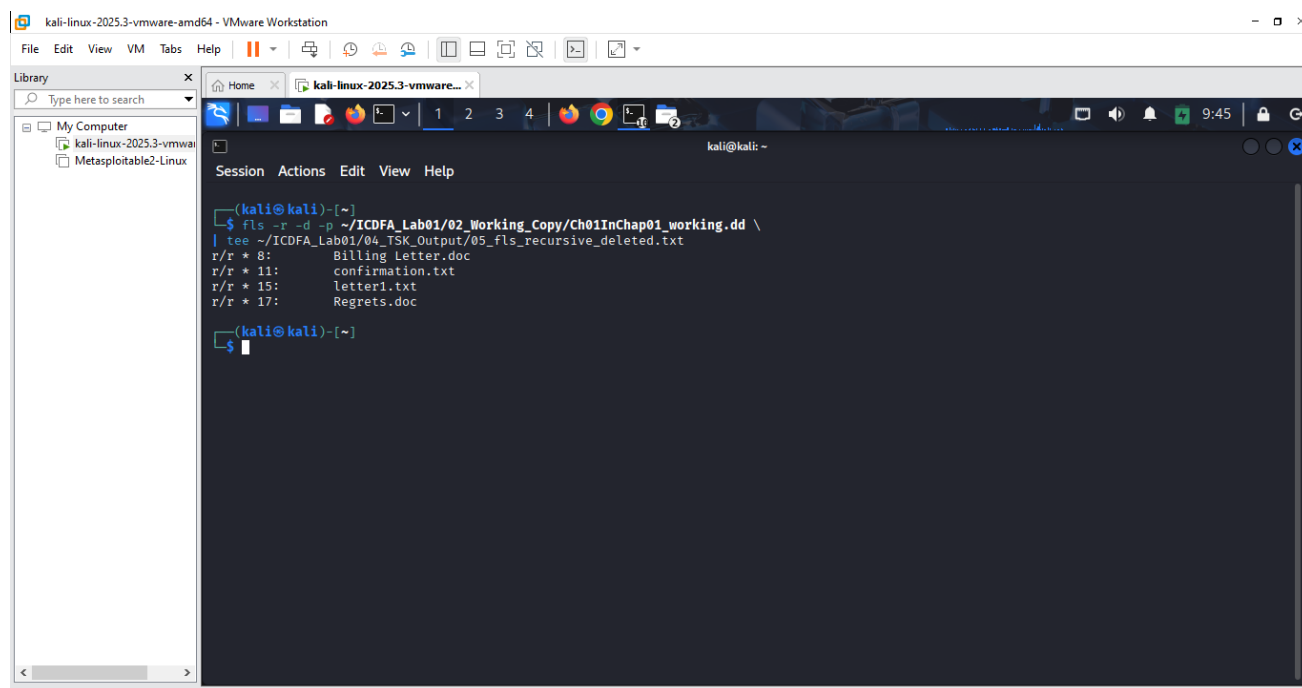


Fig 7.8 Recursive deleted file listing.

Explanation

I used fls recursively to list the deleted files in the image.

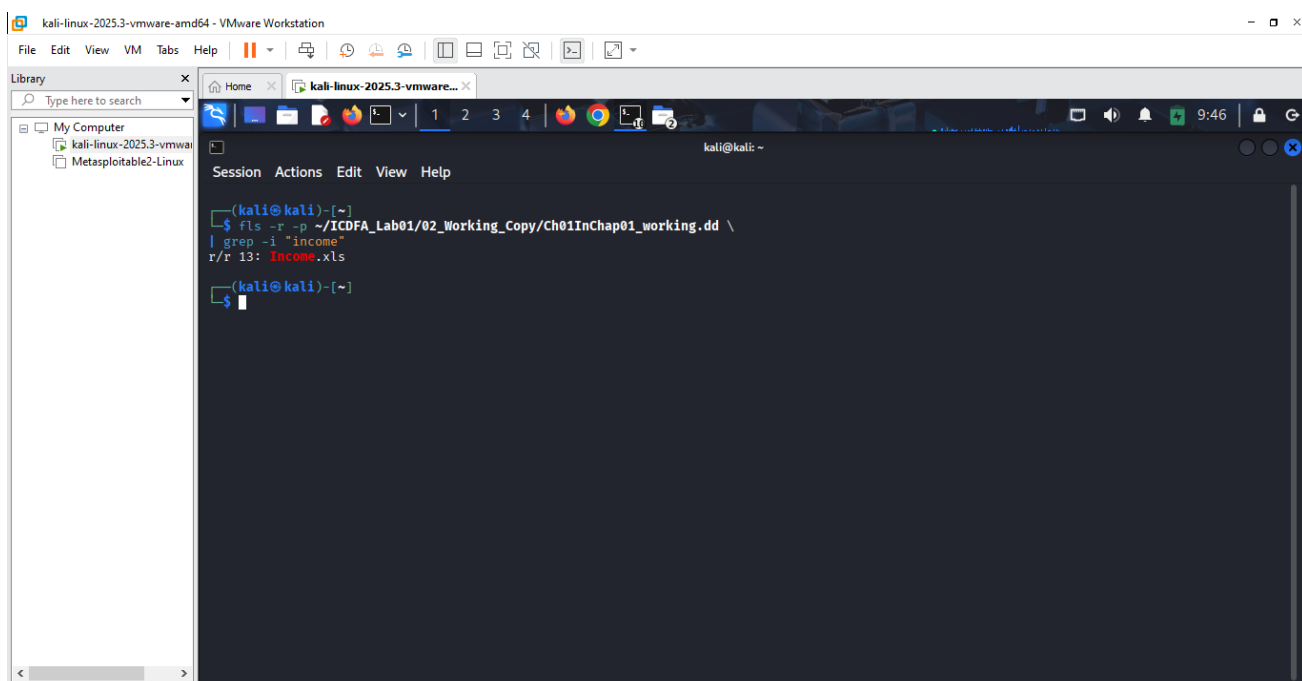


Fig 7.9 Income_xls identified.

Explanation

INCOME.XLS was located from the Sleuth Kit file listing.

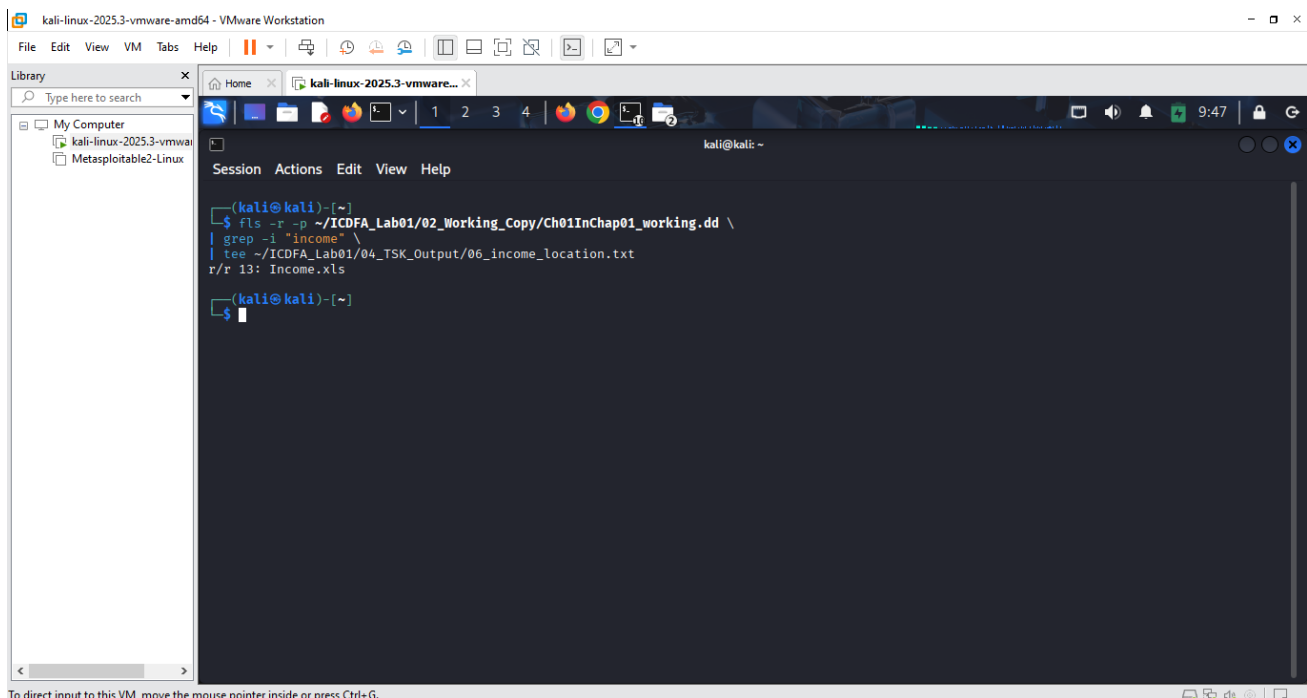


Fig 8. Income_xls identified page 1.

Explanation

This confirms that INCOME.XLS has metadata address **13**.

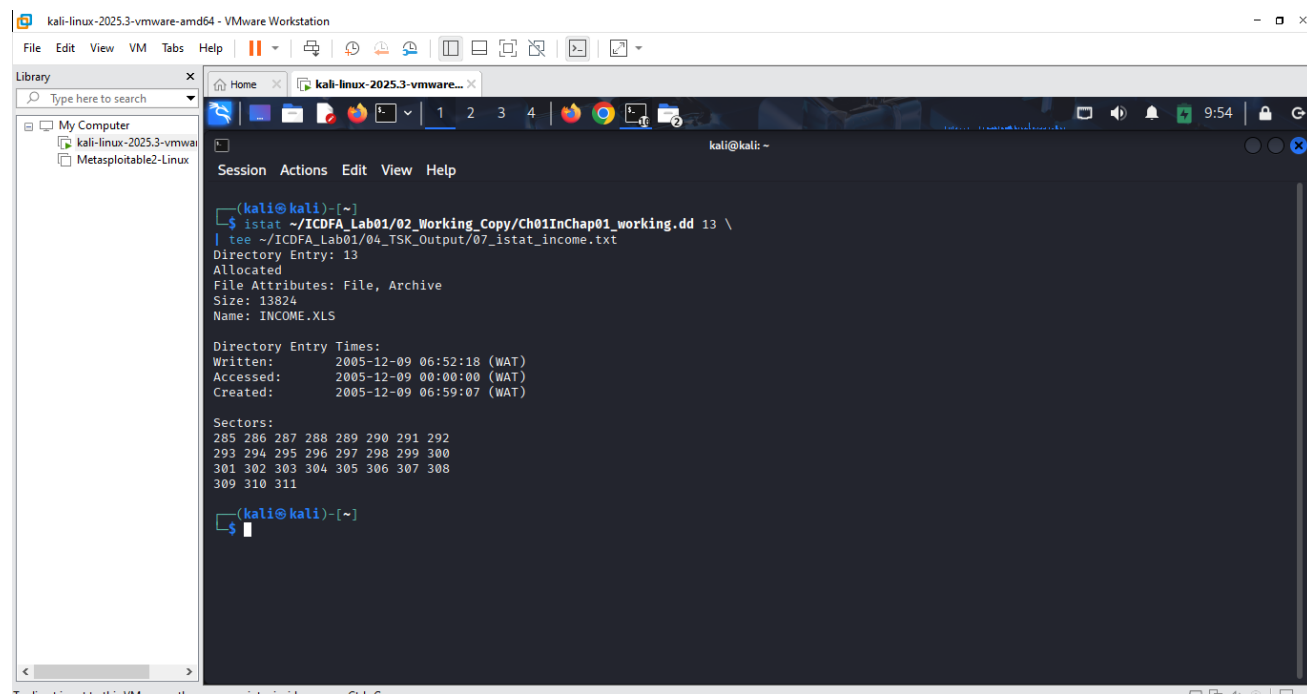


Fig 8.1 istat income metadata.

Explanation

istat showed the file size, timestamps and sectors used by INCOME.XLS.

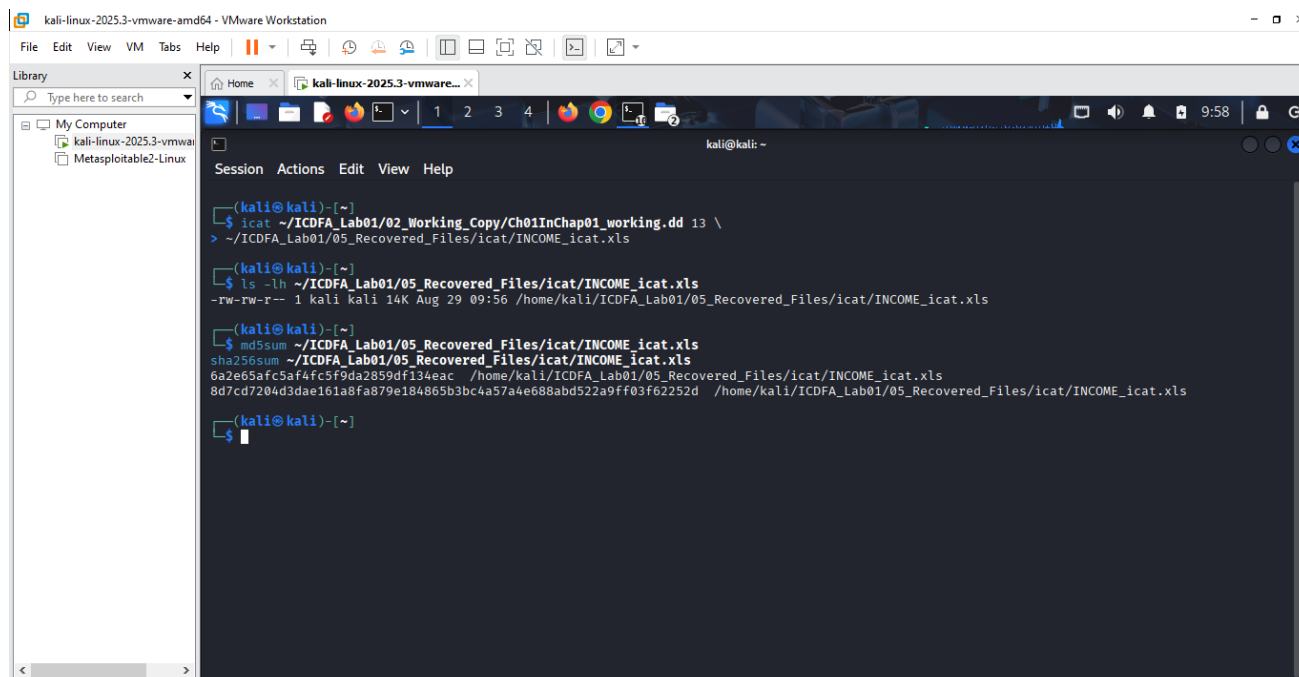


Fig 8.2 Verifying the md5sum.

Explanation

I calculated the hash of the spreadsheet recovered with icat.

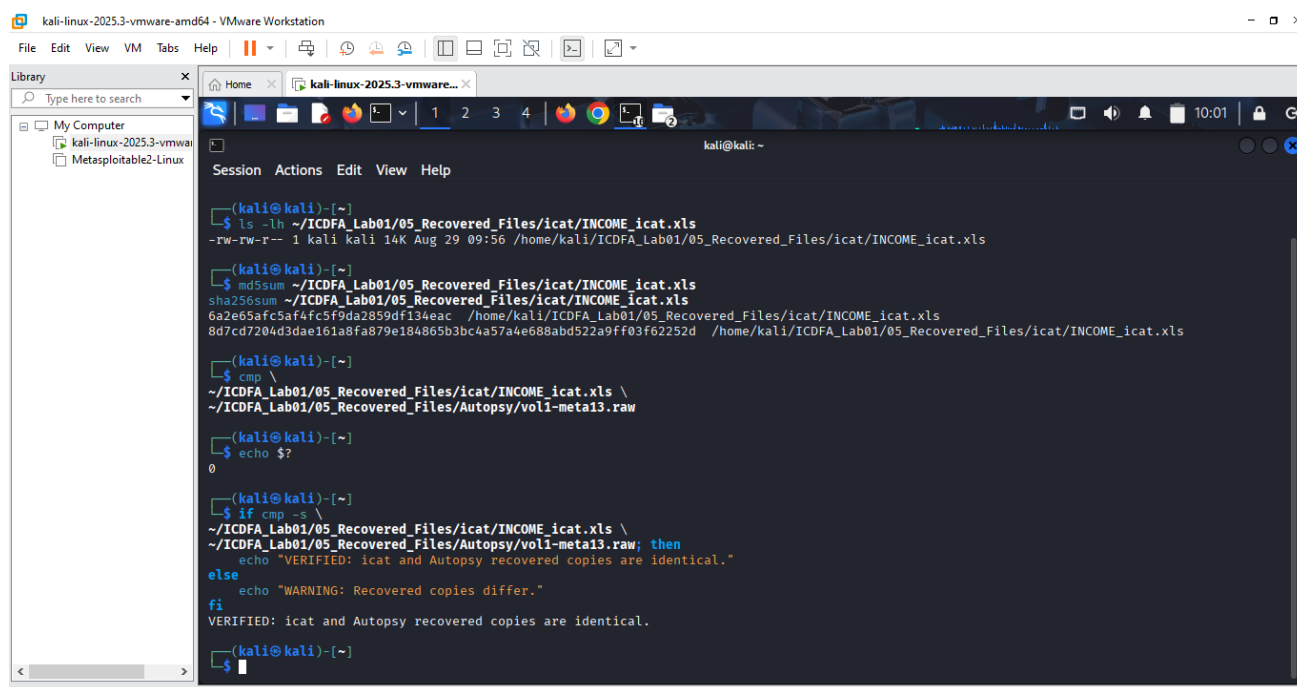
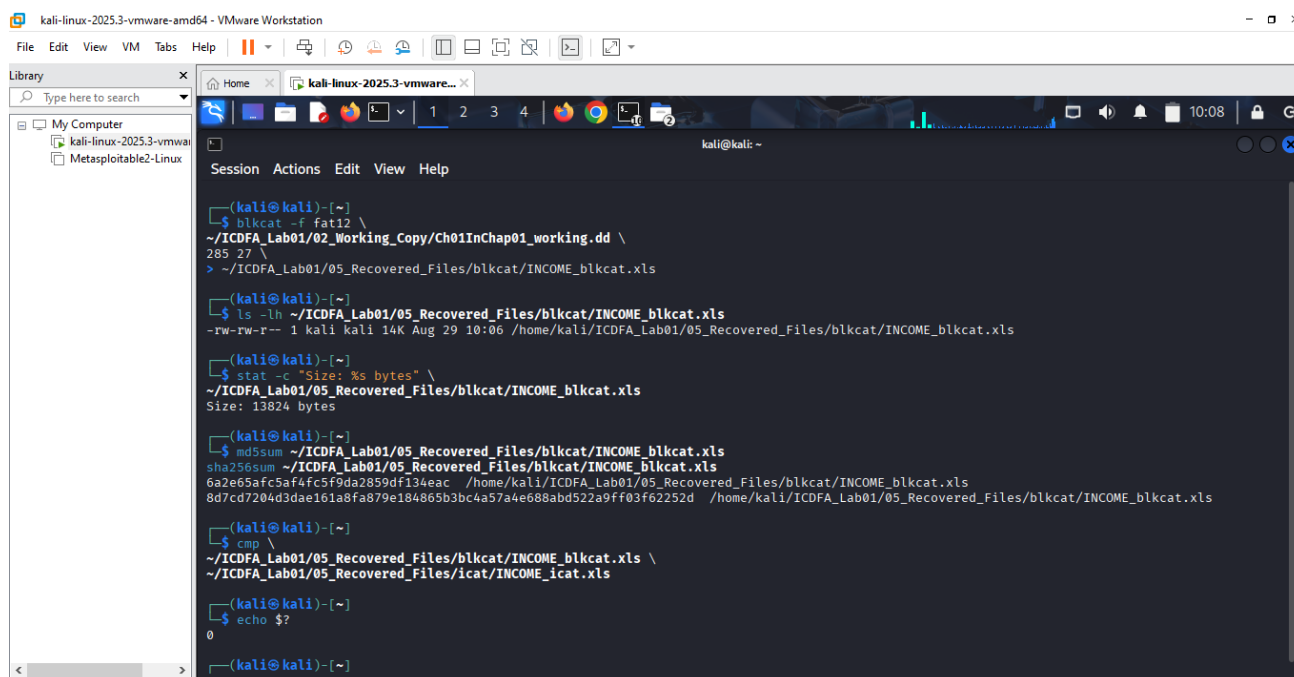


Fig 8.3 icat income recovery verified.

Explanation

The icat copy matched the Autopsy recovery, confirming both were identical.



```
(kali@kali)-[~]
$ blkcat -f fat12 \
~/ICDFA_Lab01/02_Working_Copy/Ch01InChap01_working.dd \
285 27 \
> ~/ICDFA_Lab01/05_Recovered_Files/blkcat/INCOME_blkcat.xls

(kali@kali)-[~]
$ ls -lh ~/ICDFA_Lab01/05_Recovered_Files/blkcat/INCOME_blkcat.xls
-rw-rw-r-- 1 kali kali 14K Aug 29 10:06 /home/kali/ICDFA_Lab01/05_Recovered_Files/blkcat/INCOME_blkcat.xls

(kali@kali)-[~]
$ stat -c "Size: %s bytes" \
~/ICDFA_Lab01/05_Recovered_Files/blkcat/INCOME_blkcat.xls
Size: 13824 bytes

(kali@kali)-[~]
$ md5sum ~/ICDFA_Lab01/05_Recovered_Files/blkcat/INCOME_blkcat.xls
sha256sum ~/ICDFA_Lab01/05_Recovered_Files/blkcat/INCOME_blkcat.xls
6a2e65afc5a4fc5f9da2859df134eac /home/kali/ICDFA_Lab01/05_Recovered_Files/blkcat/INCOME_blkcat.xls
8d7cd7204d3dae161a8fa879e184865b3bc4a57a4e688abd522a9ff03f62252d /home/kali/ICDFA_Lab01/05_Recovered_Files/blkcat/INCOME_blkcat.xls

(kali@kali)-[~]
$ cmp \
~/ICDFA_Lab01/05_Recovered_Files/blkcat/INCOME_blkcat.xls \
~/ICDFA_Lab01/05_Recovered_Files/icat/INCOME_icat.xls

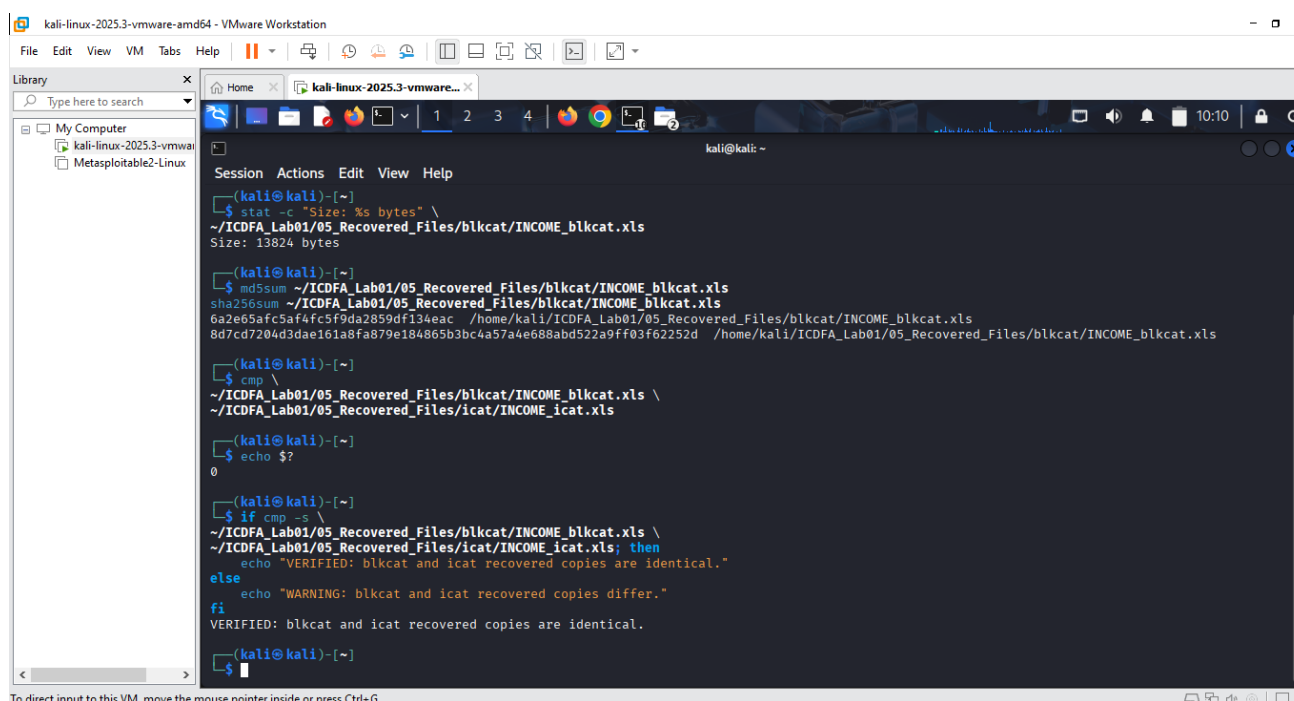
(kali@kali)-[~]
$ echo $?
0

(kali@kali)-[~]
```

Fig 8.4 blkcat income recovery verified.

Explanation

I recovered INCOME.XLS directly from its data sectors using blkcat.



```
(kali@kali)-[~]
$ stat -c "Size: %s bytes" \
~/ICDFA_Lab01/05_Recovered_Files/blkcat/INCOME_blkcat.xls
Size: 13824 bytes

(kali@kali)-[~]
$ md5sum ~/ICDFA_Lab01/05_Recovered_Files/blkcat/INCOME_blkcat.xls
sha256sum ~/ICDFA_Lab01/05_Recovered_Files/blkcat/INCOME_blkcat.xls
6a2e65afc5a4fc5f9da2859df134eac /home/kali/ICDFA_Lab01/05_Recovered_Files/blkcat/INCOME_blkcat.xls
8d7cd7204d3dae161a8fa879e184865b3bc4a57a4e688abd522a9ff03f62252d /home/kali/ICDFA_Lab01/05_Recovered_Files/blkcat/INCOME_blkcat.xls

(kali@kali)-[~]
$ cmp \
~/ICDFA_Lab01/05_Recovered_Files/blkcat/INCOME_blkcat.xls \
~/ICDFA_Lab01/05_Recovered_Files/icat/INCOME_icat.xls

(kali@kali)-[~]
$ echo $?
0

(kali@kali)-[~]
$ if cmp -s \
~/ICDFA_Lab01/05_Recovered_Files/blkcat/INCOME_blkcat.xls \
~/ICDFA_Lab01/05_Recovered_Files/icat/INCOME_icat.xls; then
echo "VERIFIED: blkcat and icat recovered copies are identical."
else
echo "WARNING: blkcat and icat recovered copies differ."
fi
VERIFIED: blkcat and icat recovered copies are identical.

(kali@kali)-[~]
$
```

Fig 8.4.1 blkcat income recovery verified page 1.

Explanation

The blkcat recovery matched the icat copy during verification.

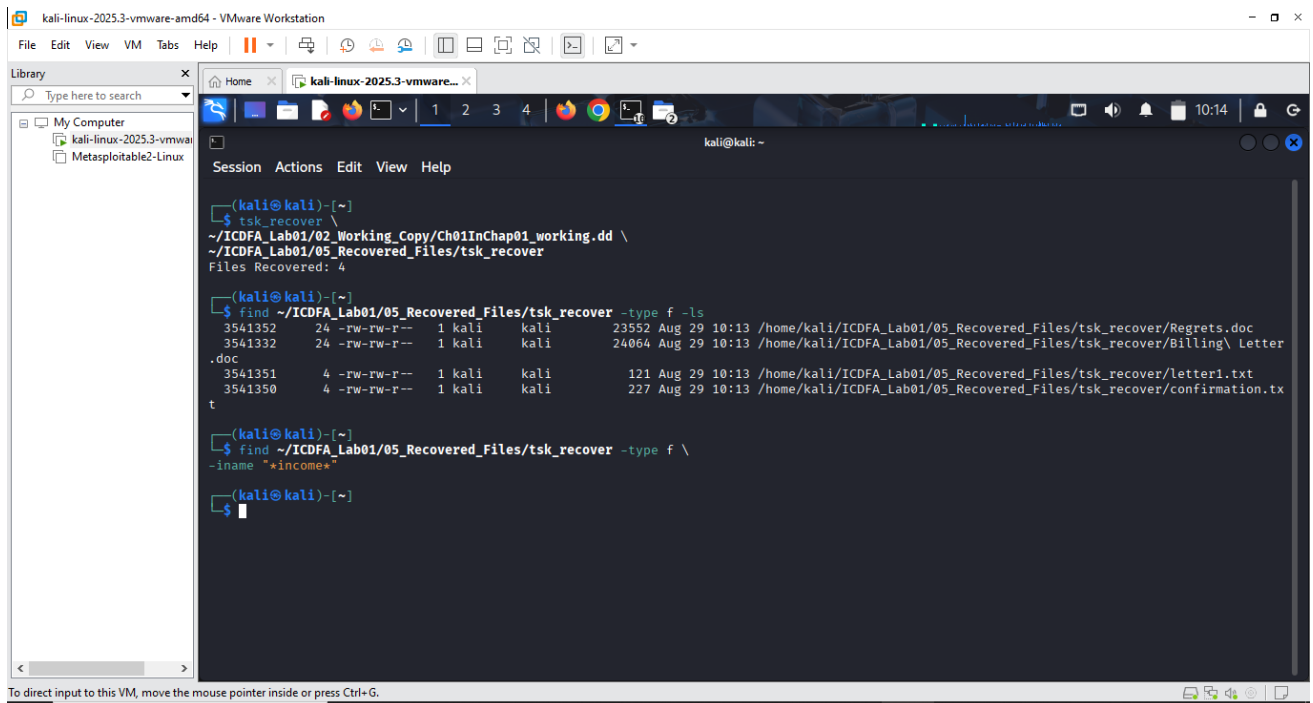


Fig 8.5 Recover deleted files.

Explanation

tsk_recover successfully recovered the deleted files from the image.

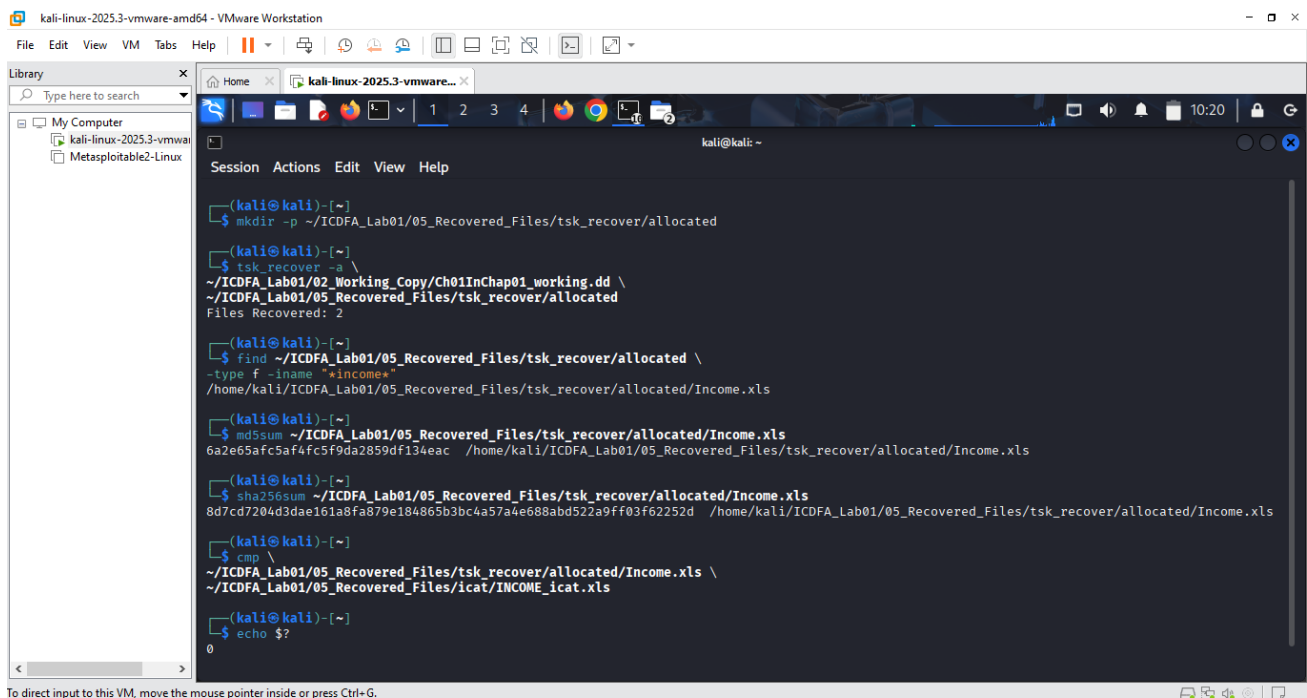


Fig 8.6 Recover income hash verified.

Explanation

I recovered the allocated Income.xls with tsk_recover and verified its hash.

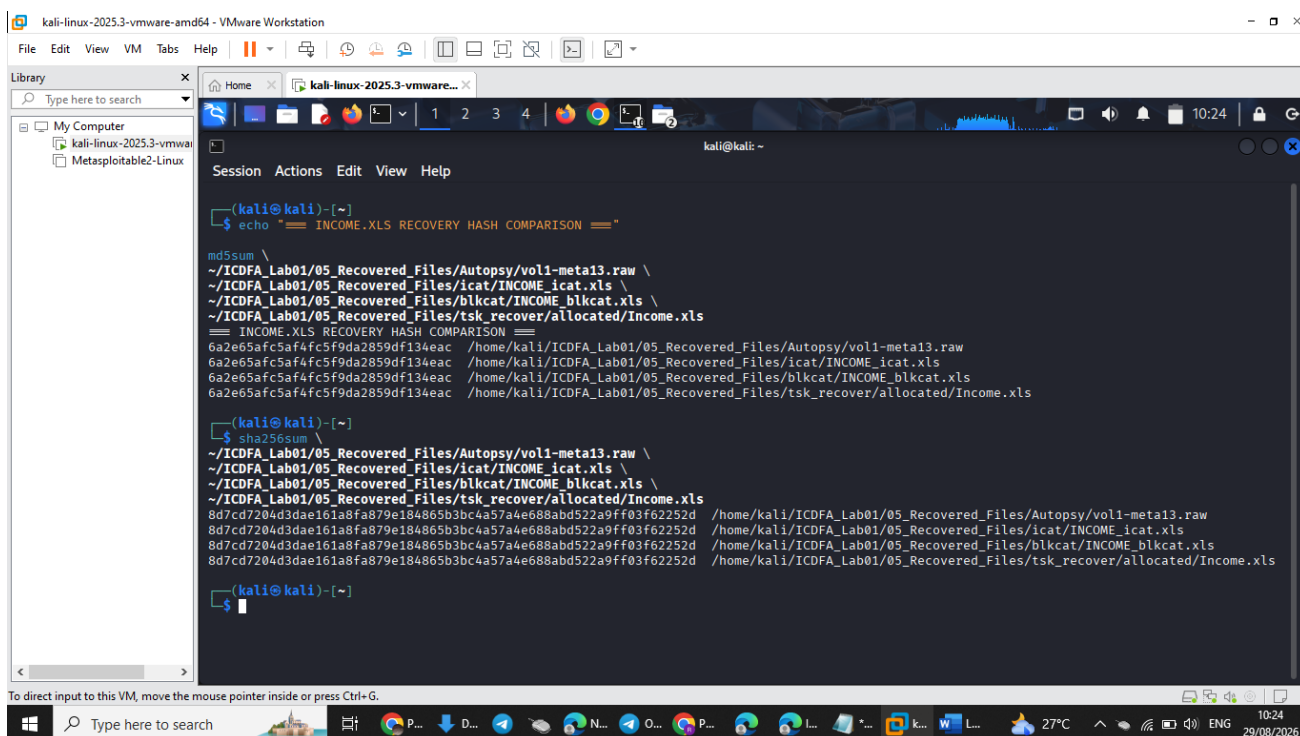


Fig 8.7 income xls all recovery hash comparison.

Explanation

I compared the hashes of all four recovered copies of INCOME.XLS.

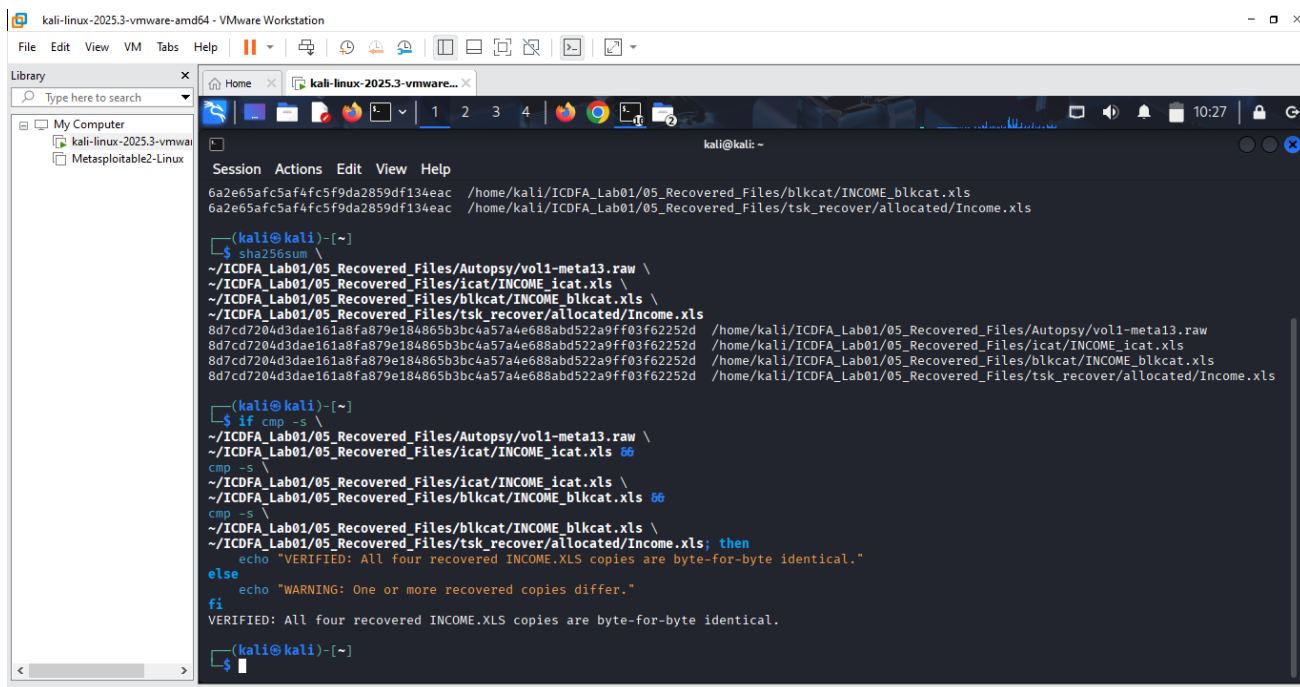


Fig 8.8 income xls byte for byte verification.

Explanation

The comparison confirmed that all recovered INCOME.XLS copies were byte-for-byte identical.

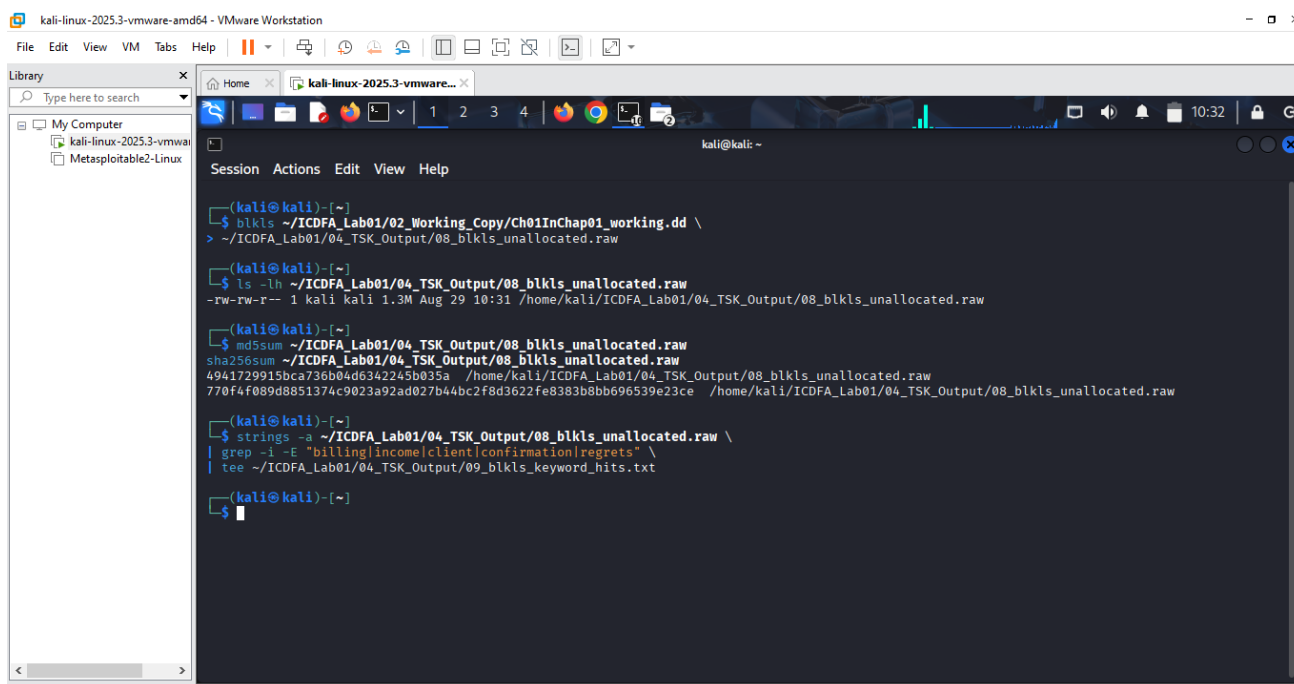


Fig 8.9 blkls unallocated analysis.

Explanation

I used blkls to extract and examine the unallocated space from the image.

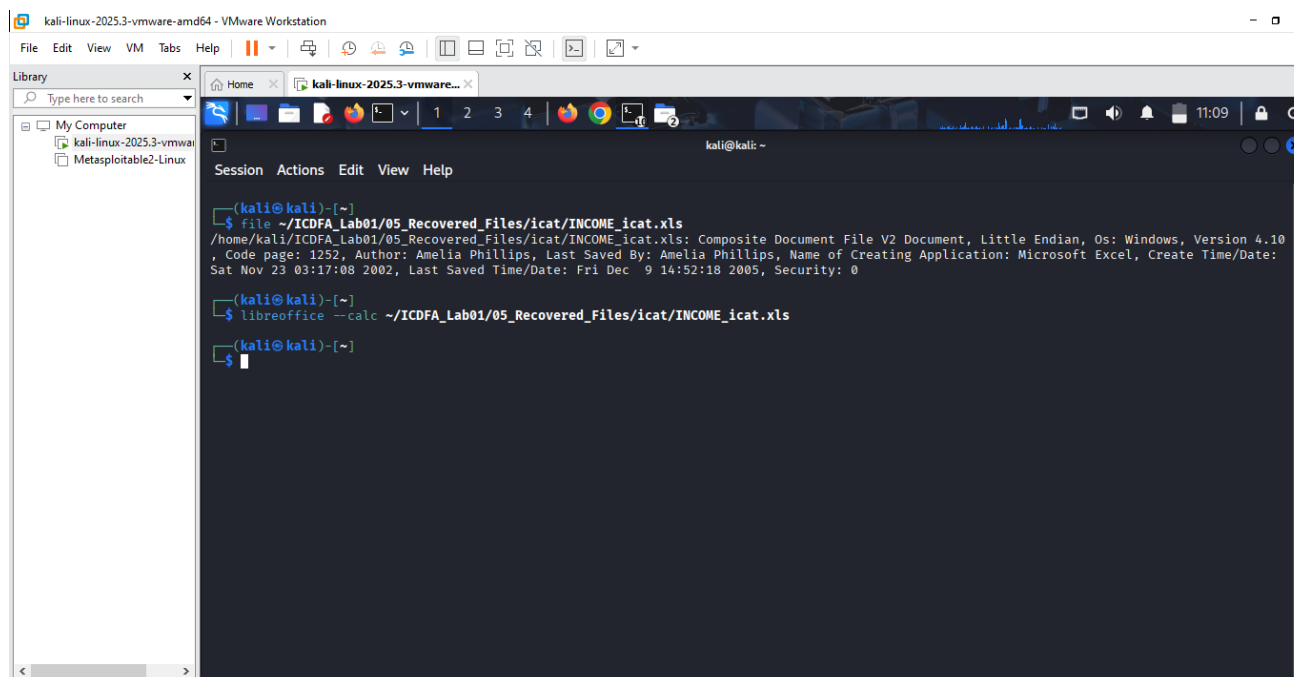
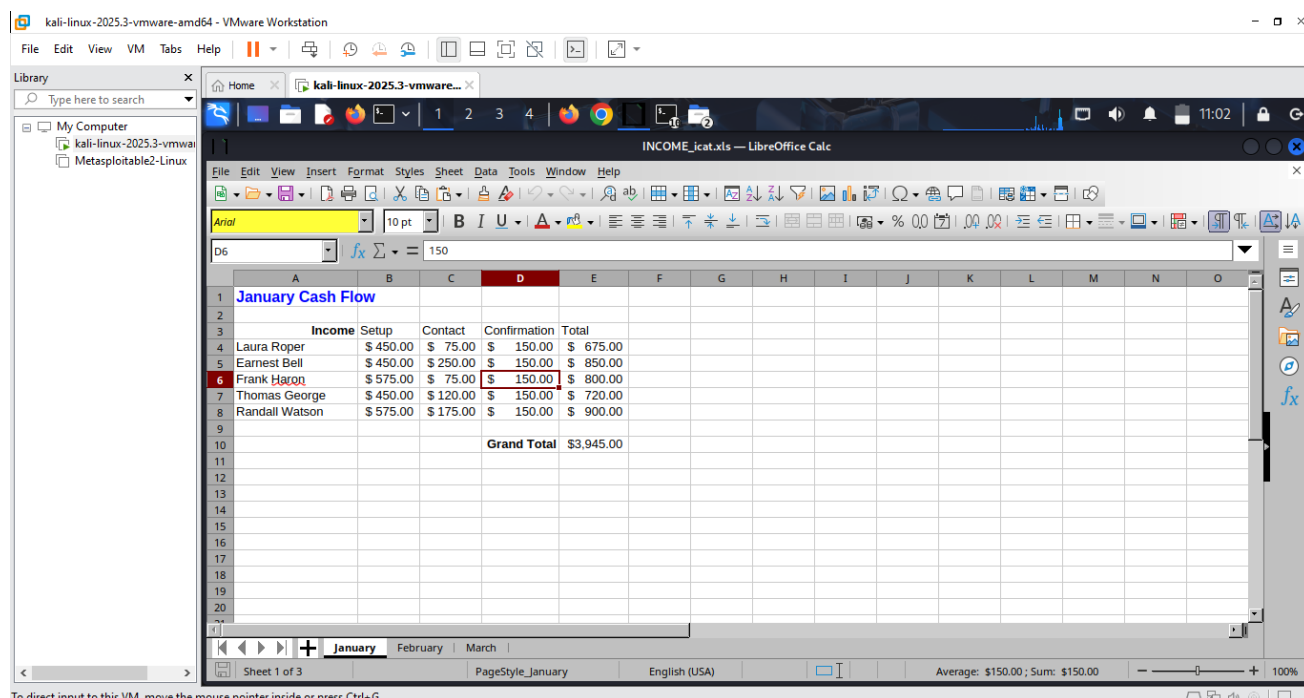


Fig 9.0 Verifying the file type.

Explanation

The file command confirmed the recovered spreadsheet's file type.



To direct input to this VM, move the mouse pointer inside or press Ctrl+G.

Fig 9.1 income xls successfully examined.

Explanation

I successfully opened the recovered INCOME.XLS in LibreOffice Calc and viewed its contents.

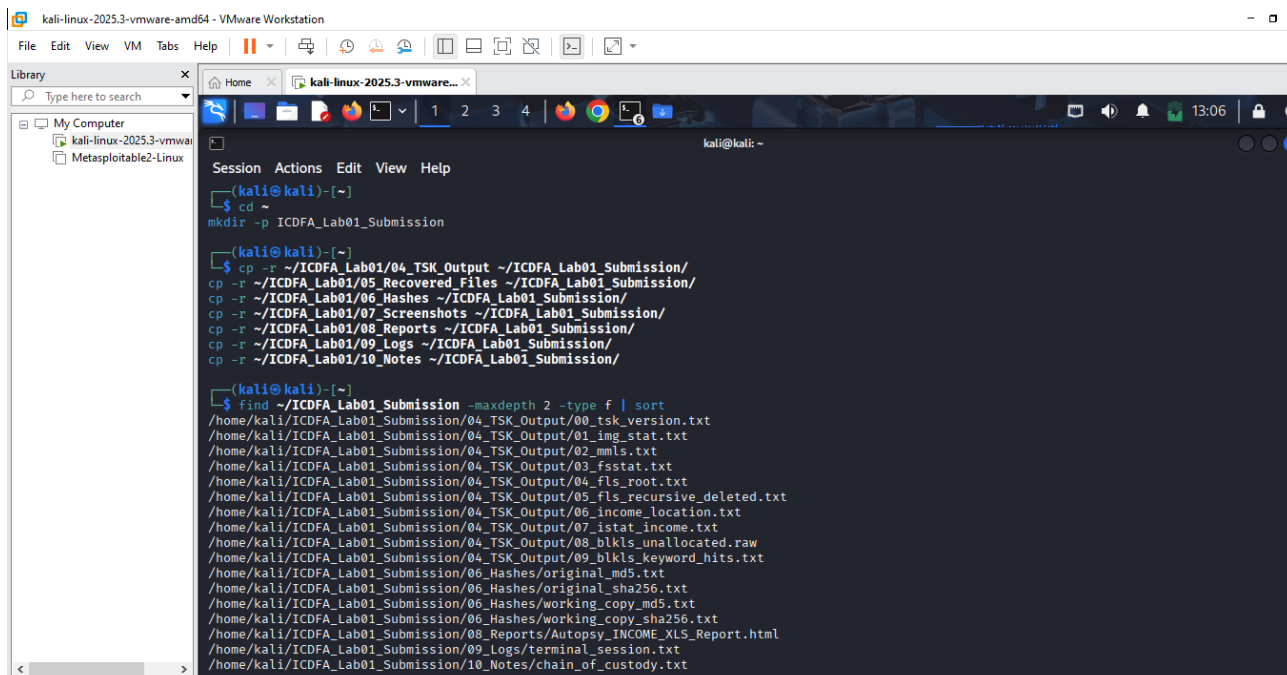
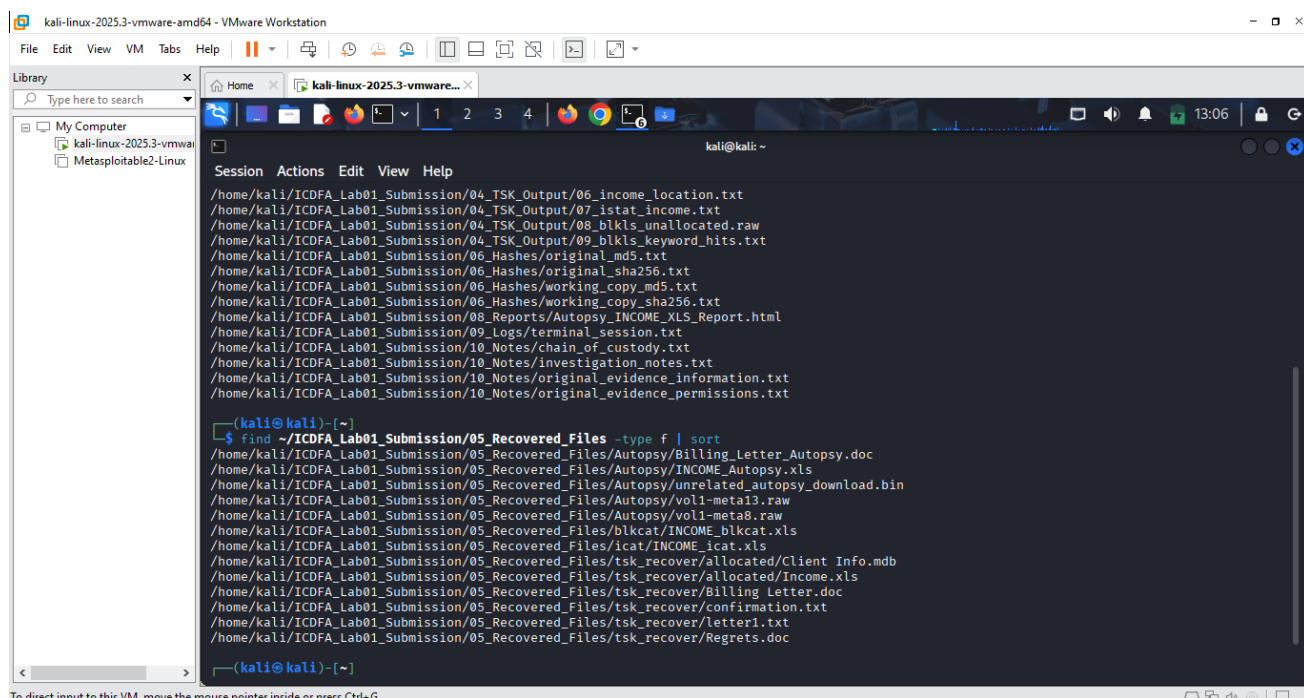


Fig 9.2 Arranging the file for Submission.



To direct input to this VM, move the mouse pointer inside or press Ctrl+G.

Fig 9.2.1 Arranging the file for Submission page 1.

Conclusion

The practical provided a complete digital forensic examination of the Ch01InChap01.dd image while ensuring that the original evidence remained protected throughout the investigation. A separate forensic working directory was created, the original image was hashed and set to read-only, and a verified working copy was used for the analysis. Using Autopsy, the image was correctly identified as a FAT12 filesystem, its integrity was verified, deleted files were identified, relevant metadata was examined, keyword searches were carried out, important evidence was recovered and documented, and a final forensic report was generated.

The Sleuth Kit was also used to independently verify the findings obtained from Autopsy. It confirmed the filesystem structure, deleted file entries, and the location of INCOME.XLS. The spreadsheet was located at metadata address 13, had a size of 13,824 bytes, and occupied data sectors 285–311. The file was successfully recovered using Autopsy, icat, blkcat, and tsf_recover. Hash verification and byte-for-byte comparison showed that all four recovered copies were identical, confirming the consistency of the recovery process. In addition, blkls was used to examine the unallocated space, and the recovered spreadsheet was successfully opened and viewed in LibreOffice Calc.

Overall, the practical showed that proper forensic procedures were followed and that the integrity of the evidence was maintained throughout the examination. It also demonstrated that the findings could be reliably reproduced using both graphical and command-line forensic tools.

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Fig 4.3.1 File Analyzing Page 1.

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Fig 4.4.1 File Analyzing and all deleted files page 1.

Fig 4.5 Selecting the Income.xls file.

Fig 4.5.1 Selecting the Income.xls file page 1.

Fig 4.6 Verifying the md5sum hash after download.

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Fig 8.5 Recover deleted files.

Fig 8.6 Recover income hash verified.

Fig 8.7 income xls all recovery hash comparison.

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Fig 8.9 blkls unallocated analysis.

Fig 9.0 Verifying the file type.

Fig 9.1 income xls successfully examined.

Fig 9.2 Arranging the file for Submission.

Fig 9.2.1 Arranging the file for Submission page 1.

References

Ch01InChap01.dd. Authorised training forensic disk image examined during the laboratory practical.
ICDFA Materials.